

UMC 0.11 um Logic and Mixed-mode Al Advanced Enhancement D5.0G3.3 Common Drain LDMOSFET SPICE Model Document

**Version 0.1 Phase 1
2012/06/25**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.1_p1	2012/06/25	Ho Cheng Hu	1. Original release 2. AC parameters are copied from l110ae_ldcd_d85g33_v012

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-mode 1P8M Metal Metal Capacitor Al
Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

D5.0G3.3 n-ch MOSFET
D5.0G3.3 p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.2.4

For detailed descriptions of these compact model equations and parameters, please refer to this material.

BSIM3V3.2.4 MOSFET Model User Manual

In BSIM3V3.2.4, there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the silicon data, which can be found in the following sections.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2009.03

Cadence SPECTRE version 7.0.1.053

Mentor Graphics ELDO version 2009.1 (v7.1_1.1)

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET

FF: Fast n-ch MOSFET and Fast p-ch MOSFET

SS: Slow n-ch MOSFET and Slow p-ch MOSFET

FNFP: Fast n-ch MOSFET and Slow p-ch MOSFET

SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

.lib "/ModelFileName.lib" CaseName

, where CaseName can be tt, ff, ss, fnsp, or snfp.

- Define device condition by 'Xx' statement;

Example : X1 Vdd Vgg Vss Vbb DeviceName w=2*wf l=lf nf=nf

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

SetOption1 options scale=0.9

include "/ModelFileName.lib.scs" section=CaseName

, where CaseName can be tt, ff, ss, fnsp, or snfp.

- Define device condition by 'xx' statement;

Example : x1 vdd vgg vss vbb DeviceName w=2*wf l=lf nf=nf

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

.lib "/ModelFileName.lib.eldo" CaseName

, where CaseName can be tt, ff, ss, fnsp, or snfp.

- Define device condition by 'Xx' statement;

Example : X1 Vdd Vgg Vss Vbb DeviceName w=2*wf l=lf nf=nf

ModelFileName for D5.0G3.3 MOSFET: l110ae_ldcd_d50g33_v011

DeviceName for D5.0G3.3 n-ch MOSFET: n_ldcd_d50g33_l110ae

DeviceName for D5.0G3.3 p-ch MOSFET: p_ldcd_d50g33_l110ae

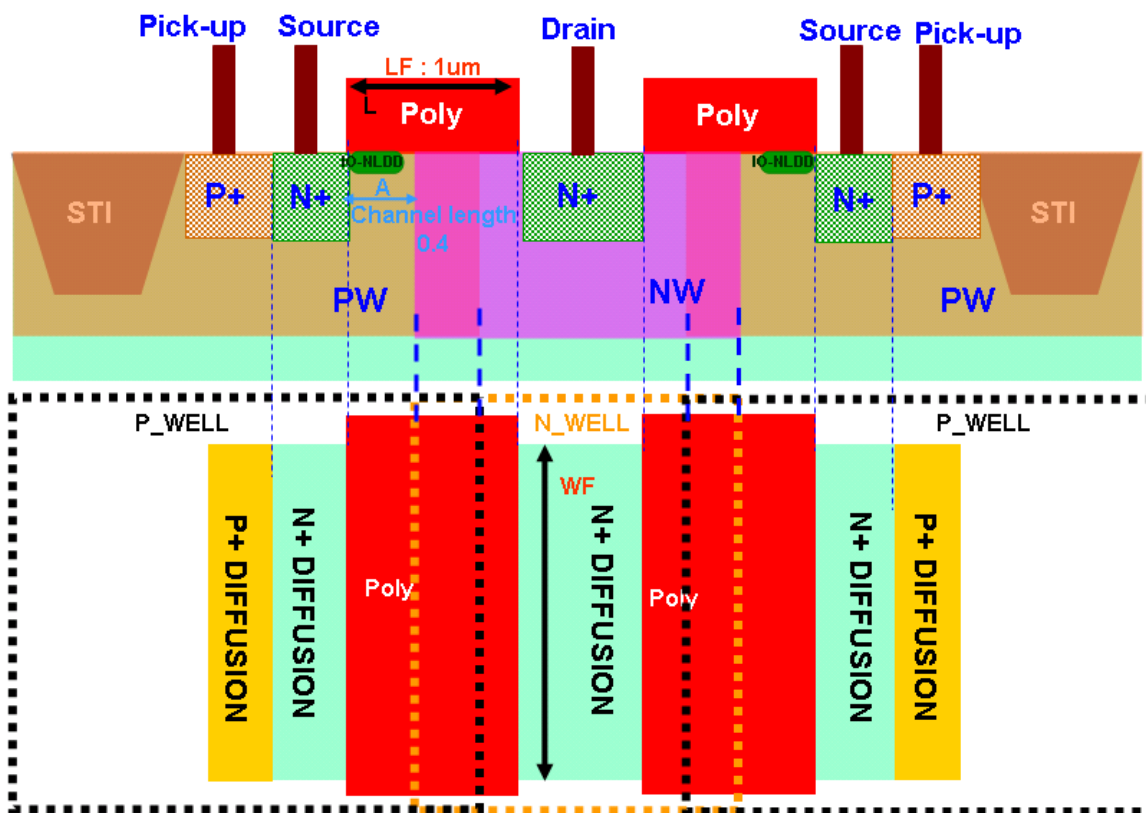


Figure 1-1 Top and Cross-sectional views of D8.5G3.3 n-ch MOSFETs unit

The area/periphery of drain and source diodes of D8.5G3.3 n-ch MOSFETs unit after 90% shrink and poly brought-up 12nm can be calculated as: $ad=(w/2+1.8um)*1.584u$, $pd=w+6.768u$, $as=w*0.498um$, $ps=w+1.992um$.

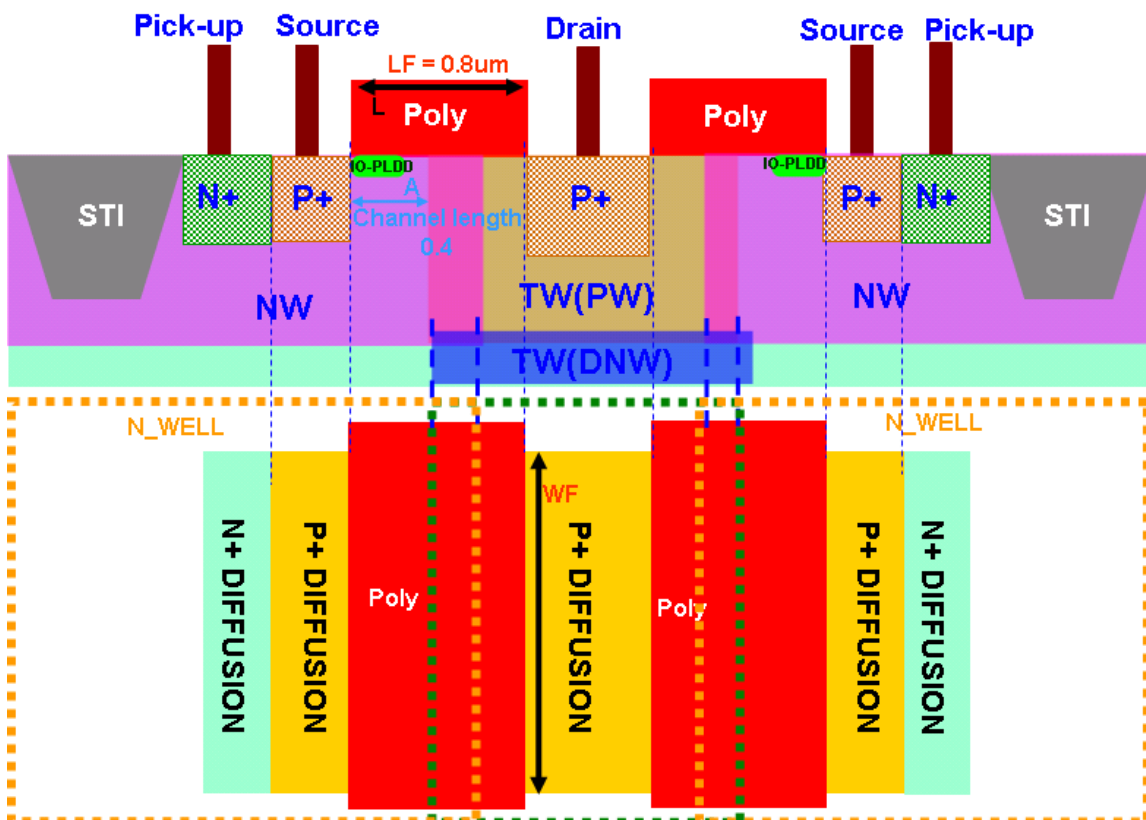


Figure 1-2 Top and Cross-sectional views of D5.0G3.3 p-ch MOSFETs unit

The area/periphery of drain and source diodes of D5.0G3.3 p-ch MOSFETs unit after 90% shrink and poly brought-up 12nm can be calculated as: $ad=(w/2+1.8um)*1.224u$, $pd=w+6.048u$, $as=w*0.498um$, $ps=w+1.992um$.

As stated above, one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and will not be guaranteed.

Geometry range (After 90% shrink) :

For D5.0G3.3 n-ch MOSFET:

$$L_{DES} = 0.9 \mu m$$

$$18 \mu m \leq W_{DES} \leq 180 \mu m$$

For D5.0G3.3 p-ch MOSFET:

$$L_{DES} = 0.72 \mu m$$

$$18 \mu m \leq W_{DES} \leq 180 \mu m$$

Voltage range:

For D5.0G3.3 n-ch MOSFET:

$$0V \leq V_{GS} \leq 3.3V (*1.1)$$

$$0V \leq V_{DS} \leq 5.0V$$

$$V_{BS} = 0V$$

For D5.0G3.3 p-ch MOSFET:

$$0V \geq V_{GS} \geq -3.3V (*1.1)$$

$$0V \geq V_{DS} \geq -5.0V$$

$$V_{BS} = 0V$$

Temperature range:

$$-40^{\circ}C \sim +125^{\circ}C$$

*Note: (1) L_{DES} denotes length of polysilicon on the diffusion unit, which is always 1 μm and 0.8 μm before 90% shrink for nMOSFET and pMOSFET, respectively,
(2) The bias difference between any two terminals of a MOSFET cannot exceed V_{CC} ,*

2 D5.0G3.3 n-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW1PA	--	--	--
Lot ID	D6G6T.1	--	--	--
Wafer Number	#03	--	--	--

2.2 Electrical Parameters

Key electrical parameters for the characterization of this device are summarized in the table below. Linear as well as saturation threshold voltages are extracted as I_D equal to $300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 - 0.54\mu\text{m})$ and $-70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 - 0.36\mu\text{m})$ for NMOS and PMOS, respectively.

Table 2-2 Electrical parameters for NMOS

All parameters are given for $T=25\text{ }^{\circ}\text{C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
-----------	-----------	------	--------	----------------	-----------------	----------------

Narrow and short channel device ($W/L=18\text{ }\mu\text{m} / 0.9\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS}=0.1\text{ V}$	V	0.723	0.730	0.730	0.723
V_{TSAT}	$V_{DS}=5\text{ V}$	V	0.707	0.714	0.714	0.707
I_{ON}	$V_{DS}=5\text{ V}$, $V_{GS}=3.3\text{ V}$	$\mu\text{A}/\mu\text{m}$	415	416	416	415
I_{OFF}	$V_{DS}=5\text{ V}$, $V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	$3.9\text{E-}14$	$2.4\text{E-}14$	$3.8\text{E-}14$	$3.9\text{E-}14$

2.3 MOSFET DC Model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

WL	0.9
180	X
144	X
90	X
36	X
18	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}				I_D vs. V_{DS}			
	Start	Stop	Step		Start	Stop	Step
V_{DS} (V)	0.1	5.0	2.55	V_{DS} (V)	0	5.0	0.05
V_{GS} (V)	-0.48	3.63	0.03	V_{GS} (V)	0.9	3.3	0.48
V_{BS} (V)	0	0	0	V_{BS} (V)	0	0	0

2.3.3 Model verification plots

W in the plots refer to W_{DES} .

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25°C.

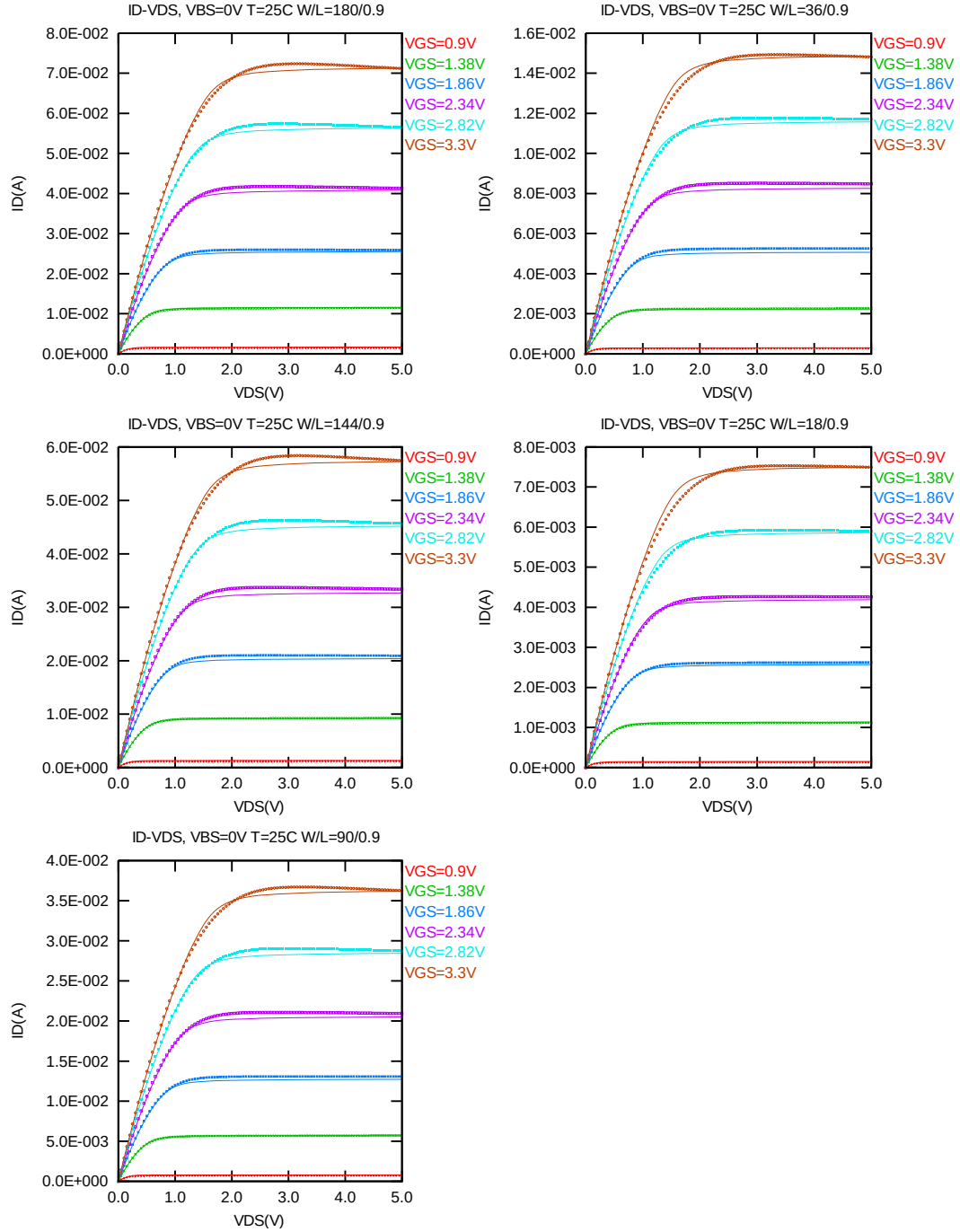


Figure 2-1 I_D vs. V_{DS} at $V_{BS} = 0$ V for NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output conductance curves for selected devices at 25°C.

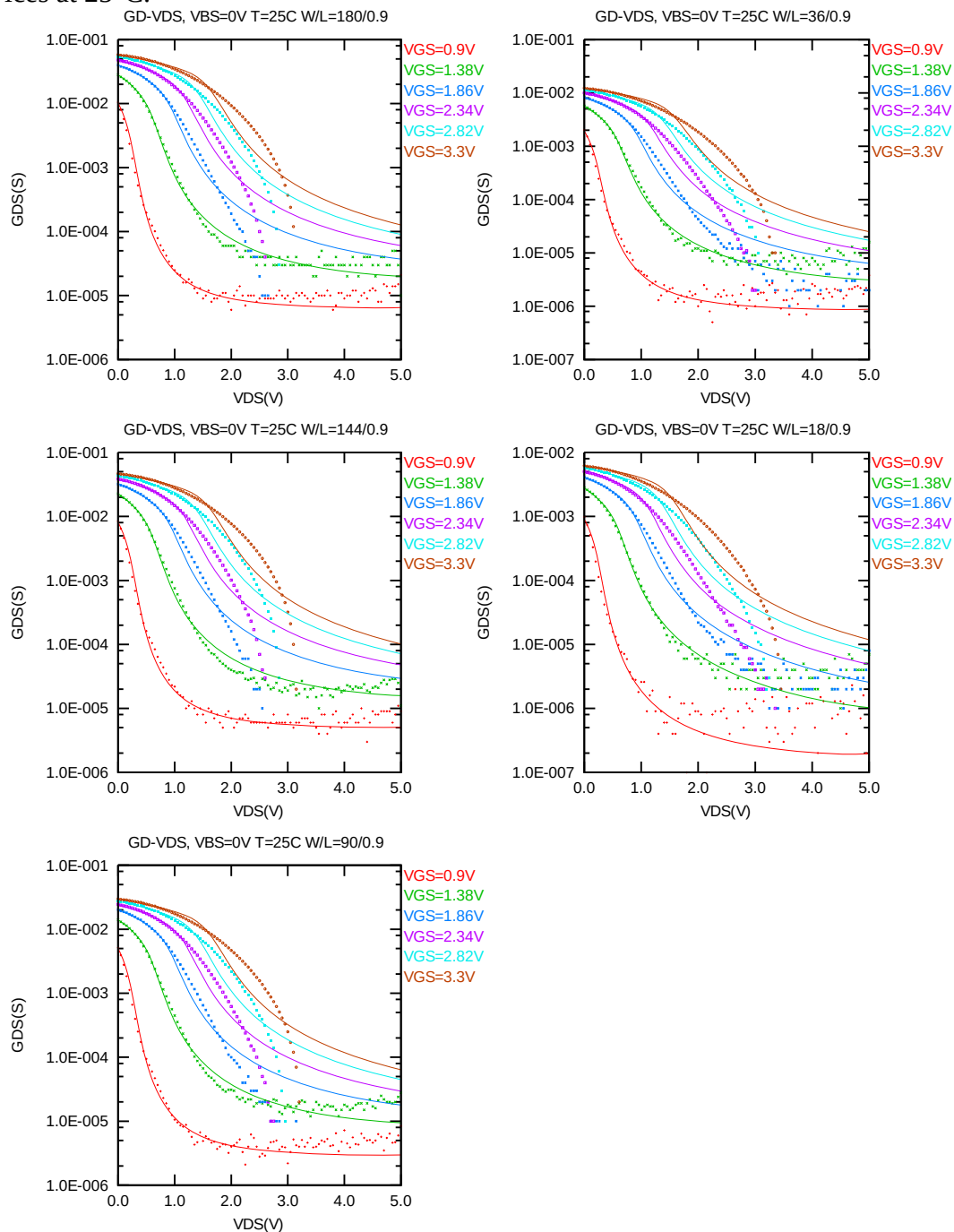


Figure 2-2 G_{DS} vs. V_{DS} at $V_{BS} = 0$ V for NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25°C.

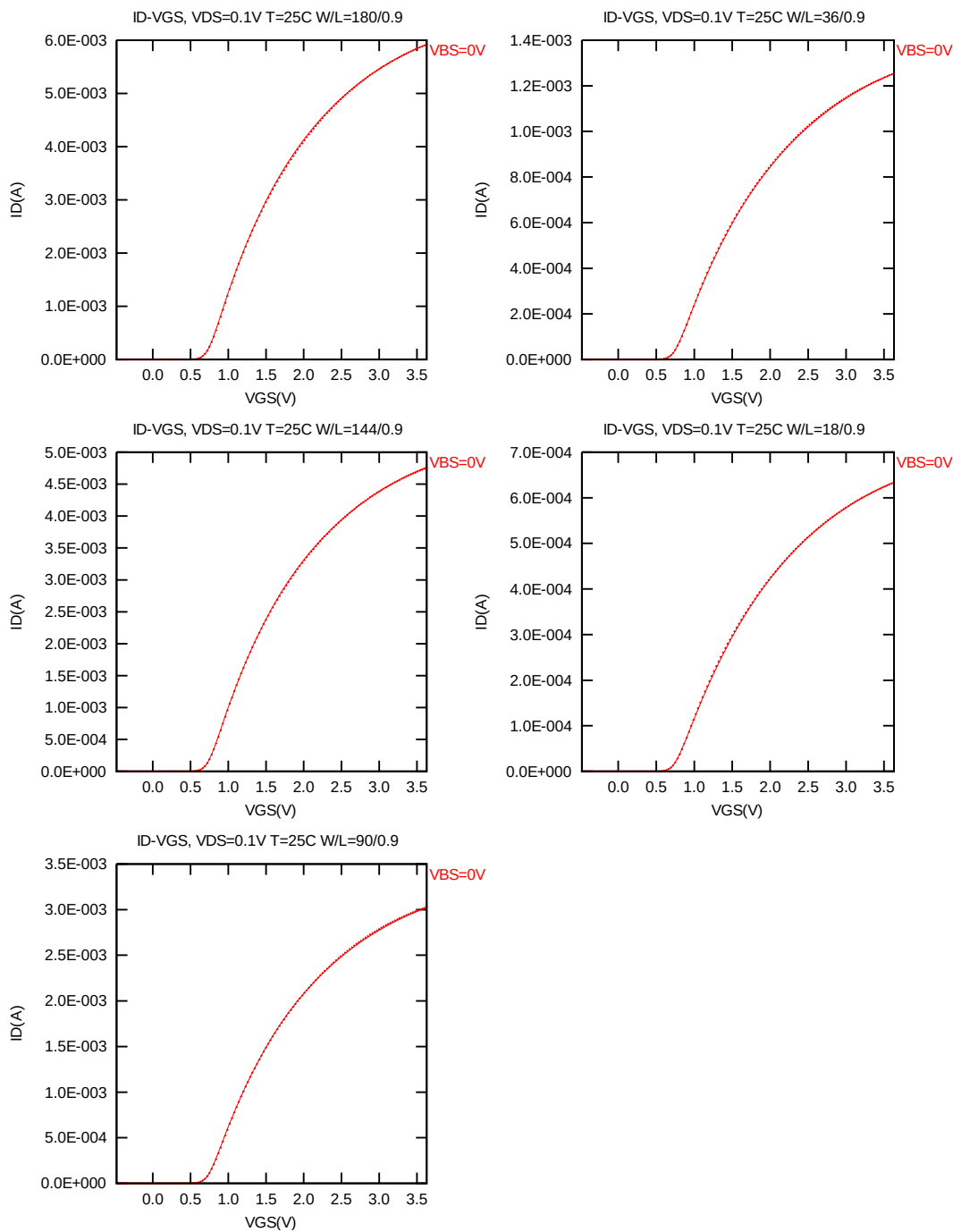


Figure 2-3 I_D vs. V_{GS} at $V_{DS} = 0.1V$ for NMOS

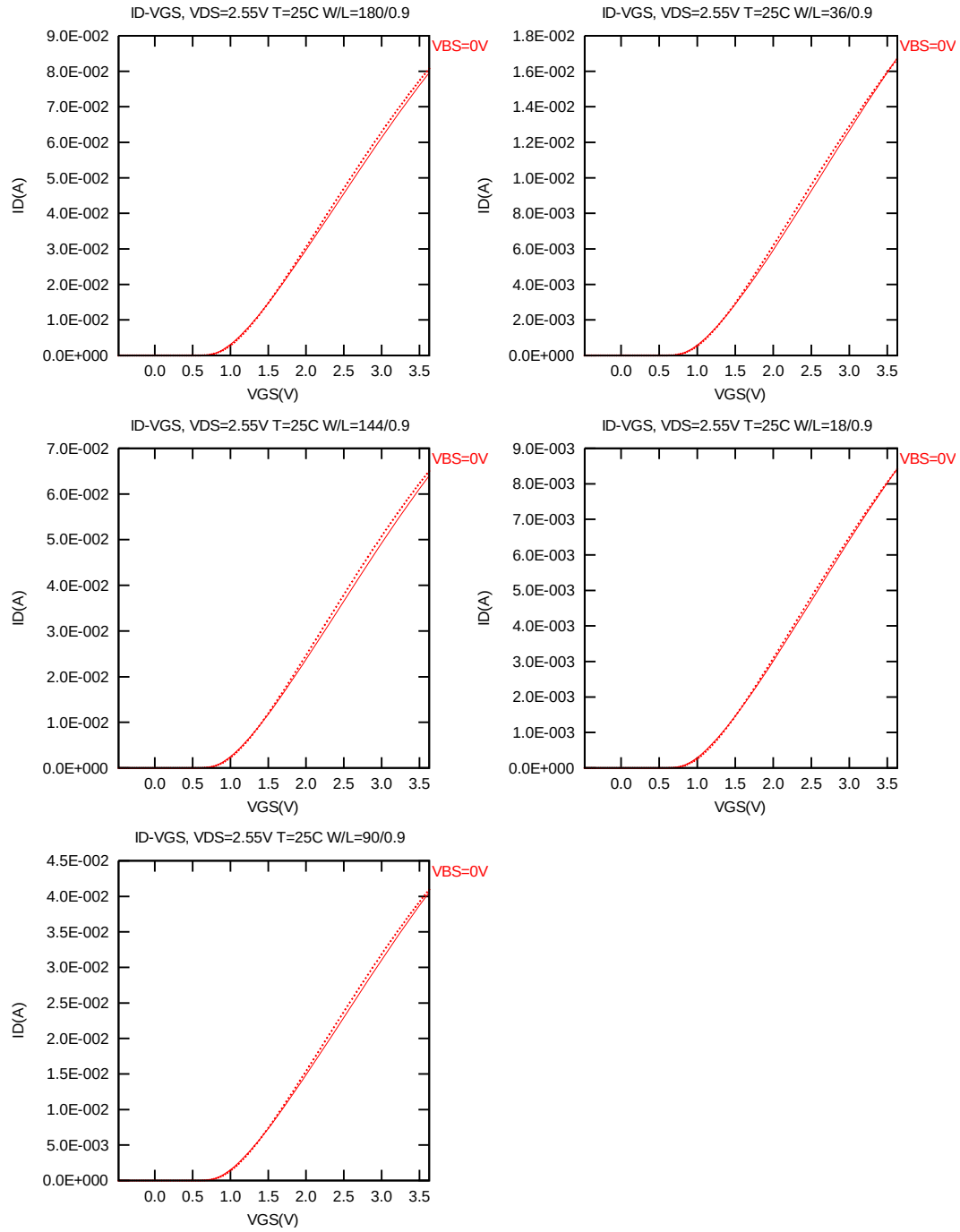


Figure 2-4 I_D vs. V_{GS} at $V_{DS} = 2.55$ V for NMOS

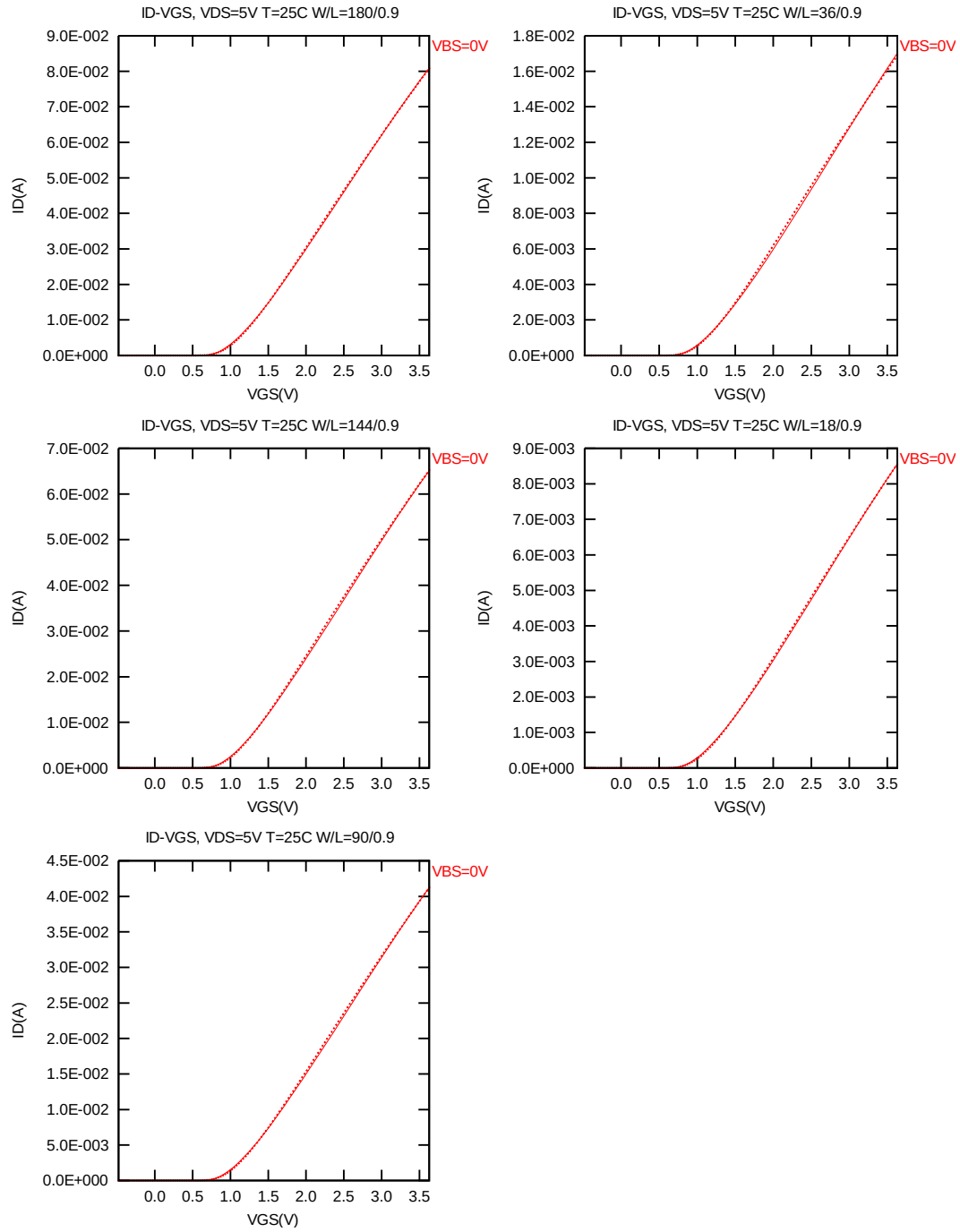


Figure 2-5 I_D vs. V_{GS} at $V_{DS} = 5.0$ V for NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25°C.

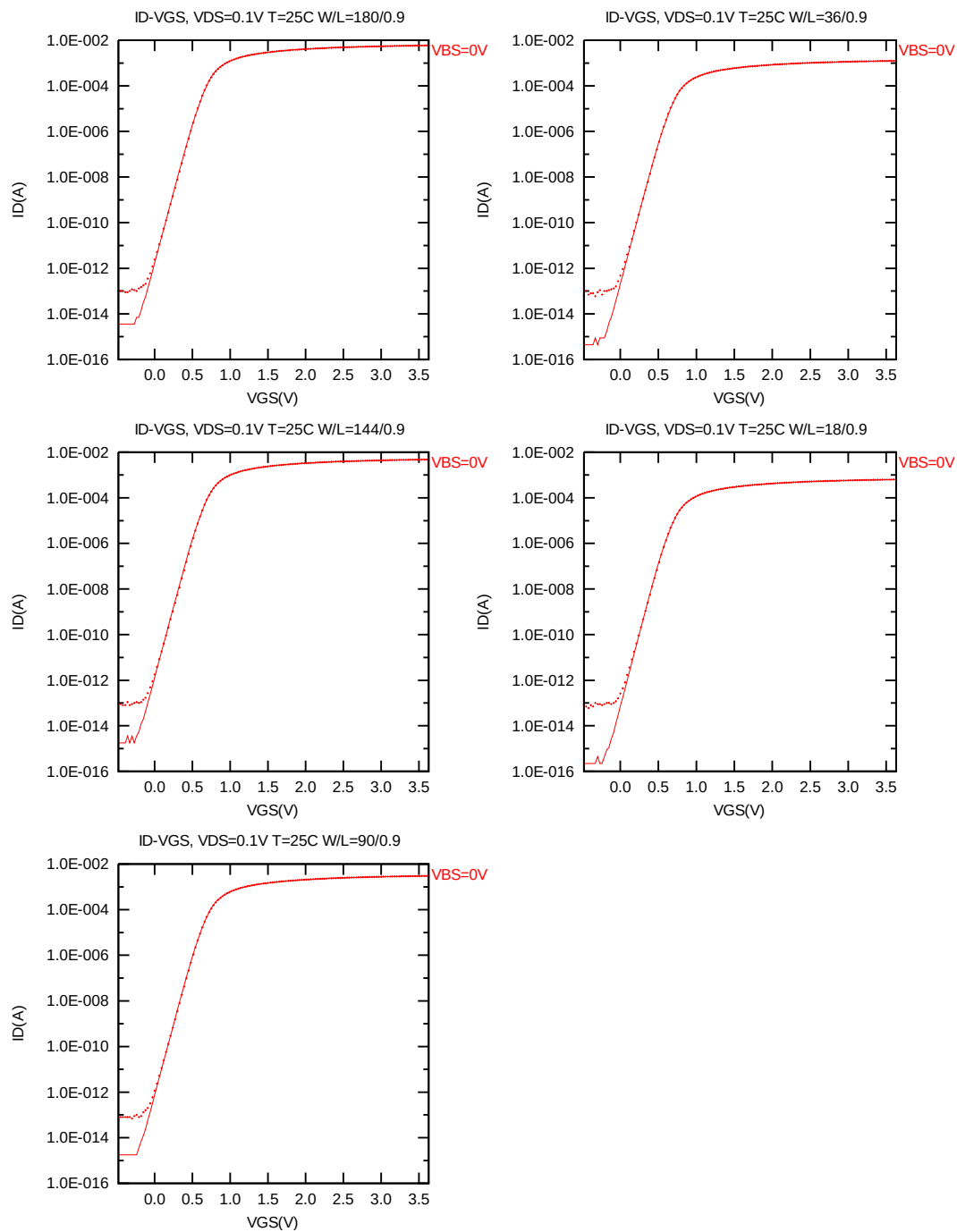


Figure 2-6 Log scaled I_D vs. V_{GS} at $V_{DS} = 0.1$ V for NMOS

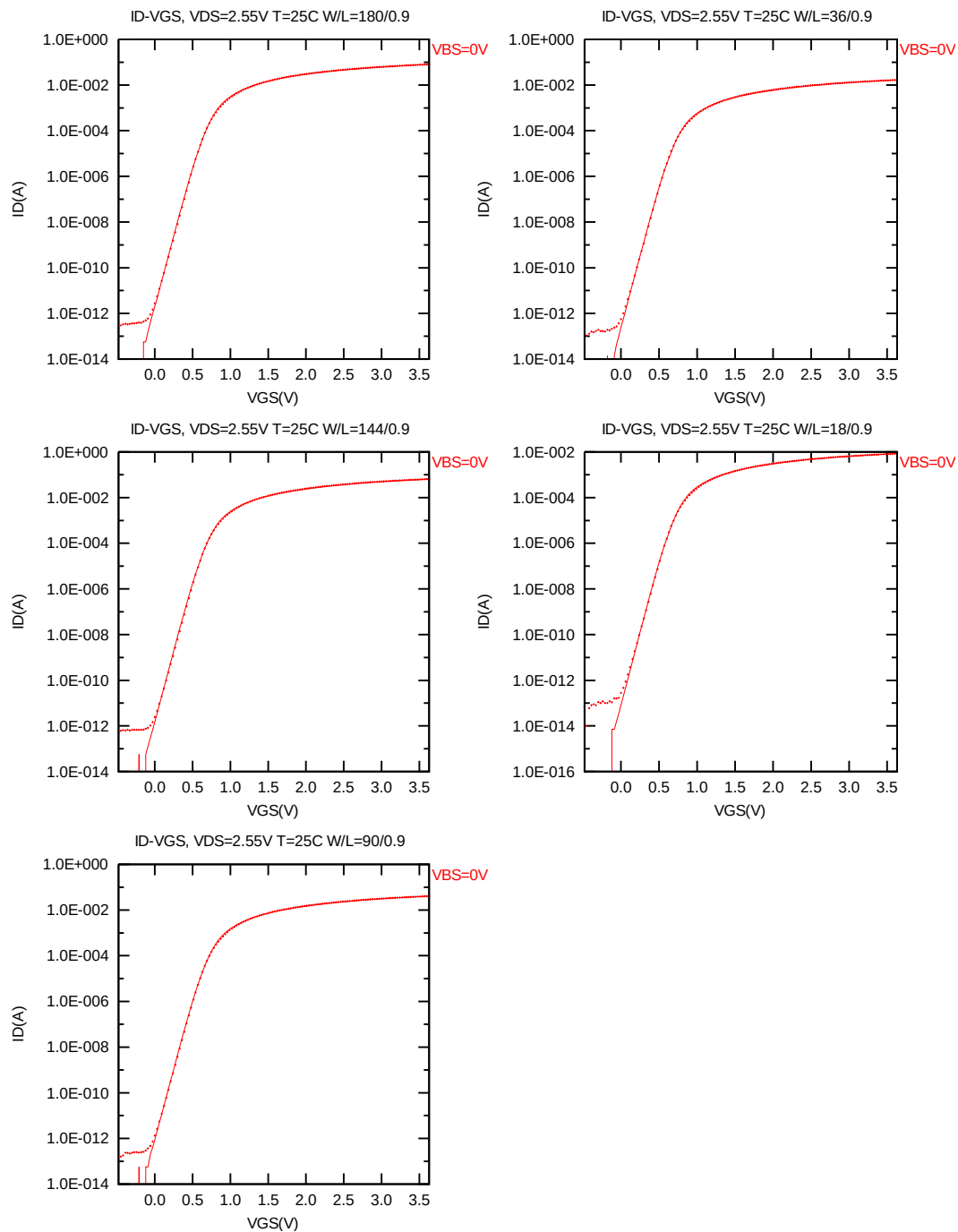


Figure 2-7 Log scaled I_D vs. V_{GS} at $V_{DS} = 2.55V$ for NMOS

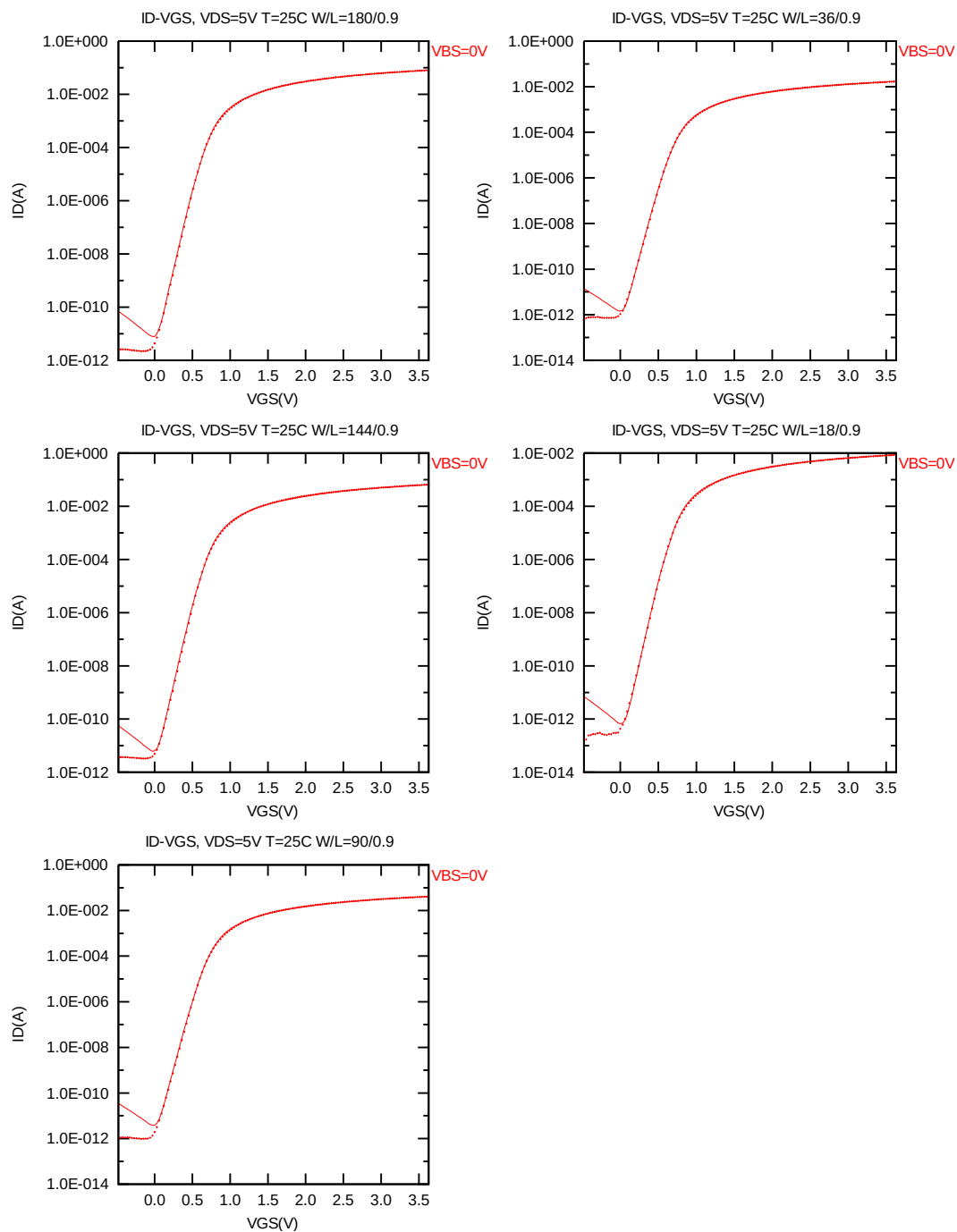


Figure 2-8 Log scaled I_D vs. V_{GS} at $V_{DS} = 5.0$ V for NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25°C.

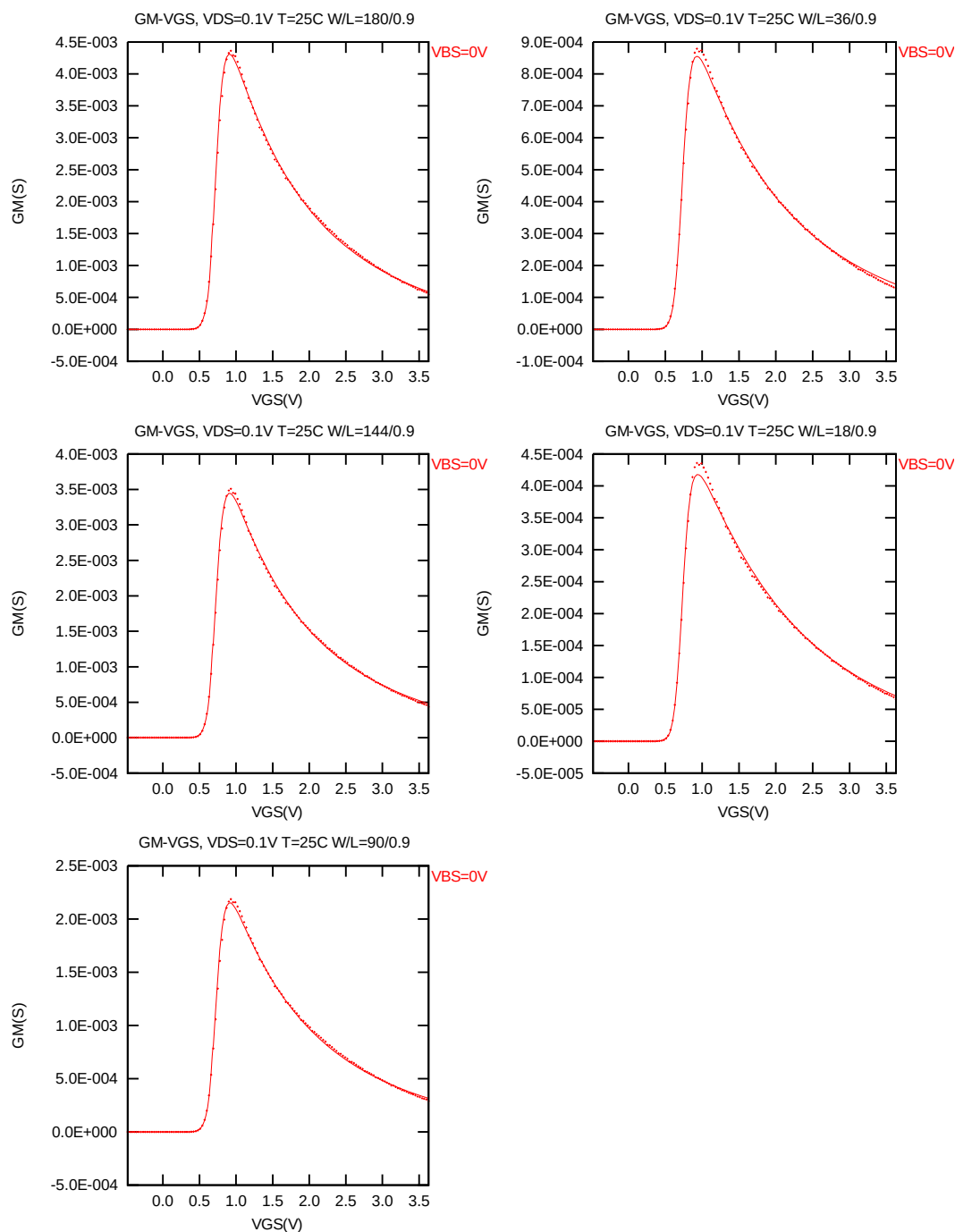


Figure 2-9 G_M vs. V_{GS} at $V_{DS} = 0.1$ V for NMOS

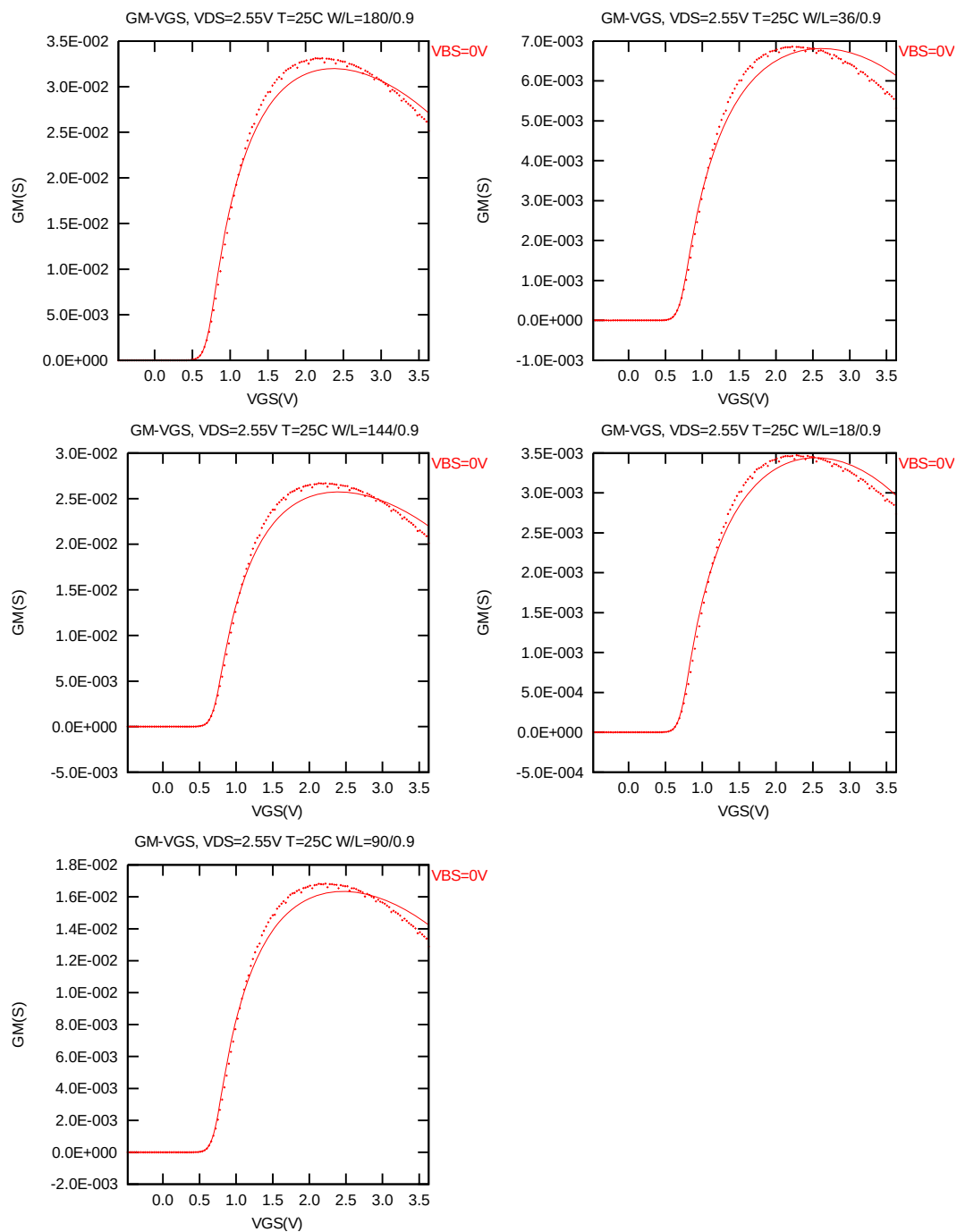


Figure 2-10 G_M vs. V_{GS} at $V_{DS} = 2.55$ V for NMOS

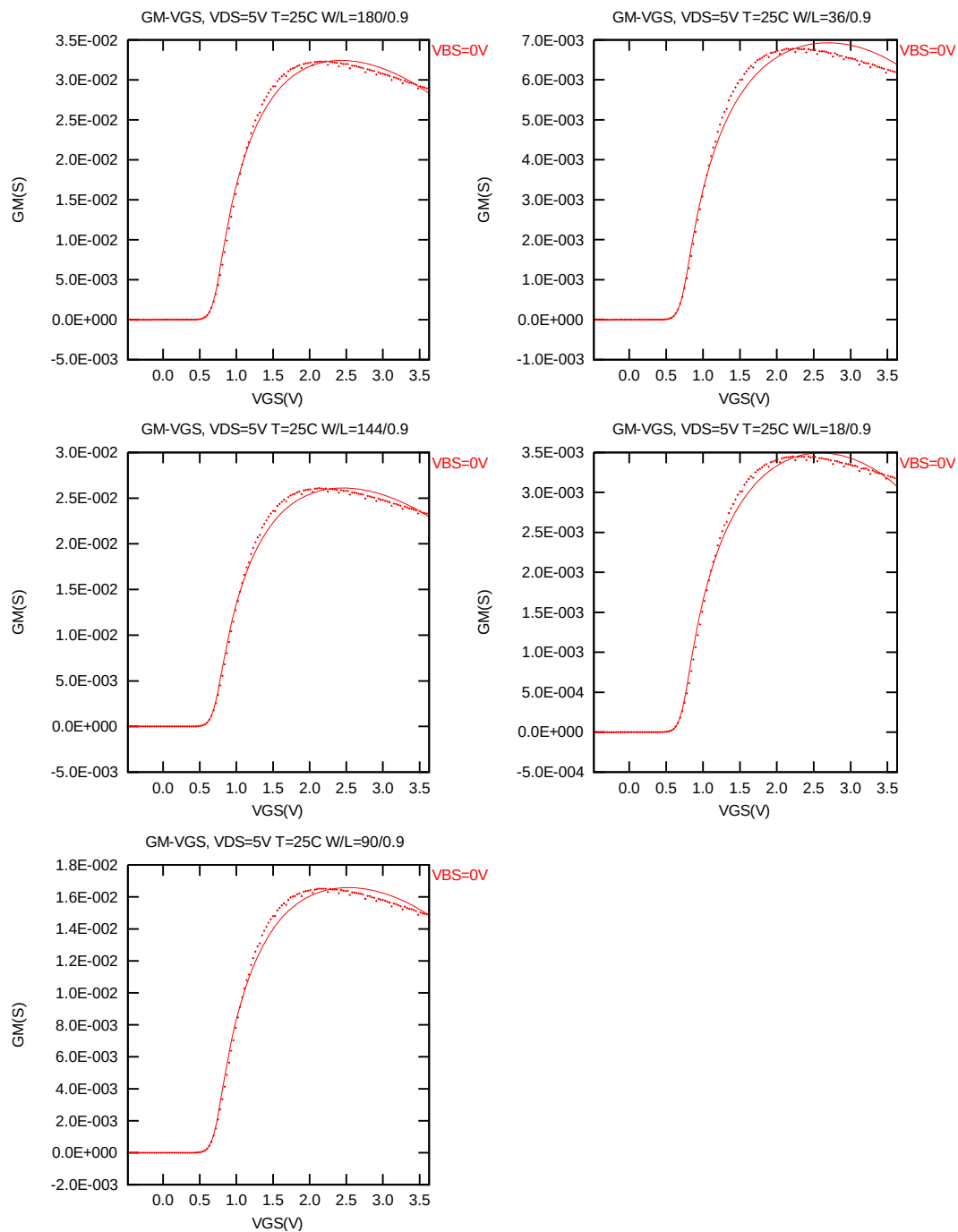


Figure 2-11 G_M vs. V_{GS} at $V_{DS} = 5.0$ V for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} , respectively.

The following plots show the channel width dependence of V_{TLIN} and V_{TSAT} extracted at $V_{DS} = 0.1$ V, $V_{DS} = 5.0$ V, respectively, for $V_{BS} = 0$ V at 25°C.

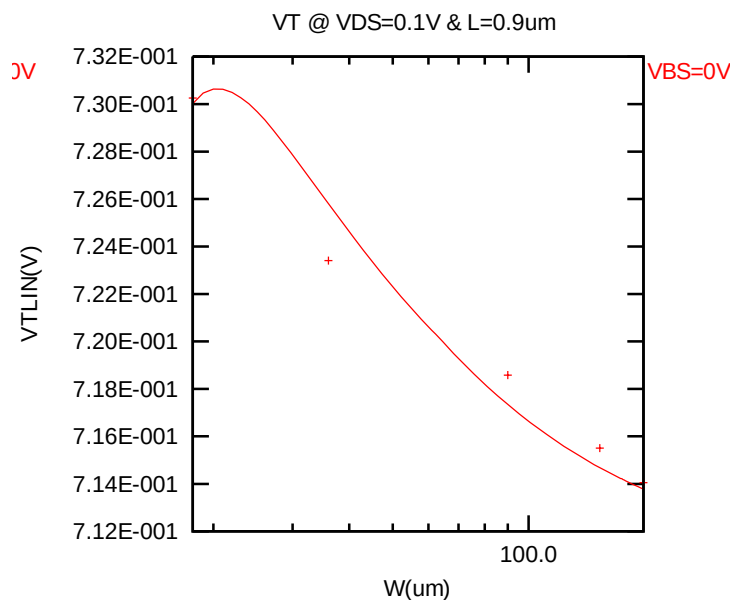


Figure 2-12 V_{TLIN} vs. channel width for low drain bias ($V_{DS} = 0.1$ V) for NMOS

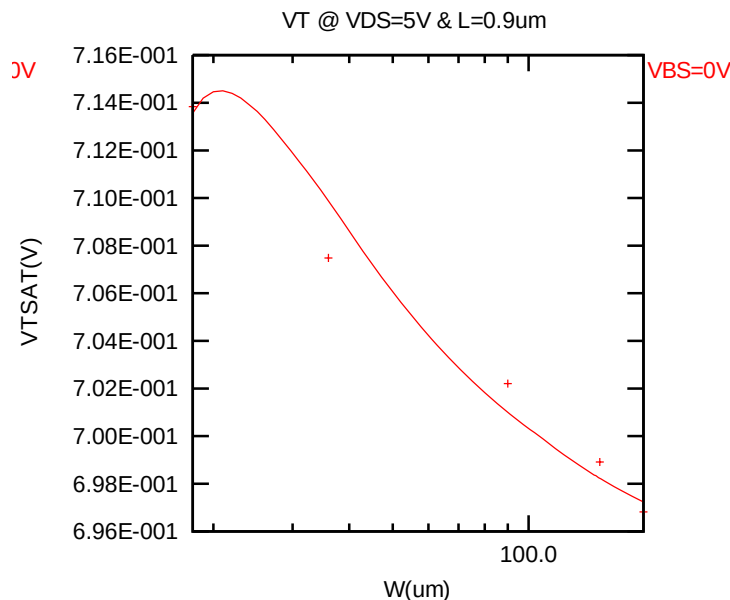


Figure 2-13 V_{TSAT} vs. channel width for high drain bias ($V_{DS} = 5.0$ V) for NMOS

The L and W in the plots refer to L_{POLY} and W_{DES} , respectively.

The following plots show the channel width dependence of I_{DLIN} and I_{ON} extracted at $V_{DS} = 0.1$ V, $V_{GS} = 3.3$ V and $V_{DS} = 5.0$ V, $V_{GS} = 3.3$ V, respectively, for $V_{BS} = 0$ V at 25°C.

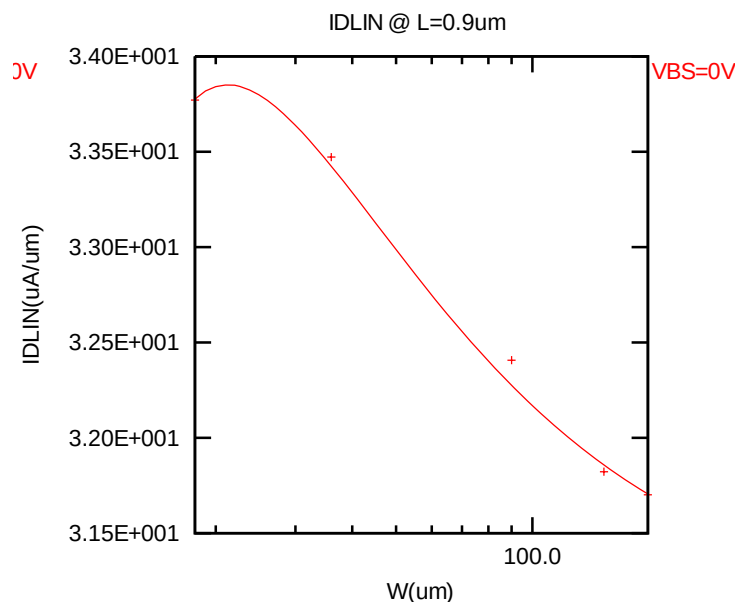


Figure 2-14 Linear current I_{DLIN} vs. channel width for NMOS

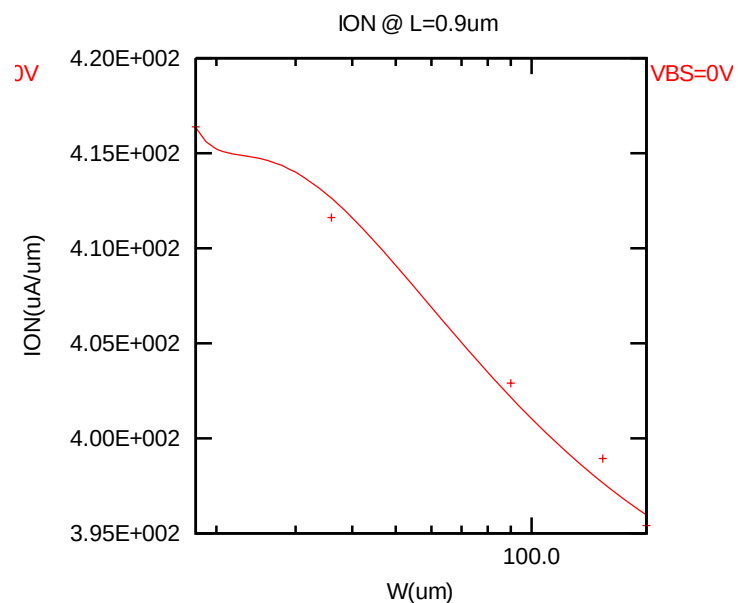


Figure 2-15 Saturation current I_{ON} vs. channel width for NMOS

The L and W in the plots refer to L_{POLY} and W_{DES} , respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = 5.0$ V and $V_{GS} = V_{BS} = 0$ V at 25°C.

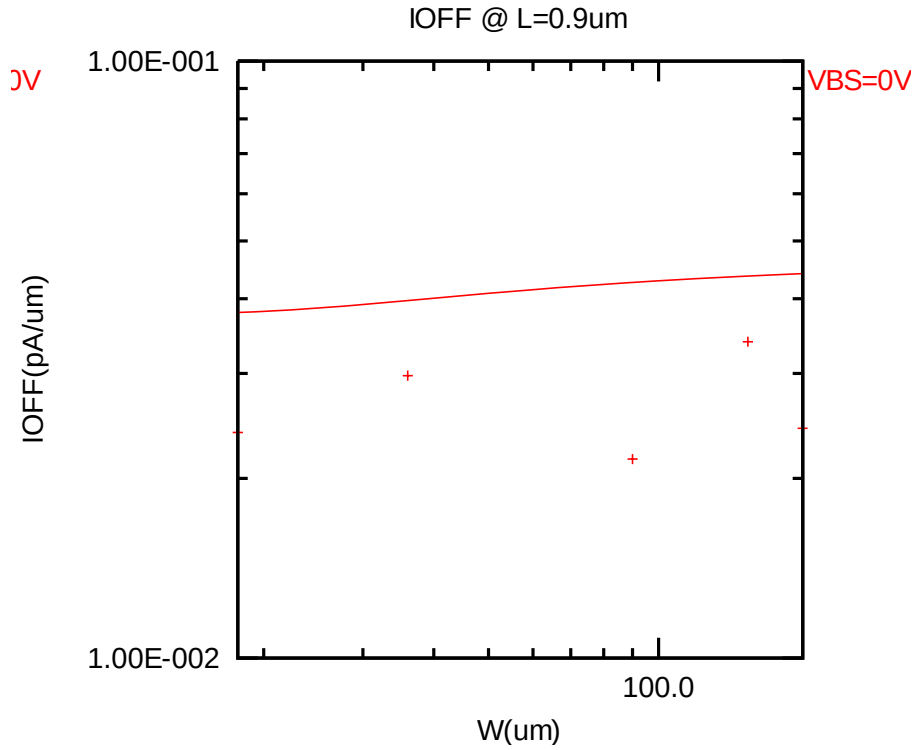


Figure 2-16 I_{OFF} vs. channel width for NMOS

V_{TLIN} and V_{TSAT} vs. temperature

The following plots show the temperature dependence of V_{TLIN} and V_{TSAT} extracted at $V_{DS} = 0.1$ V, $V_{DS} = 5.0$ V, respectively, for $V_{BS} = 0$ V at $W_{DES} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

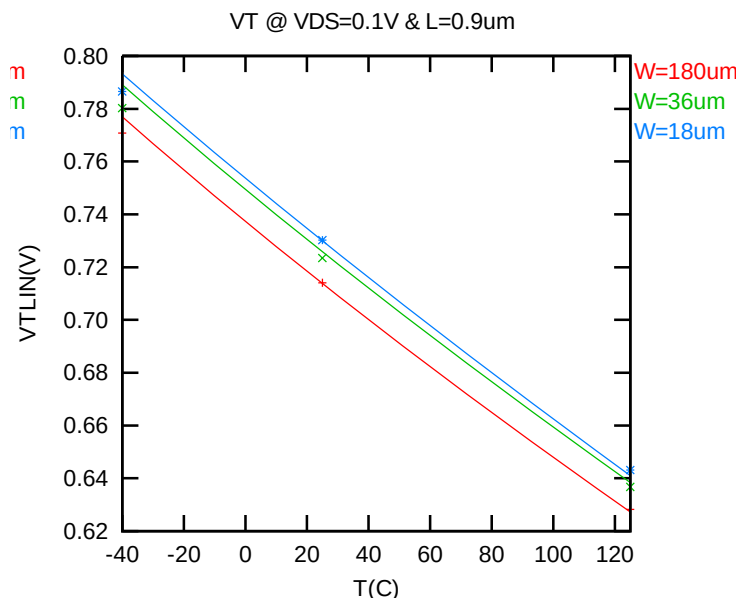


Figure 2-17 V_{TLIN} vs. temperature for NMOS

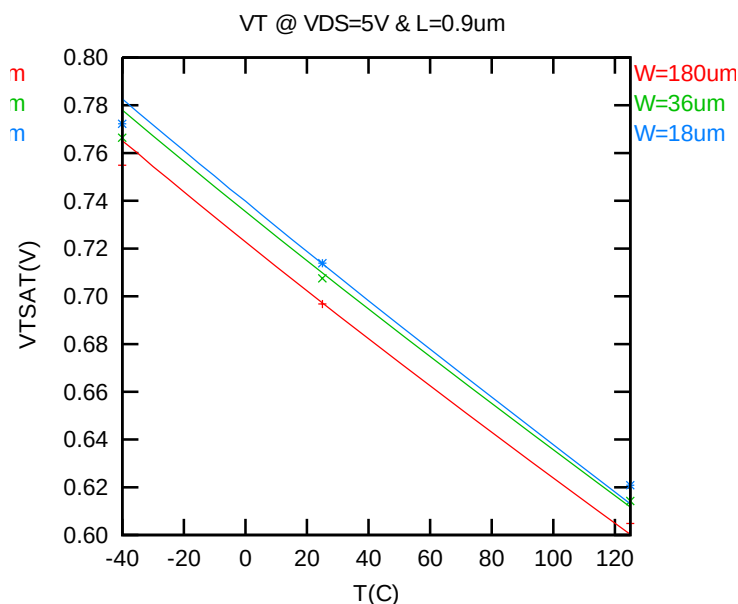


Figure 2-18 V_{TSAT} vs. temperature for NMOS

I_{DLIN} vs. temperature

The following plots show the temperature dependence of I_{DLIN} extracted at $V_{DS} = 0.1$ V, $V_{GS} = 3.3$ V, $V_{BS} = 0$ V at $W_{DES} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

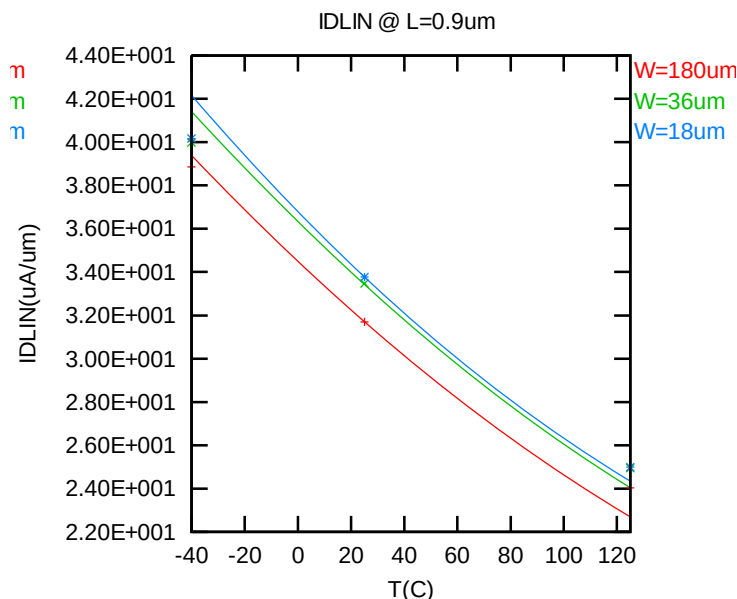


Figure 2-19 Linear current vs. temperature for NMOS

I_{DSAT} vs. temperature

The following plots show the temperature dependence of I_{DSAT} extracted at $V_{DS} = 5.0$ V, $V_{GS} = 3.3$ V, $V_{BS} = 0$ V at $W_{DES} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

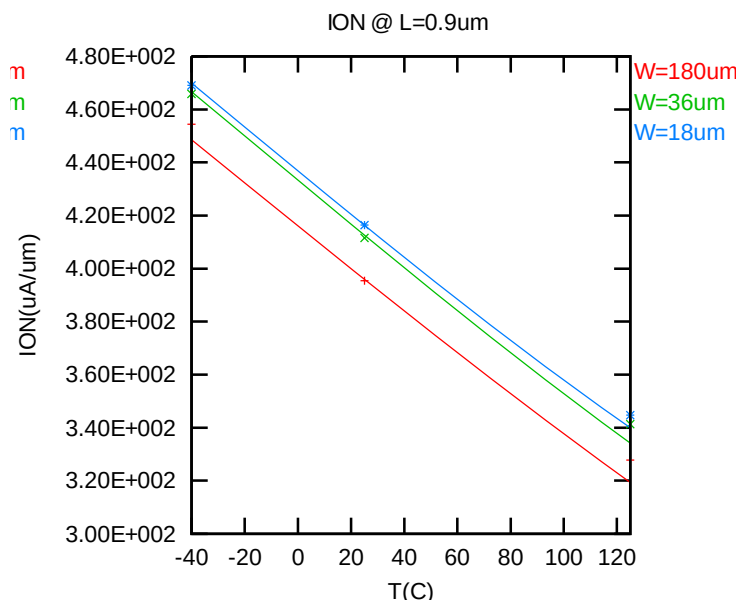


Figure 2-20 Saturation current vs. temperature for NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{DS} = 5.0$ V, $V_{GS} = V_{BS} = 0$ V at $W_{DES} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

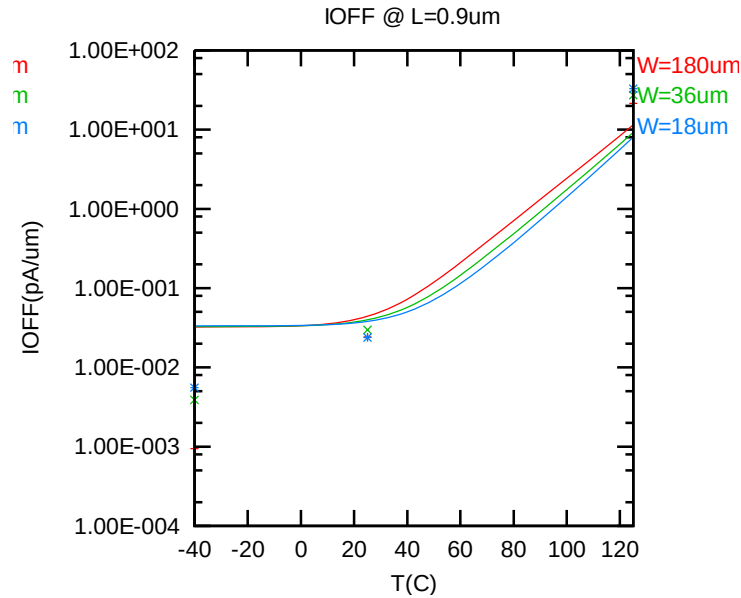


Figure 2-21 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

Model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- b. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- c. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$
- d.

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- e. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- f. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 2-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

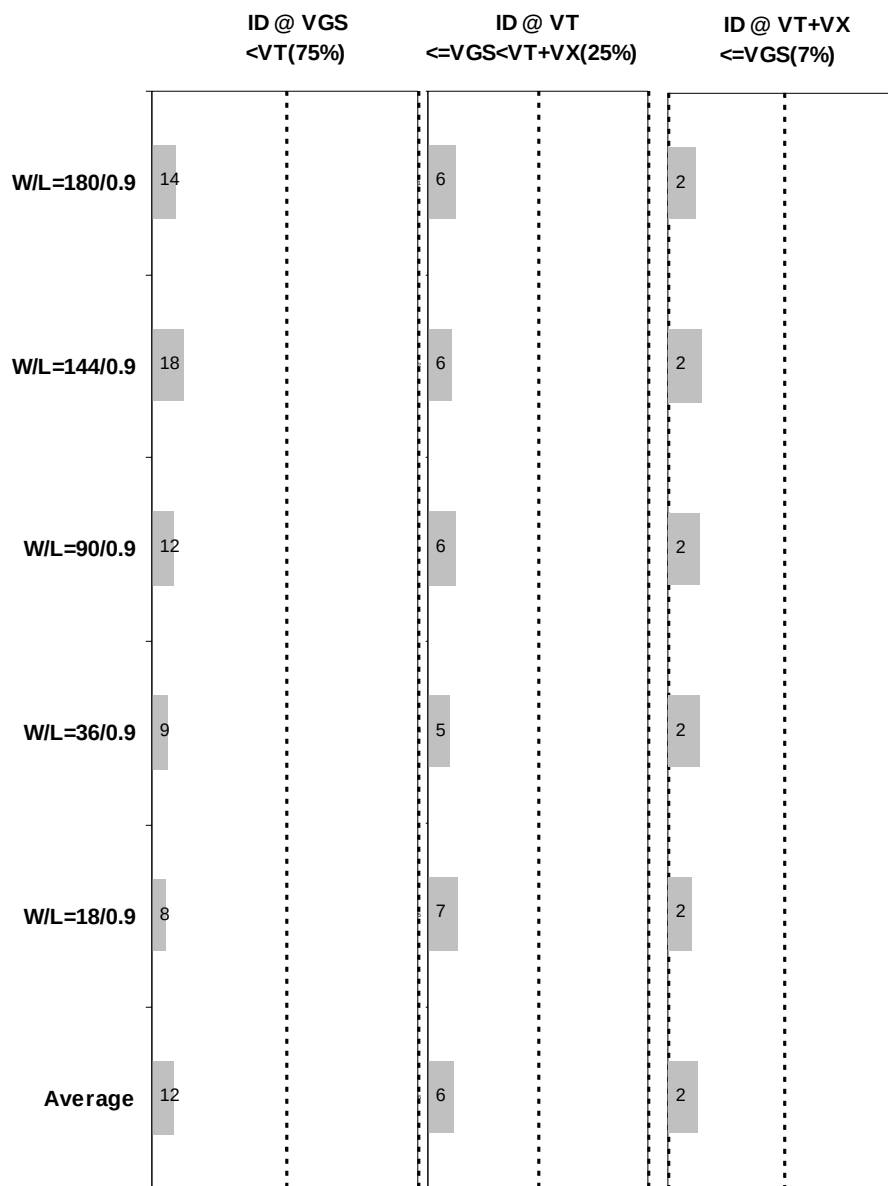


Figure 2-22 I_D accuracy at 25°C for NMOS

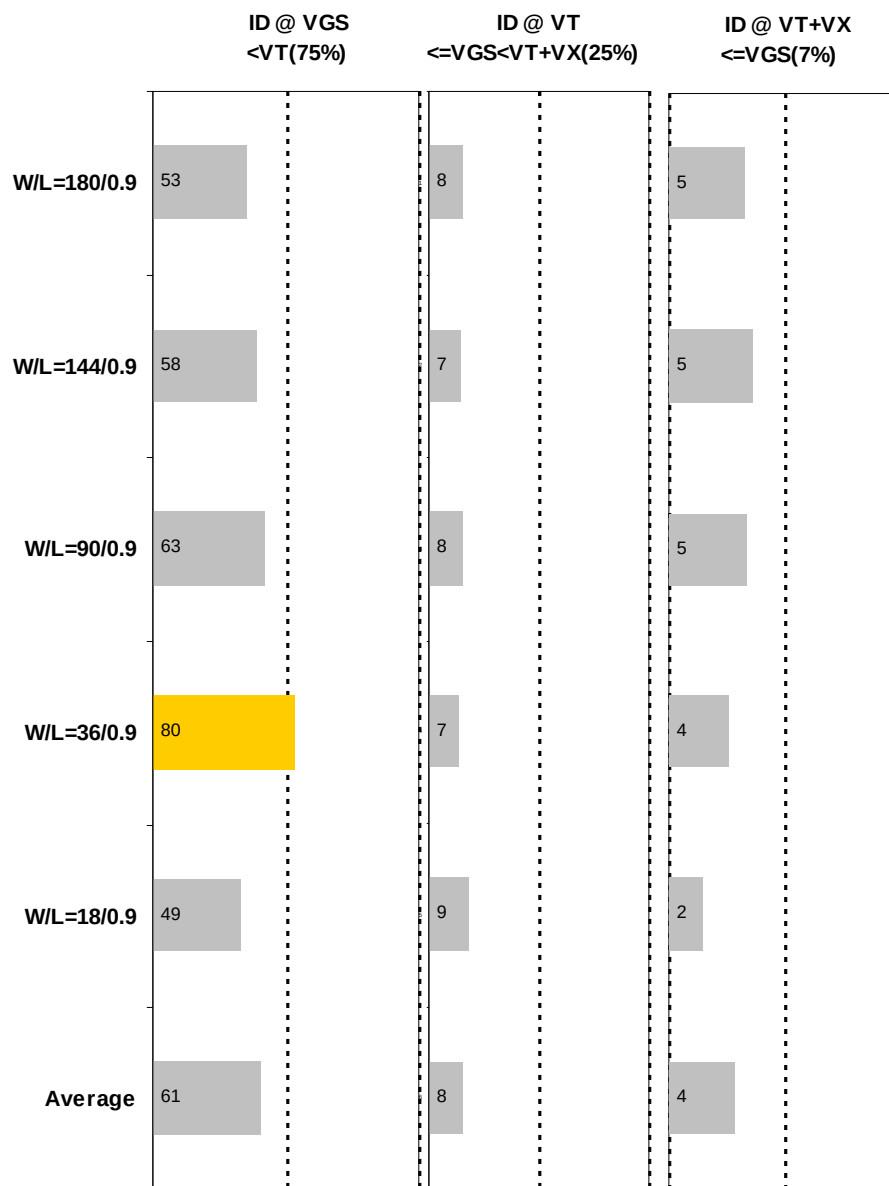


Figure 2-23 I_D accuracy at 125°C for NMOS

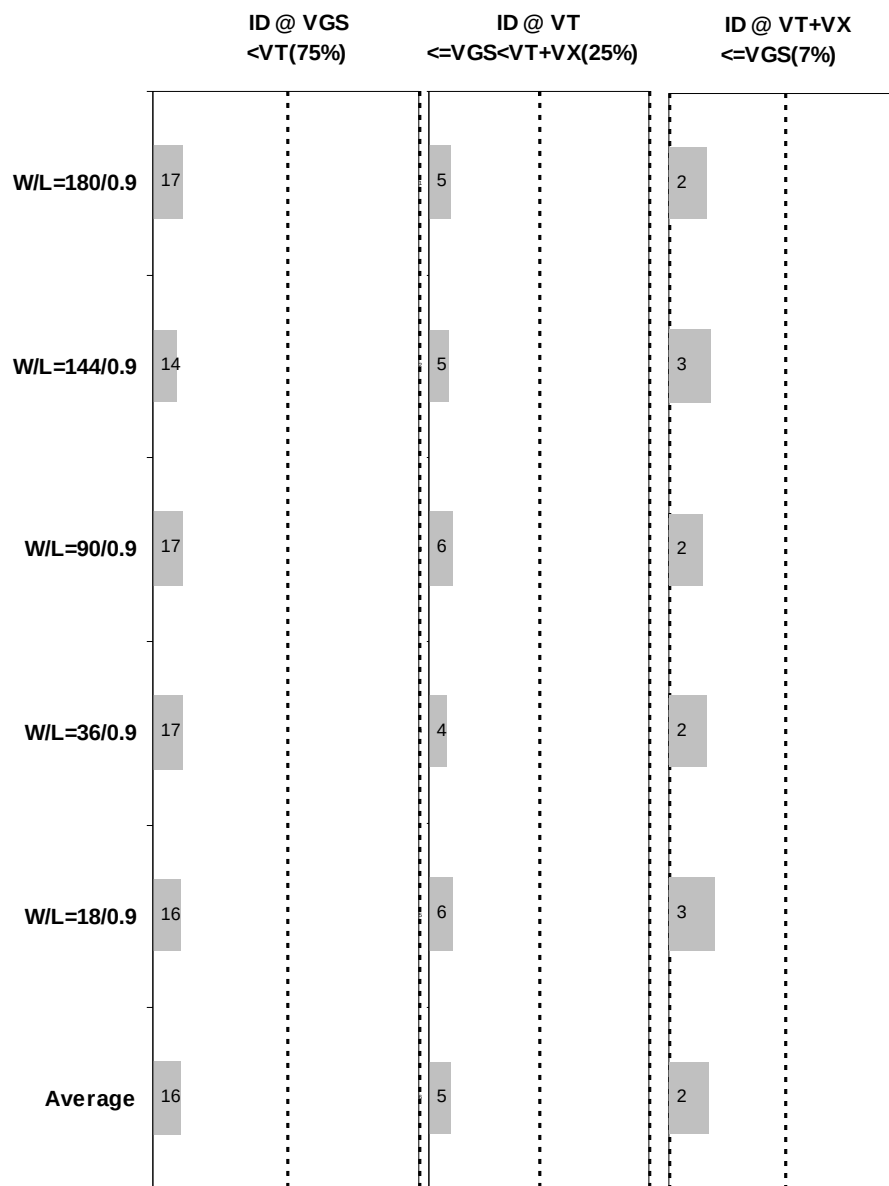


Figure 2-24 I_D accuracy at -40°C for NMOS

G_M and G_{DS} accuracy

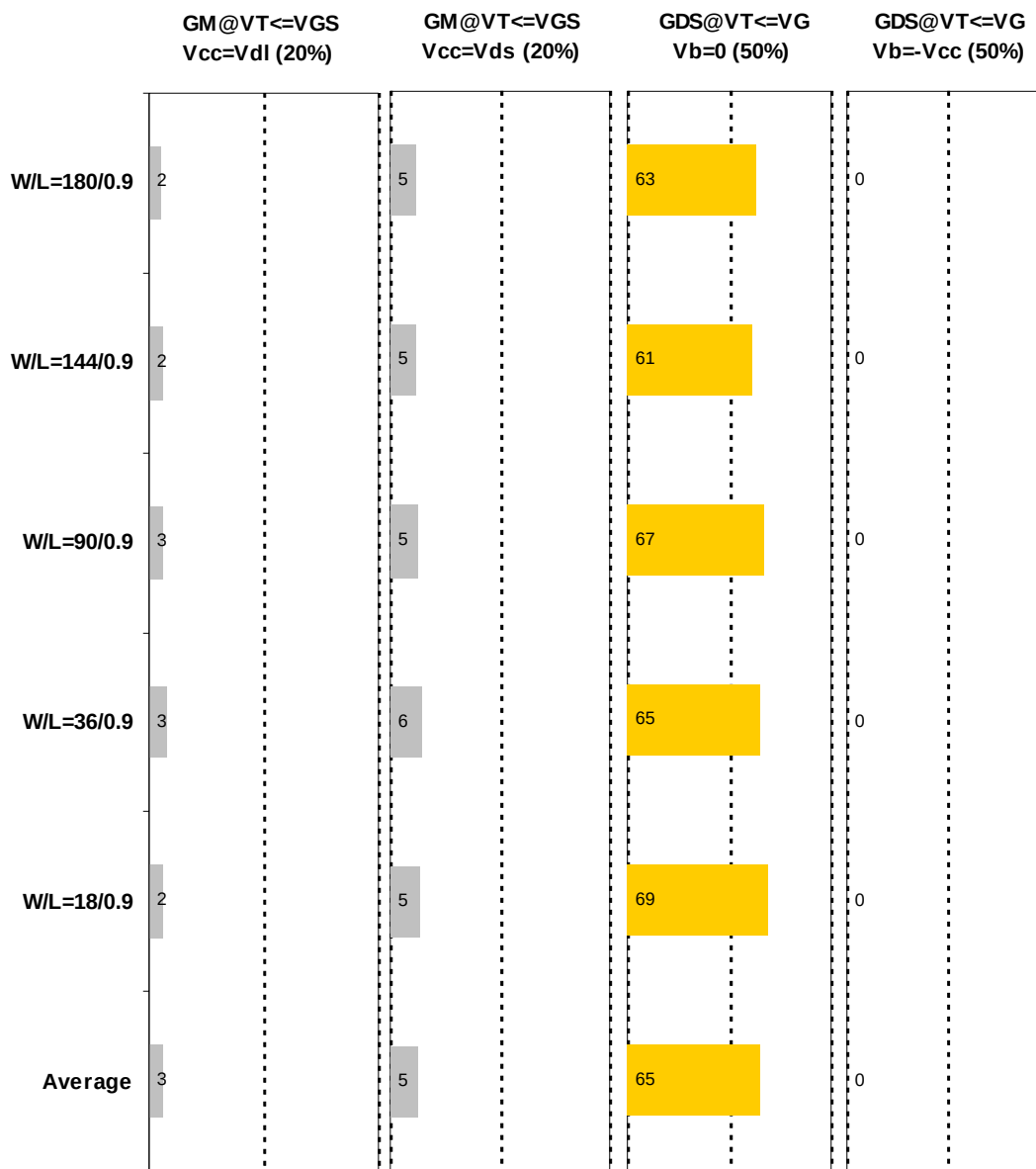


Figure 2-25 G_M and G_{DS} accuracy at 25°C for NMOS

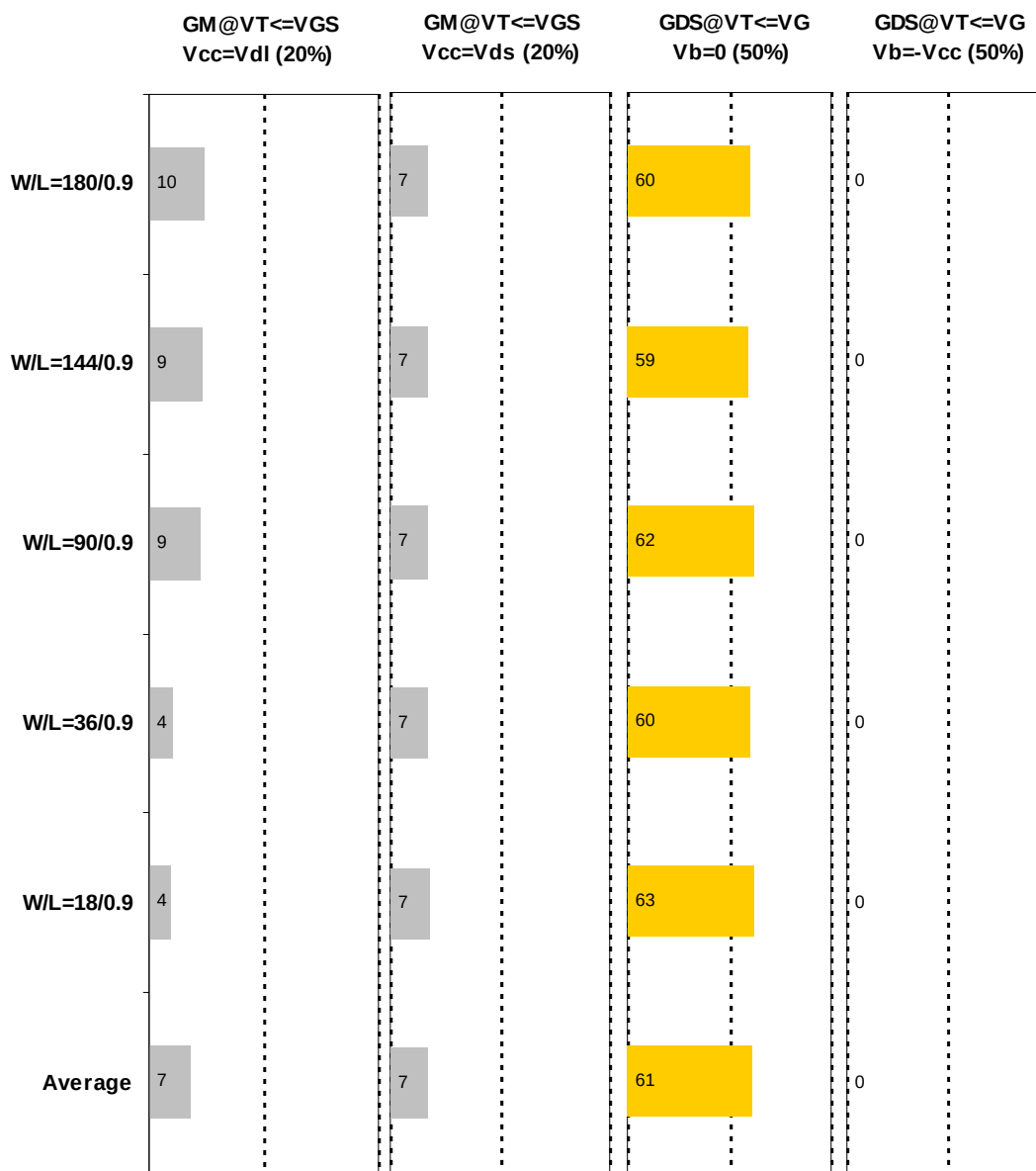


Figure 2-26 G_M and G_{DS} accuracy at 125°C for NMOS

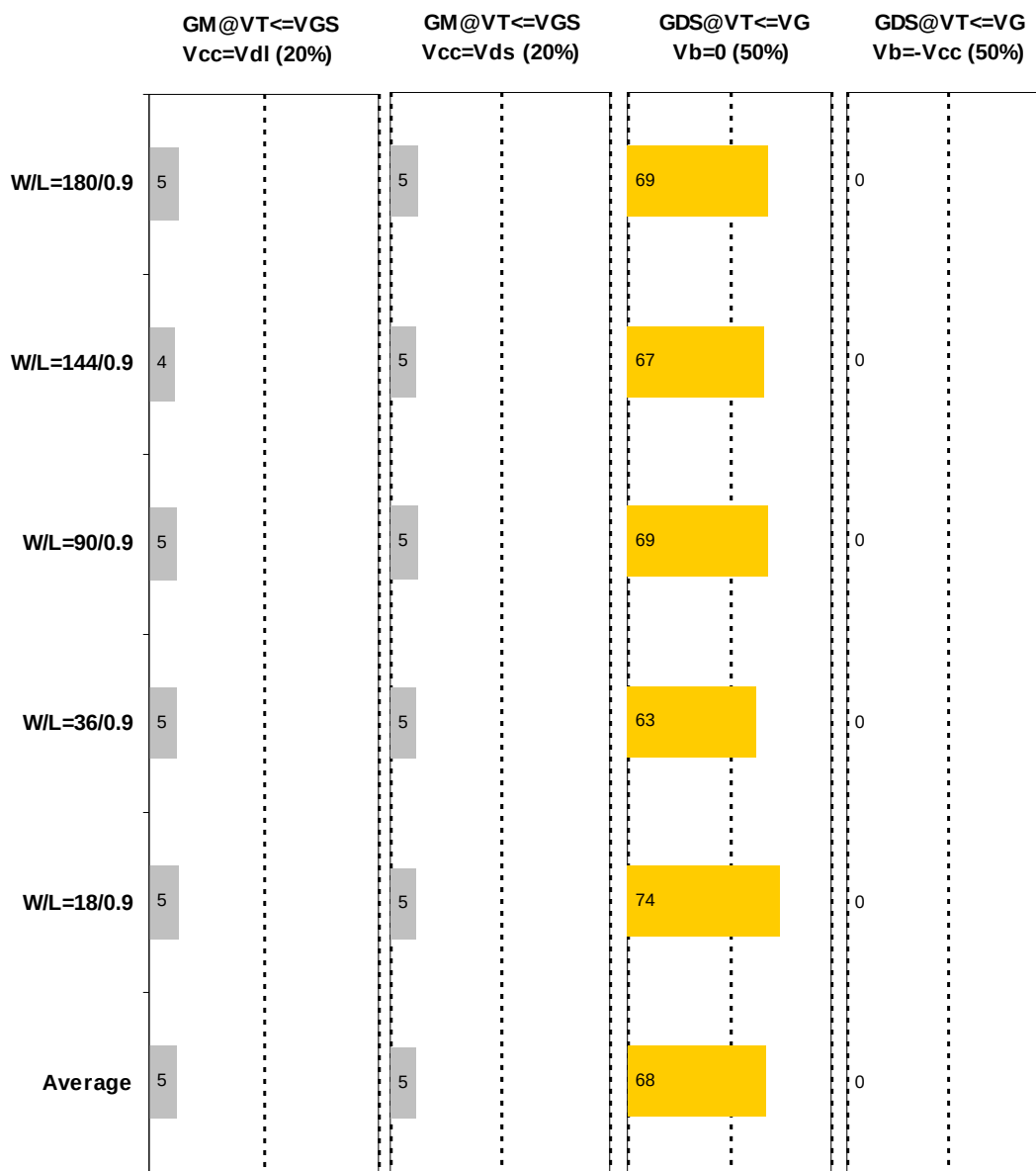


Figure 2-27 G_M and G_{DS} accuracy at -40°C for NMOS

2.4 MOSFET Capacitance Model

2.4.1 Test devices

Following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Large area measurement structure for NMOS C_{gc} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	1	10
	1	50
	1	100

2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitors are summarized in Table 2-8 below.

Table 2-7 Measurement conditions for NMOS capacitance model

V_{BIAS} [V]	From -3.6 to 3.6 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.4.3 Width and length AC model accuracy

Width and length AC model accuracy

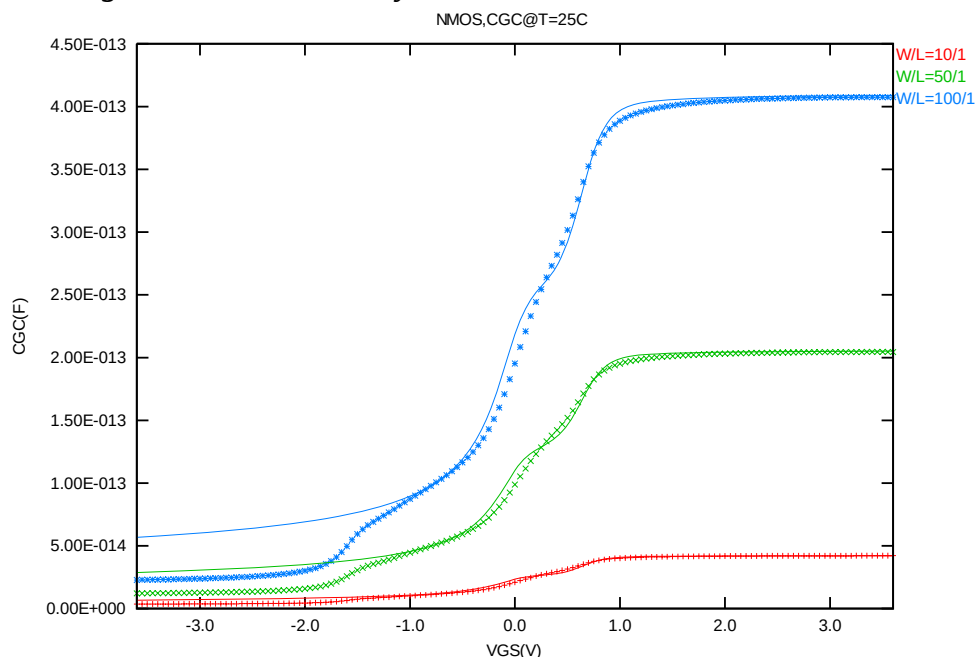


Figure 2-28 Measured and modeled C_{GC} curves for NMOSFET test structures

2.5 Parasitic Source / Drain Junction Capacitance Model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-8 Junction capacitor structures for NMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	2.62E-08	3.87E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	2.62E-08	2.82E-02	Finger diode

2.5.2 Measurement conditions

Junction capacitance was measured under the bias conditions in the following table.

Table 2-9 Junction capacitor measurement conditions for NMOS

V _{BIAS} [V]	VH: pwell & VL: n+ From 0 to 8.5 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,150

2.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from silicon data.

Table 2-10 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	1.10E-04	CJSW	F/m	9.36E-10	CJSWG	F/m	1.68E-09
PB	V	2.82E-01	PBSW	V	5.78E-01	PBSWG	V	6.71E-01
MJ	--	2.23E-01	MJSW	--	2.90E-01	MJSWG	--	4.00E-01
TCJ	1/K	2.12E-03	TCJSW	1/K	1.15E-03	TCJSWG	1/K	8.00E-04
TPB	V/K	8.34E-04	TPBSW	V/K	1.46E-03	TPBSWG	V/K	2.50E-03

C_J , C_{JSW} vs. V_J plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

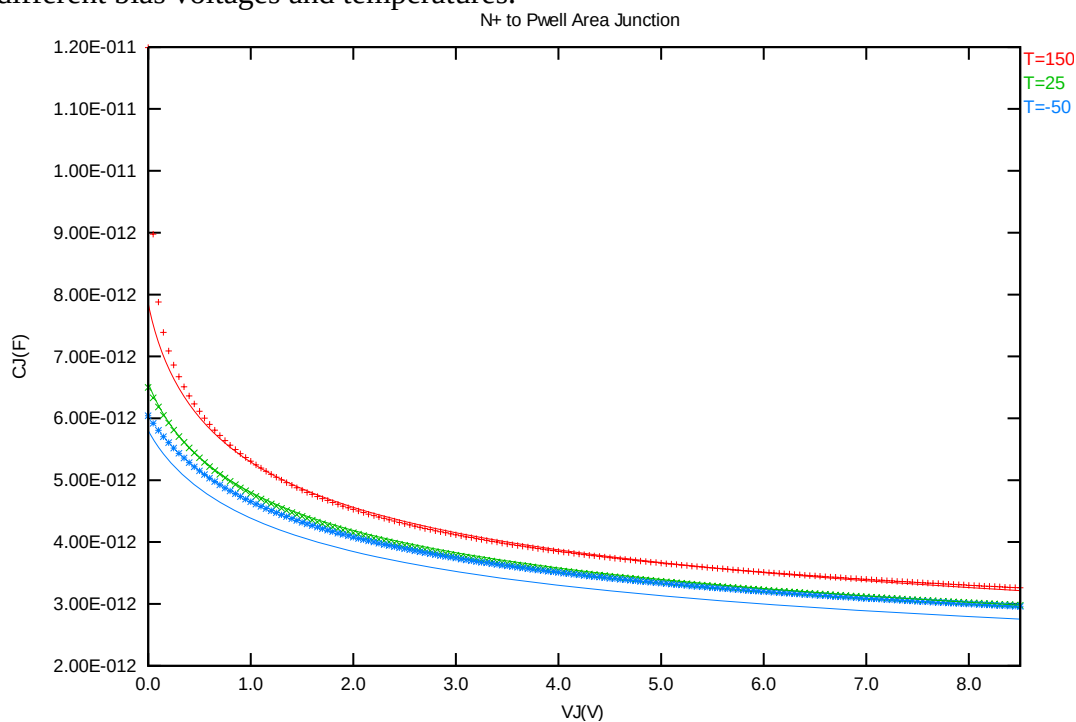


Figure 2-29 Bottom area junction capacitance at -50°C , 25°C and 150°C for NMOS

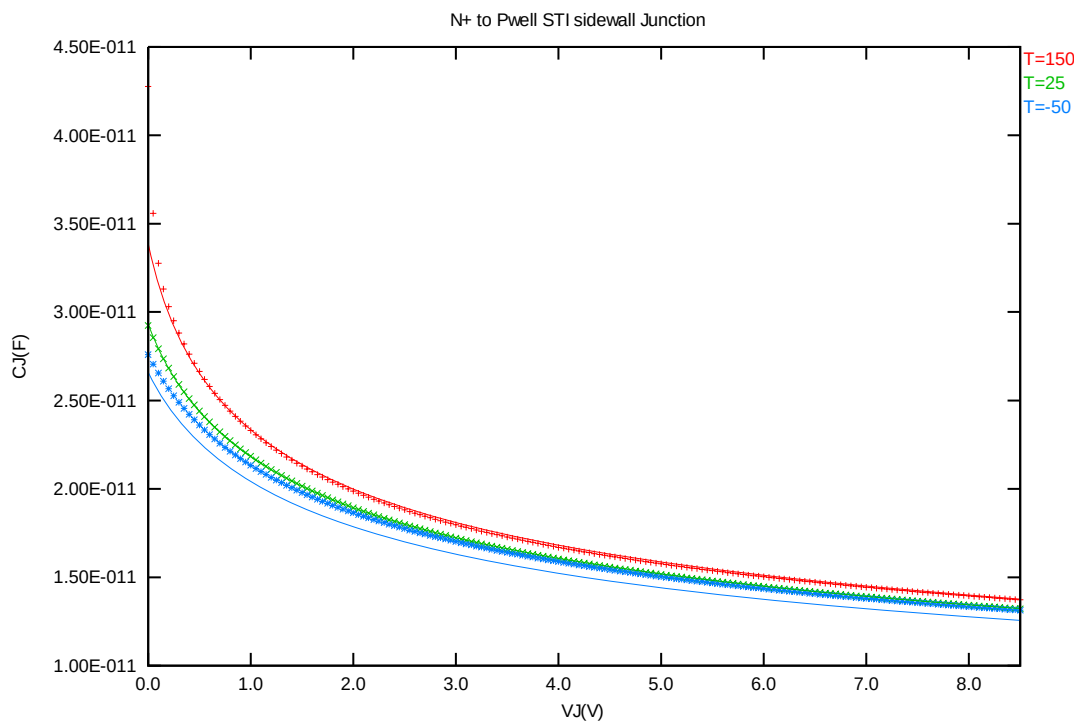


Figure 2-30 STI bounded sidewall junction capacitance at -50°C , 25°C and 150°C for NMOS

3 D5.0G3.3 p-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW1PA	--	--	--
Lot ID	D6G6T.1	--	--	--
Wafer Number	#03	--	--	--

3.2 Electrical Parameters

Key electrical parameters for the characterization of this device are summarized in the table below. Linear as well as saturation threshold voltages are extracted as I_D equal to $300\text{nA} \cdot W_{\text{DES}} \cdot 0.9^5 / (L_{\text{DES}} \cdot 0.9 - 0.54\mu\text{m})$ and $-70\text{nA} \cdot W_{\text{DES}} \cdot 0.9^5 / (L_{\text{DES}} \cdot 0.9 - 0.36\mu\text{m})$ for NMOS and PMOS, respectively.

Table 3-2 Electrical parameters for PMOS

All parameters are given for $T=25^\circ\text{C}$, $V_{\text{BS}}=0\text{ V}$ unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
-----------	-----------	------	--------	----------------	-----------------	----------------

Narrow and short channel device ($W/L=18\mu\text{m} / 0.72\mu\text{m}$)

Wide and short channel device ($W/L=18\mu\text{m} / 0.72\mu\text{m}$)

V_{TLIN}	$V_{\text{DS}} = -0.1\text{ V}$	V	-0.675	-0.672	-0.674	-0.675
V_{TSAT}	$V_{\text{DS}} = -5\text{ V}$	V	-0.663	-0.659	-0.661	-0.663
I_{ON}	$V_{\text{DS}} = -5\text{ V}$, $V_{\text{GS}} = -3.3\text{ V}$	$\mu\text{A}/\mu\text{m}$	-160.7	-164.0	-163.5	-160.7
I_{OFF}	$V_{\text{DS}} = -5\text{ V}$, $V_{\text{GS}} = 0\text{ V}$	$\text{A}/\mu\text{m}$	-2.9E-13	-1.3E-12	-2.9E-13	-2.9E-13

3.3 MOSFET DC Model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 3-3 Geometry of PMOS devices measured for parameter extraction

WL	0.72
180	X
144	X
90	X
36	X
18	X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.1	-5.0	-2.55
V_{GS} (V)	0.48	-3.63	-0.03
V_{BS} (V)	0	0	0

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-5.0	-0.05
V_{GS} (V)	-0.9	-3.3	-0.48
V_{BS} (V)	0	0	0

3.3.3 Model verification plots

W in the plots refer to W_{DES} .

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25°C.

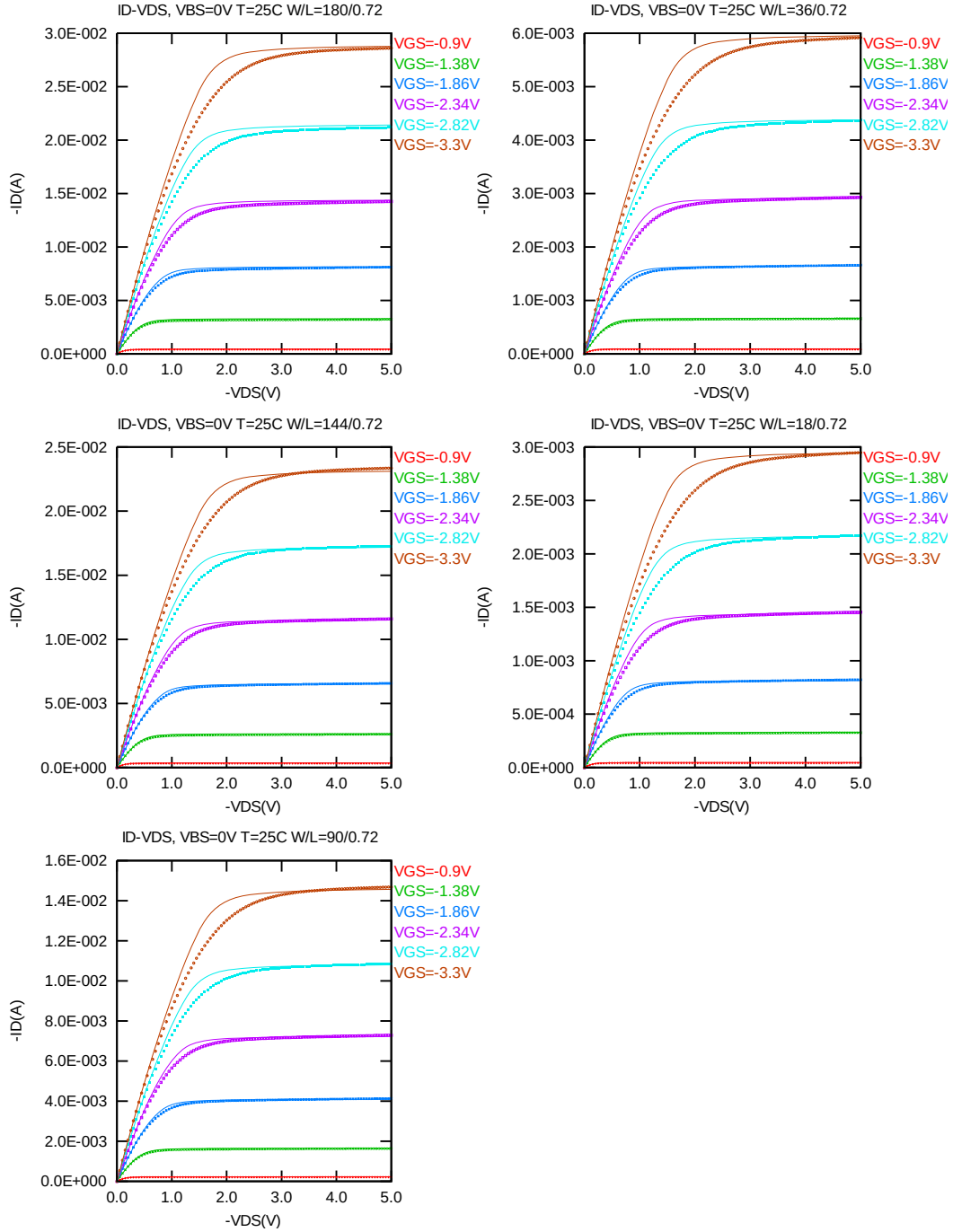


Figure 3-1 I_D vs. V_{DS} at $V_{BS} = 0V$ for PMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output conductance curves for selected devices at 25°C.

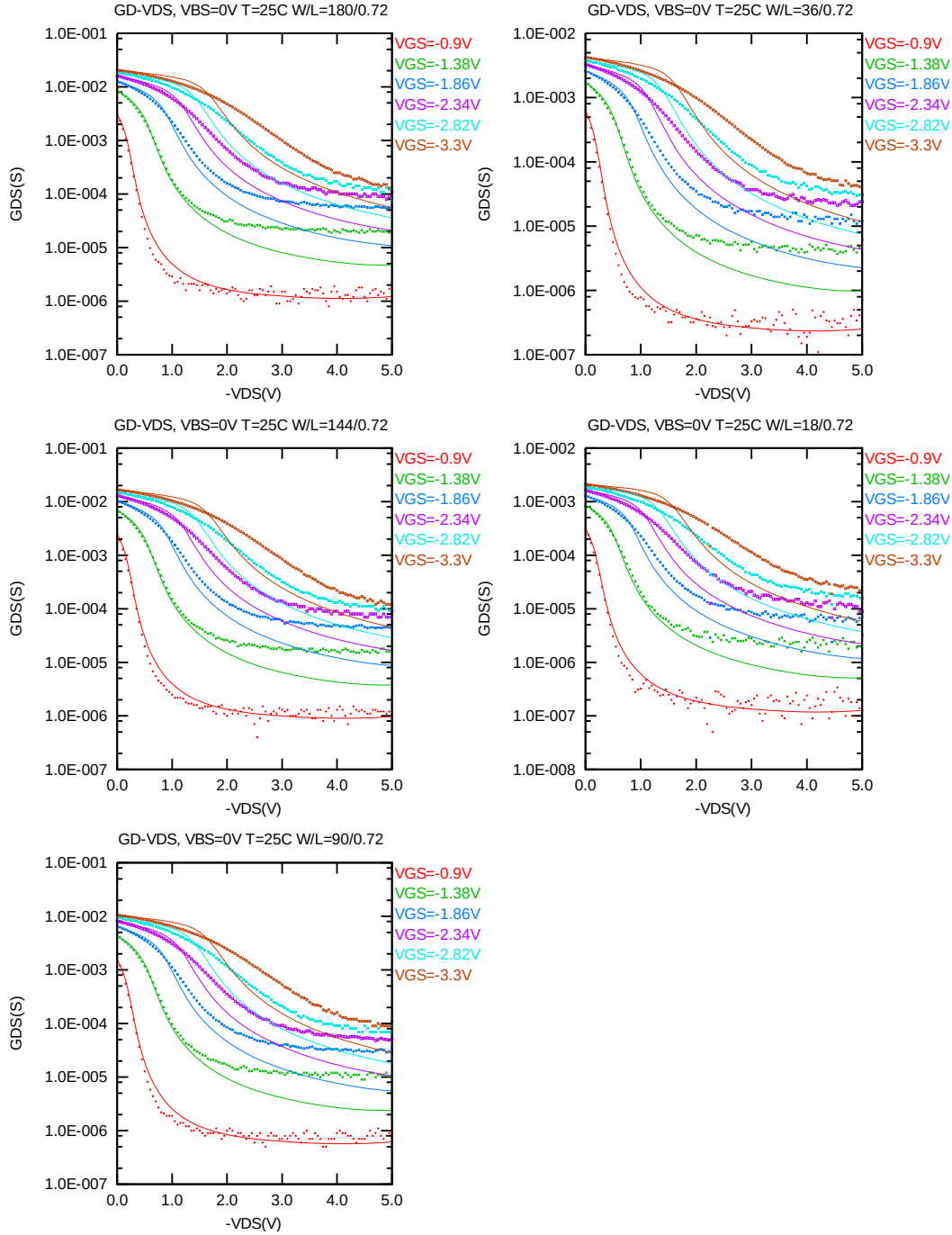


Figure 3-2 G_{DS} vs. V_{DS} at $V_{BS} = 0$ V for PMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25°C.

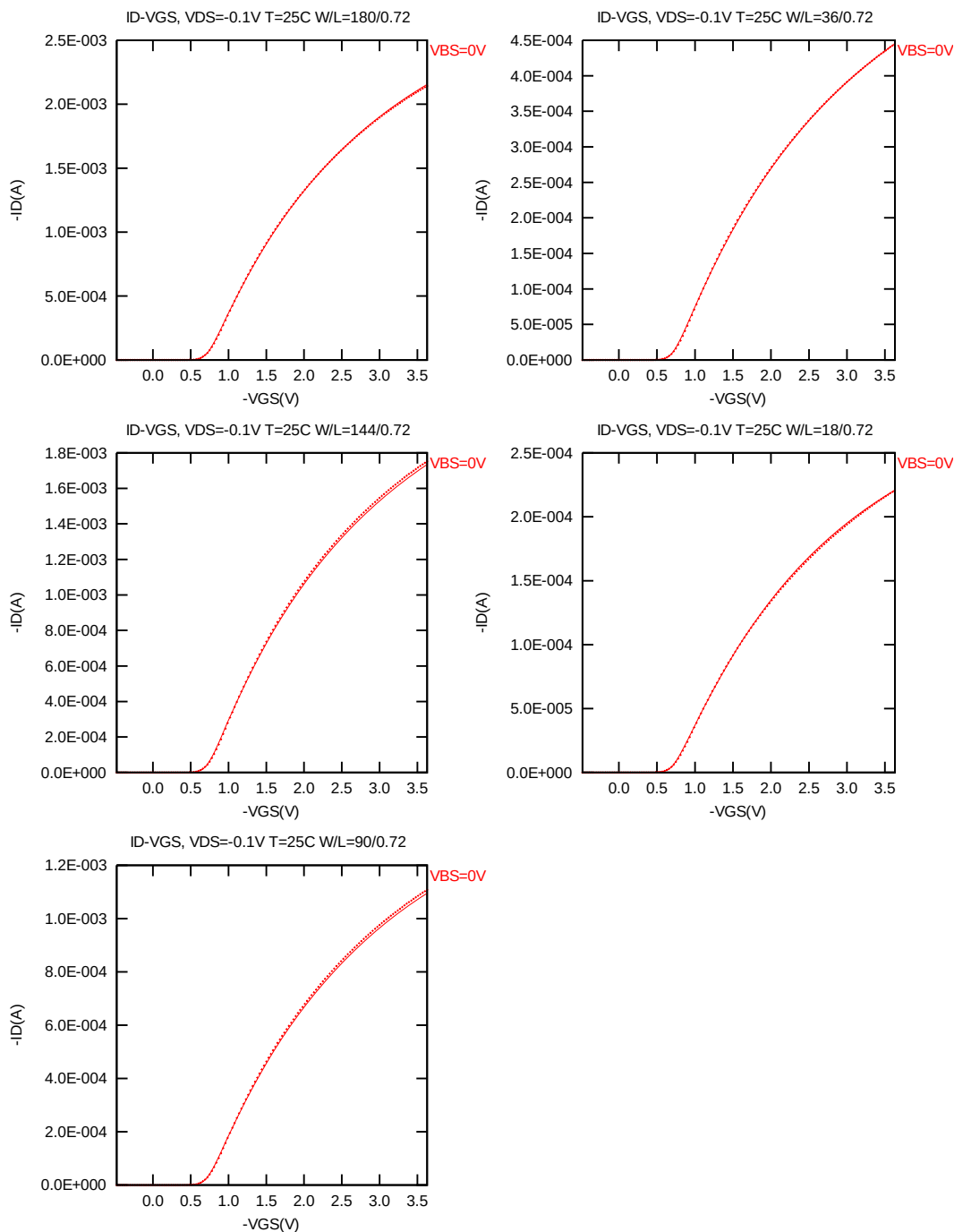


Figure 3-3 I_D vs. V_{GS} at $V_{DS} = -0.1$ V for PMOS

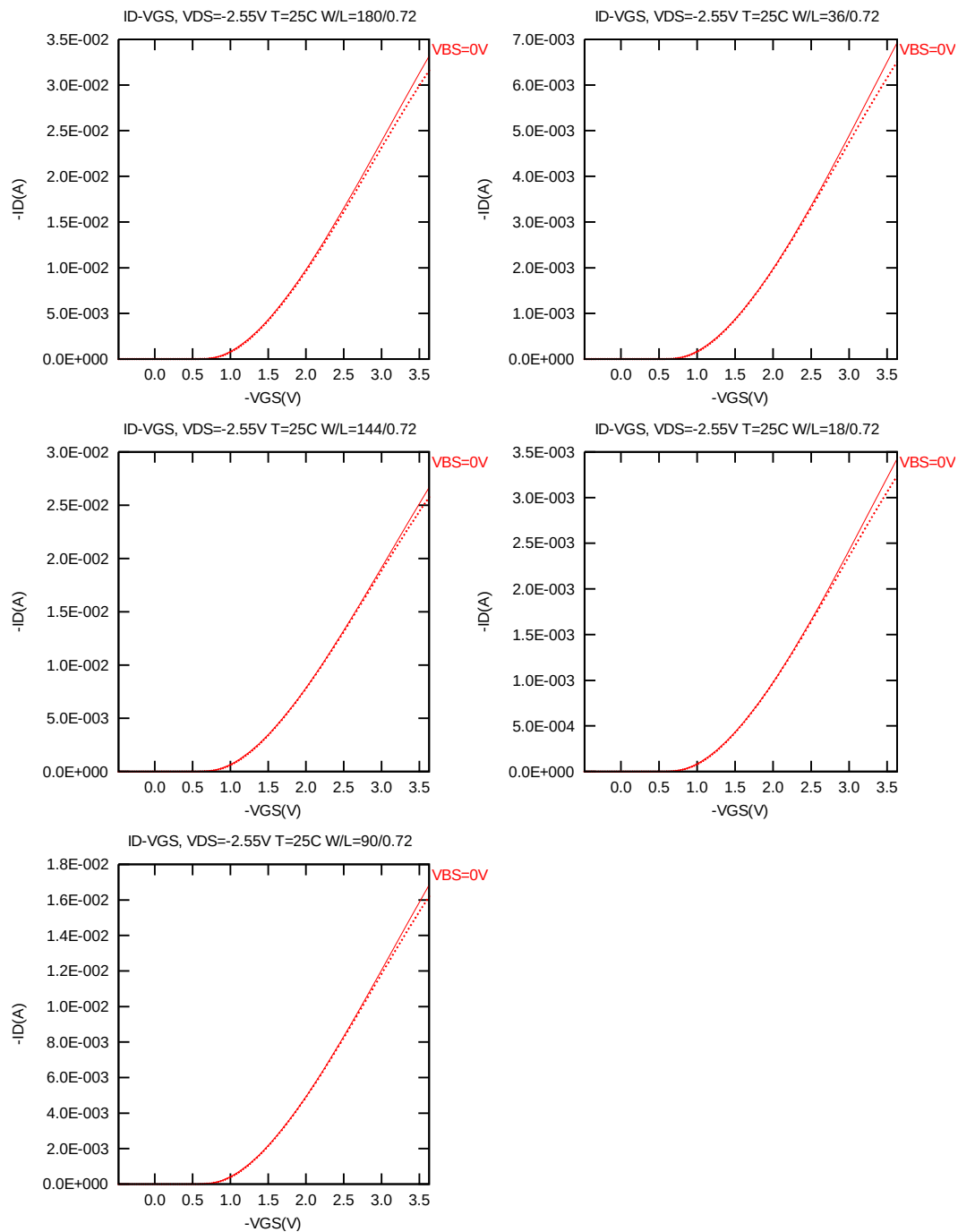


Figure 3-4 I_D vs. V_{GS} at $V_{DS} = -2.55V$ for PMOS

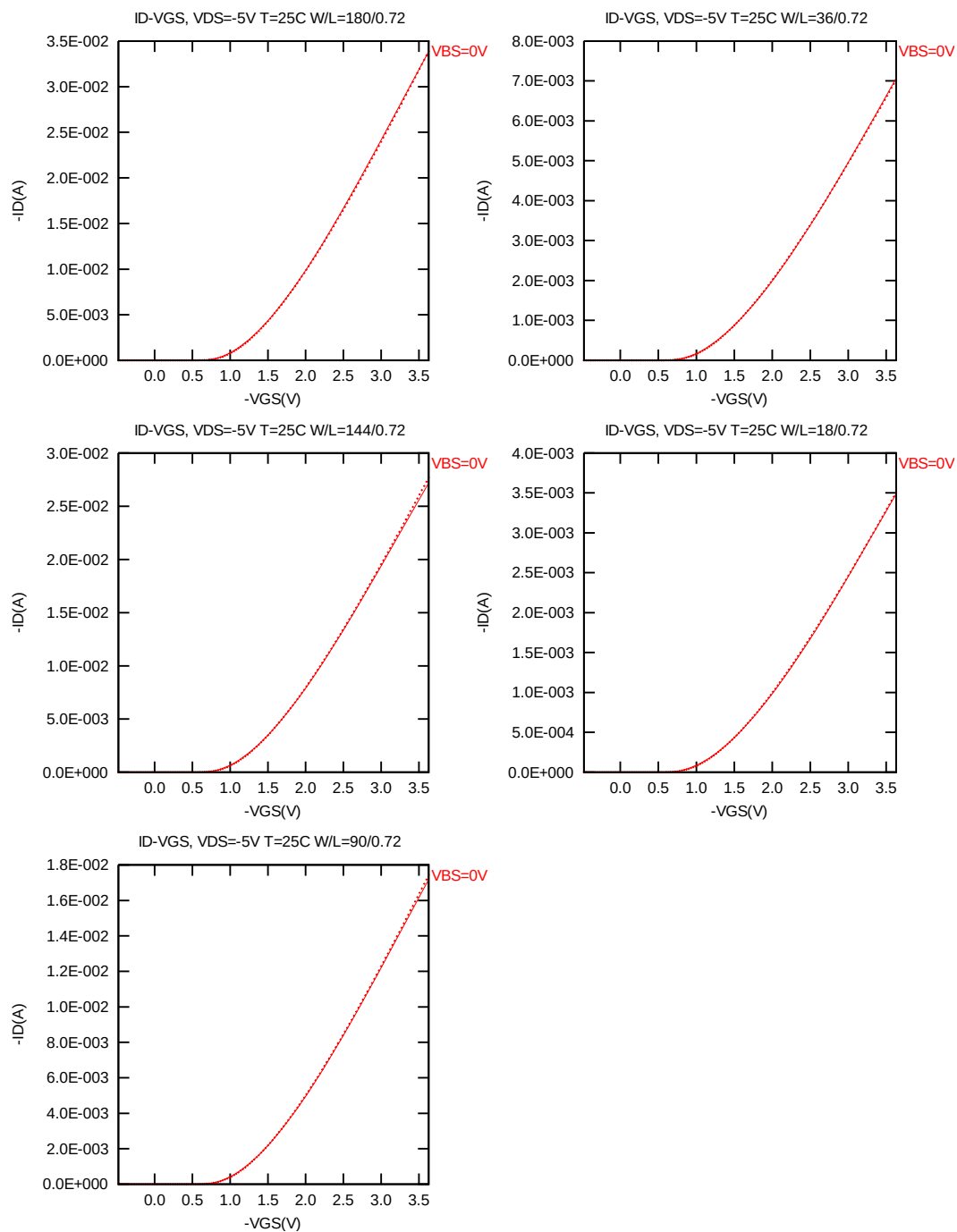


Figure 3-5 I_D vs. V_{GS} at $V_{DS} = -5.0$ V for PMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25°C.

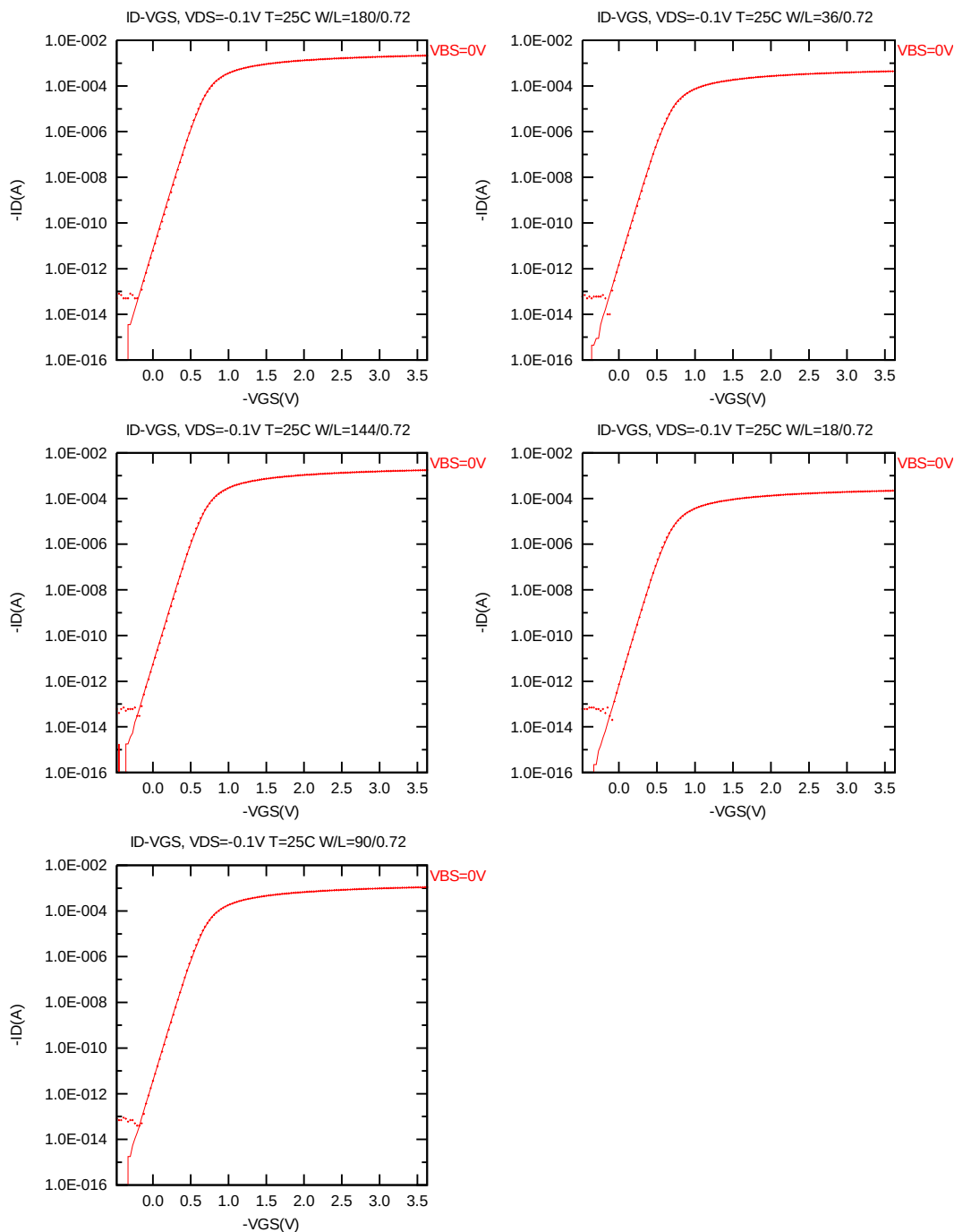


Figure 3-6 Log scaled I_D vs. V_{GS} at $V_{DS} = -0.1$ V for PMOS

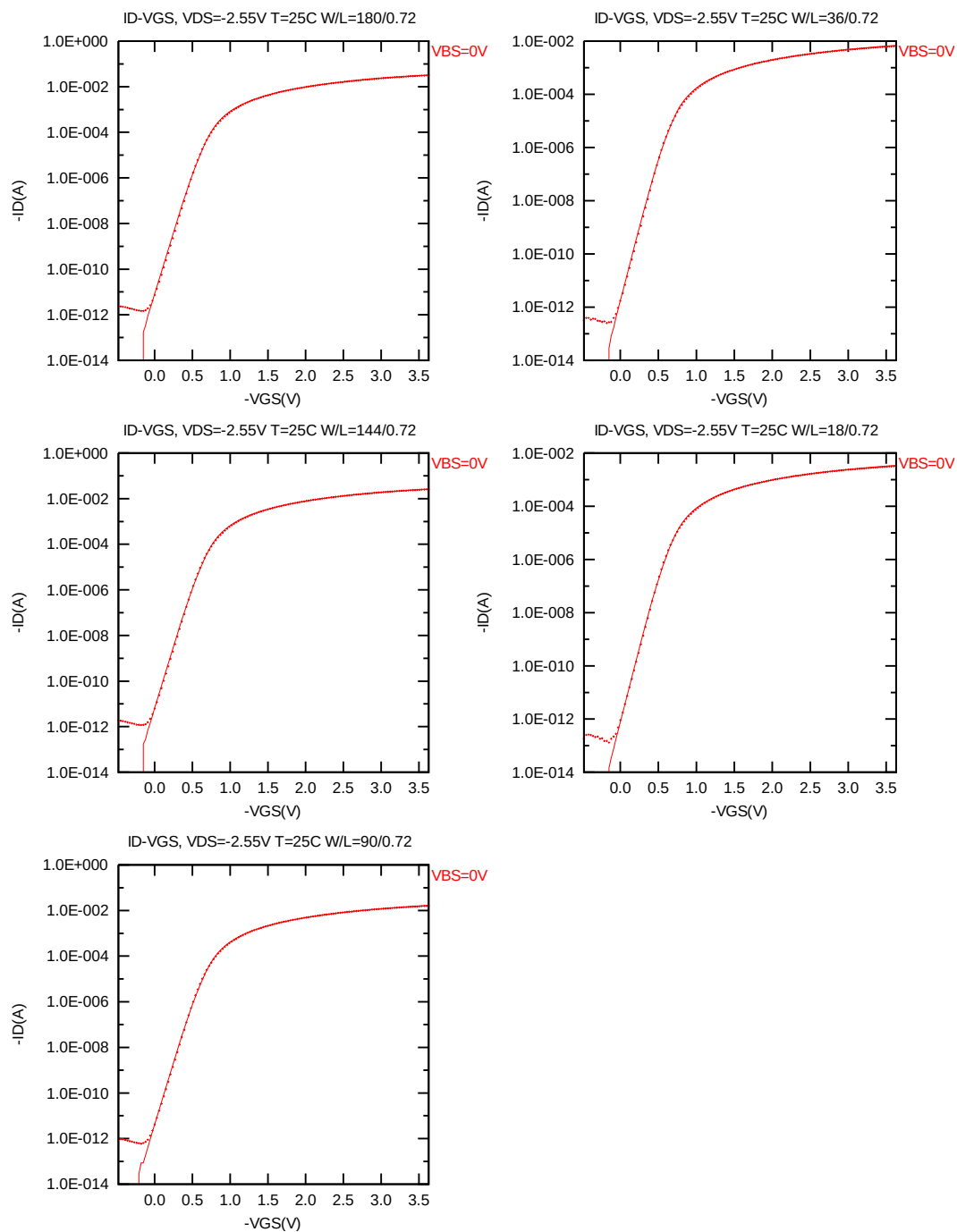


Figure 3-7 Log scaled I_D vs. V_{GS} at $V_{DS} = -2.55 V$ for PMOS

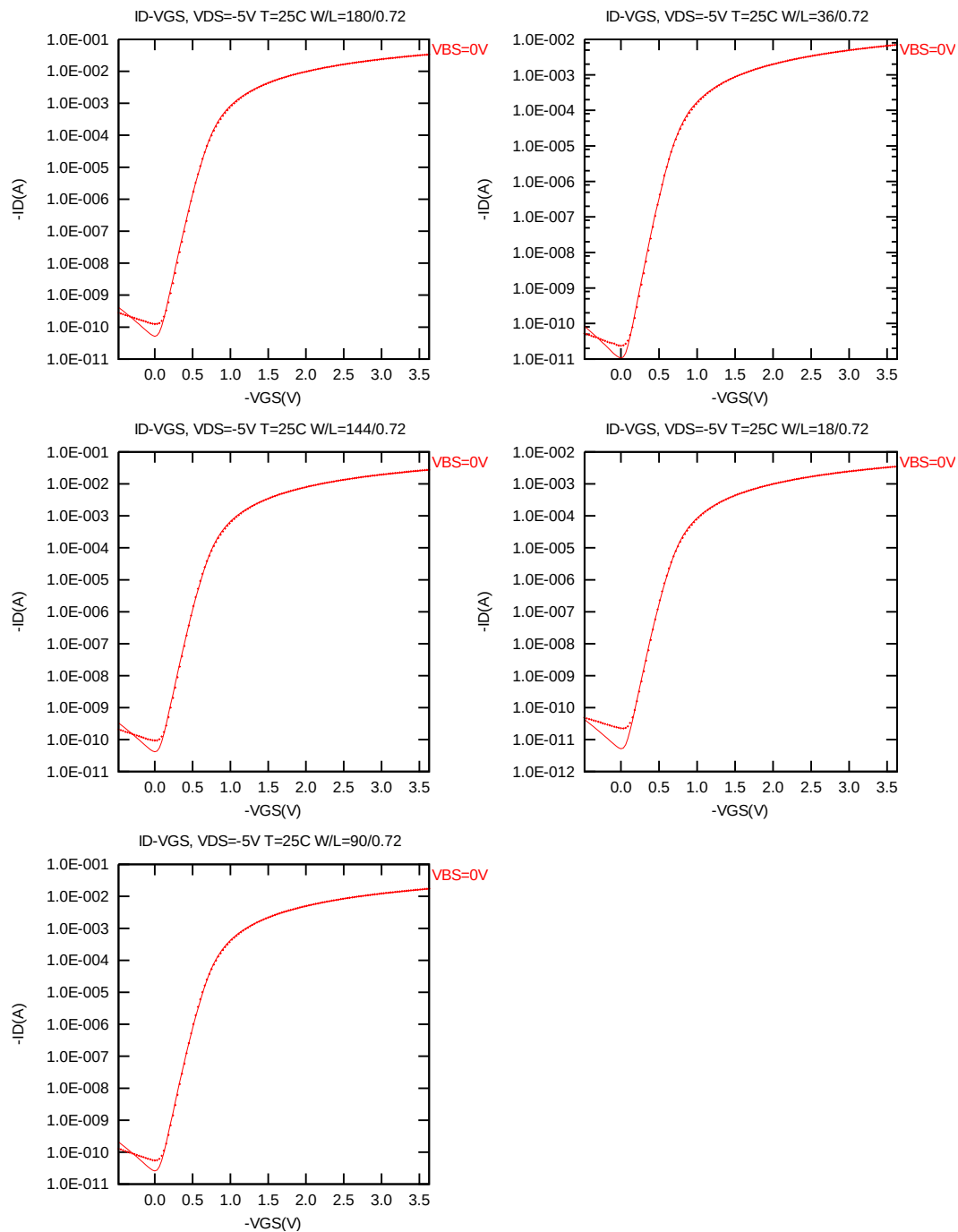


Figure 3-8 Log scaled I_D vs. V_{GS} at $V_{DS} = -5.0$ V for PMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25°C.

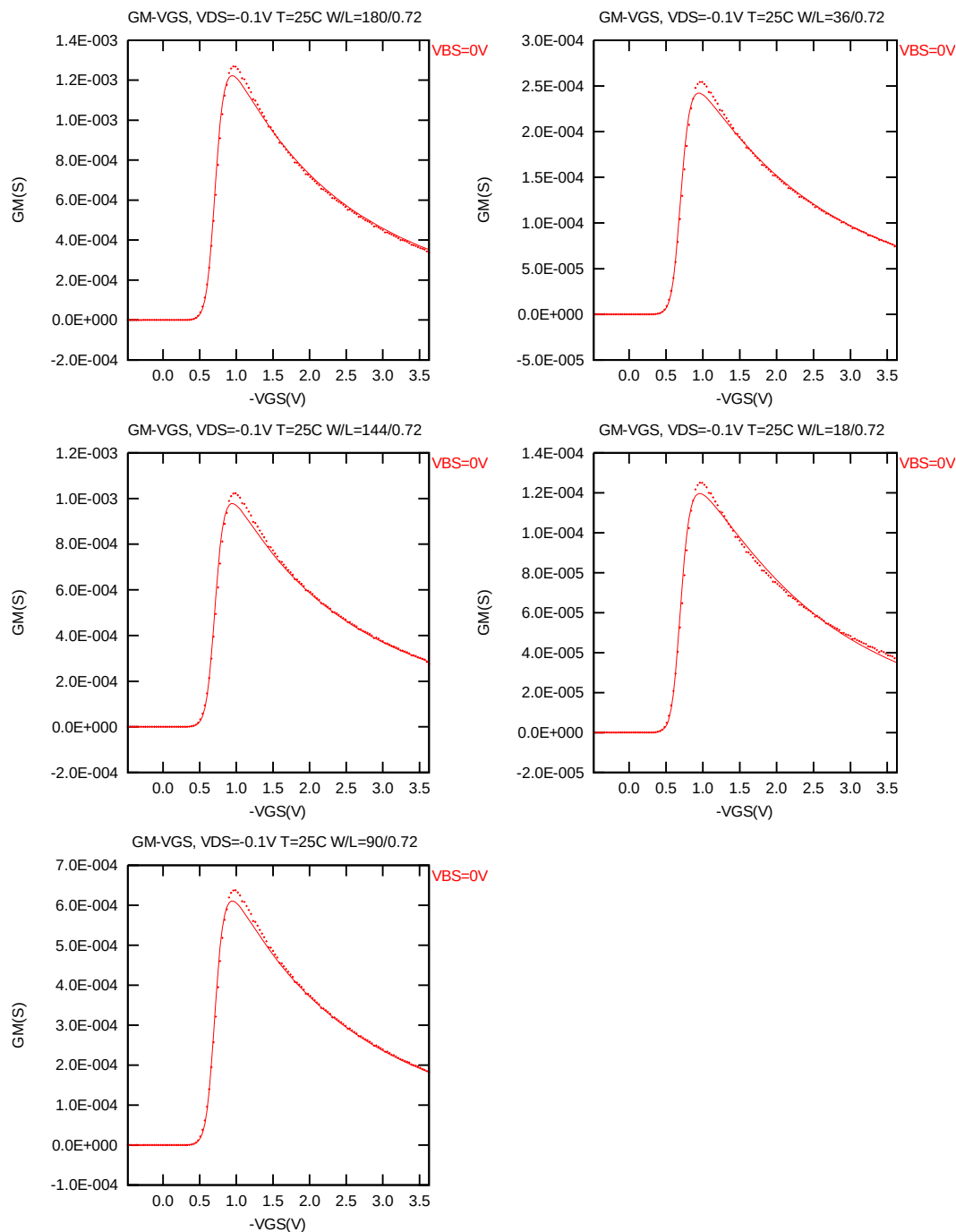


Figure 3-9 G_M vs. V_{GS} at $V_{DS} = -0.1$ V for PMOS

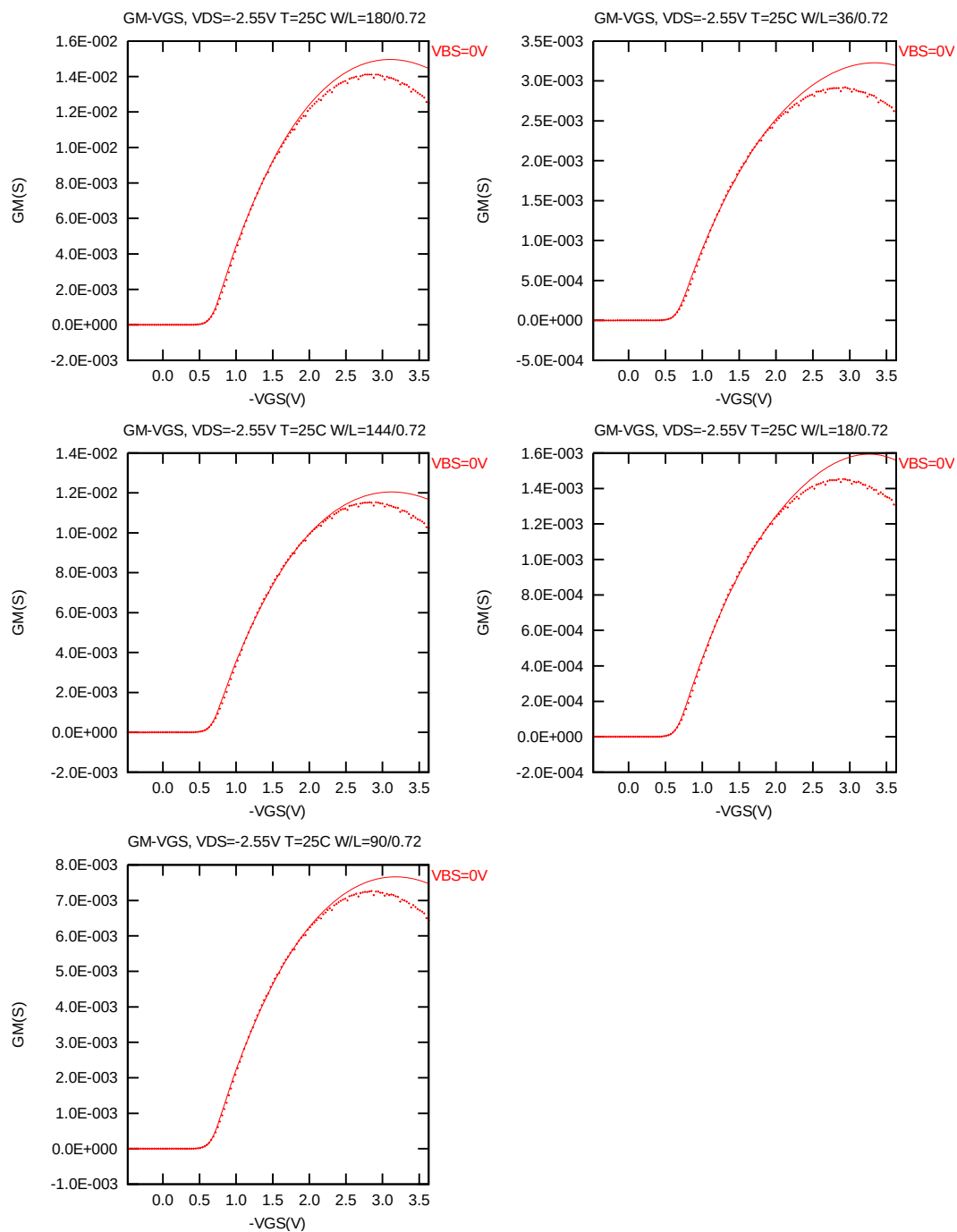


Figure 3-10 G_M vs. V_{GS} at $V_{DS} = -2.55$ V for PMOS

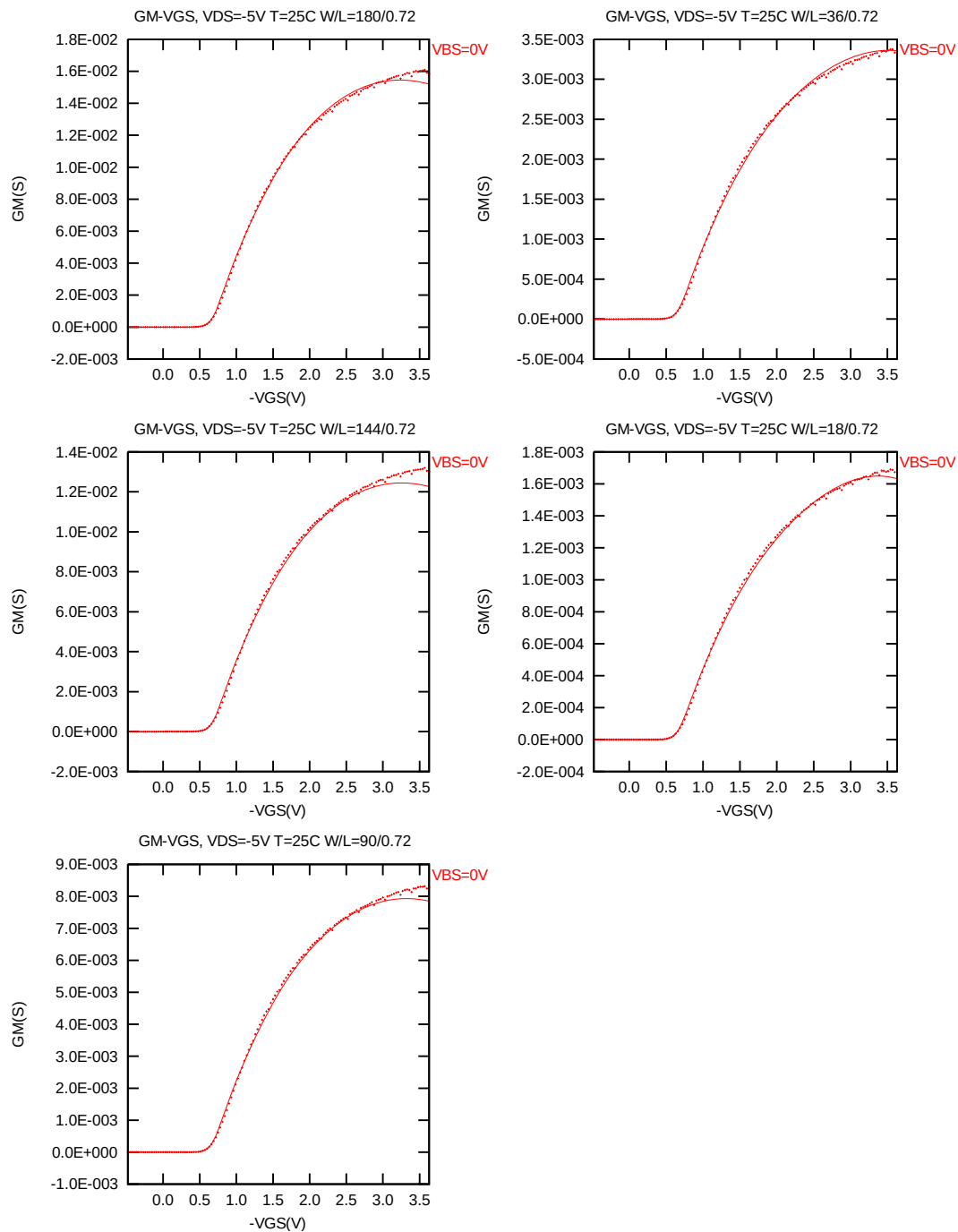


Figure 3-11 G_M vs. V_{GS} at $V_{DS} = -5.0$ V for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} , respectively.

The following plots show the channel width dependence of V_{TLIN} and V_{TSAT} extracted at $V_{DS} = -0.1$ V, $V_{DS} = -5.0$ V, respectively, for $V_{BS} = 0$ V at 25°C.

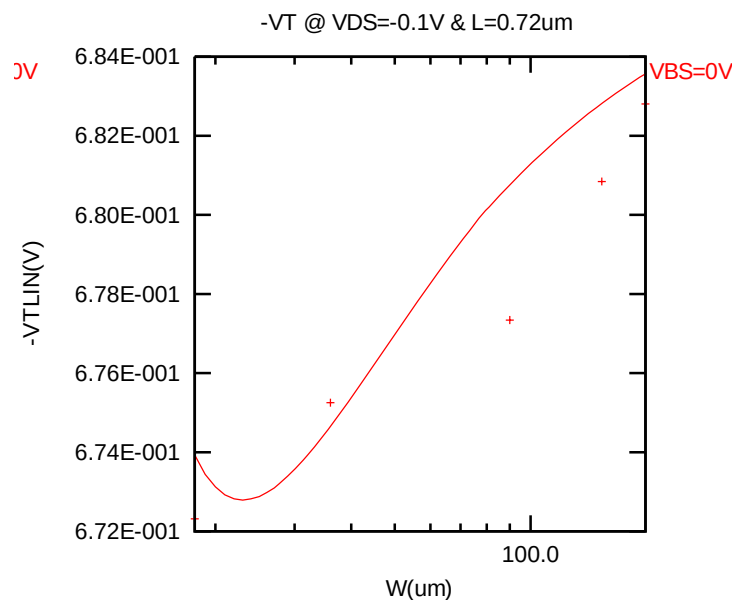


Figure 3-12 V_{TLIN} vs. channel width for low drain bias ($V_{DS} = -0.1$ V) for PMOS

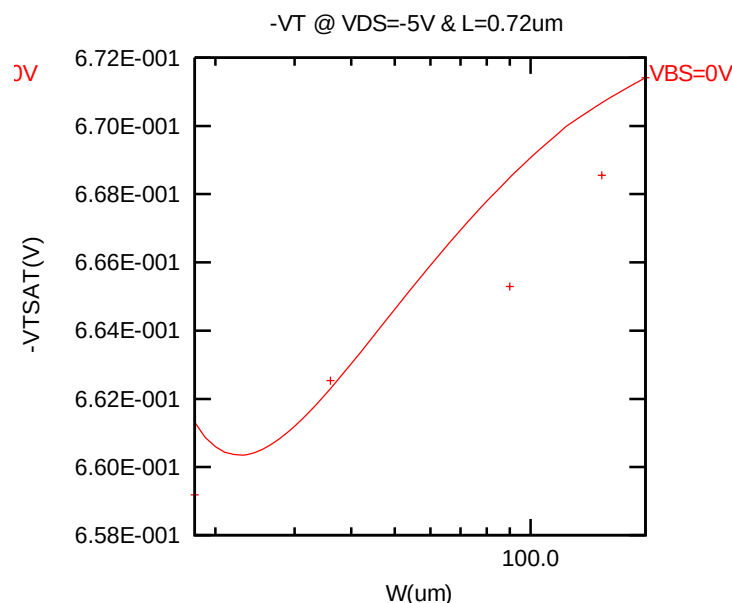


Figure 3-13 V_{TSAT} vs. channel width for high drain bias ($V_{DS} = -5.0$ V) for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} , respectively.

The following plots show the channel width dependence of I_{DLIN} and I_{ON} extracted at $V_{DS} = -0.1$ V, $V_{GS} = -3.3$ V and $V_{DS} = -5.0$ V, $V_{GS} = -3.3$ V, respectively, for $V_{BS} = 0$ V at 25°C.

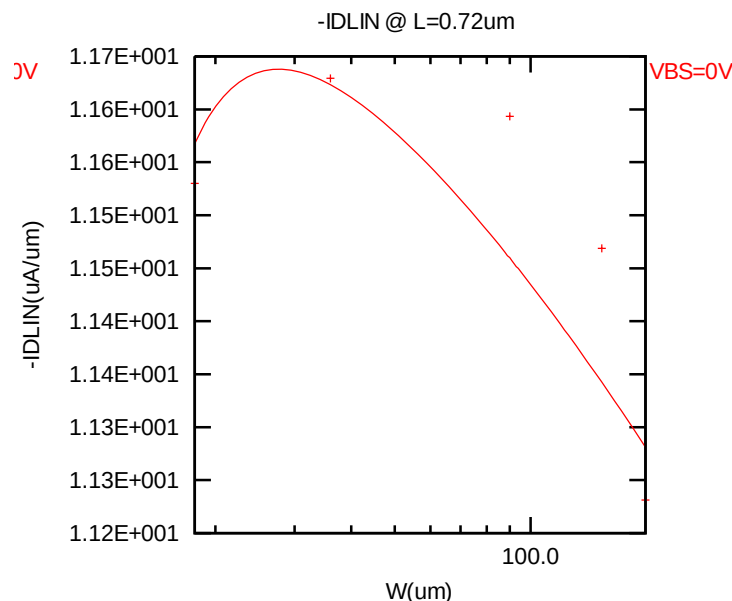


Figure 3-14 Linear current I_{DLIN} vs. channel width for PMOS

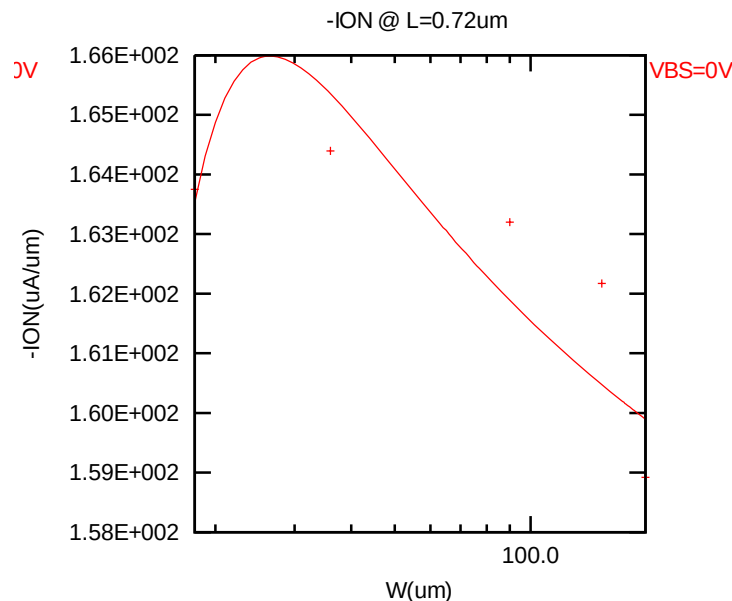


Figure 3-15 Saturation current I_{ON} vs. channel width for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} , respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = -5.0$ V and $V_{GS} = V_{BS} = 0$ V at 25°C.

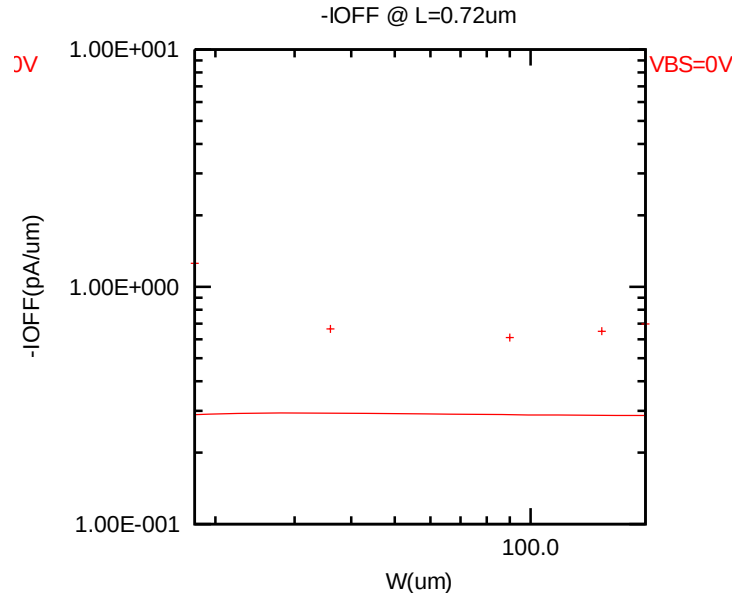


Figure 3-16 I_{OFF} vs. channel width for PMOS

V_{TLIN} and V_{TSAT} vs. temperature

The following plots show the temperature dependence of V_{TLIN} and V_{TSAT} extracted at $V_{DS} = -0.1$ V, $V_{DS} = -5.0$ V, respectively, for $V_{BS} = 0$ V at $W_{DES} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

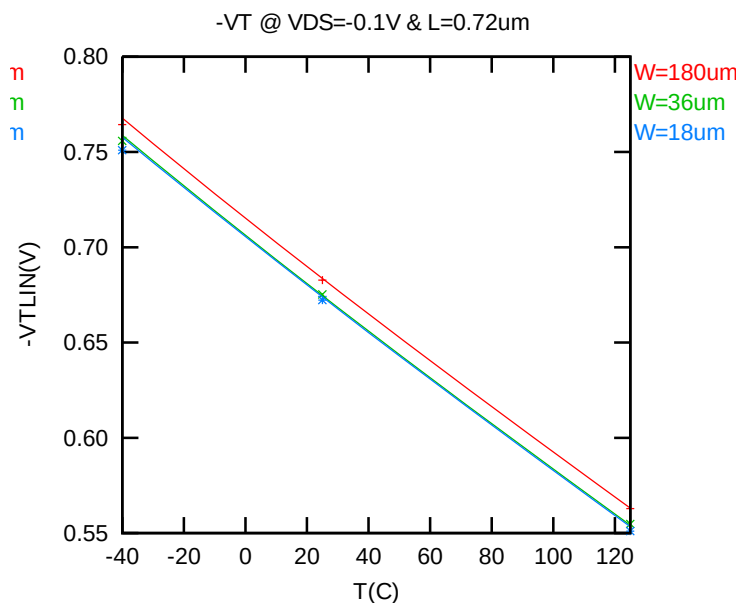


Figure 3-17 V_{TLIN} vs. temperature for PMOS

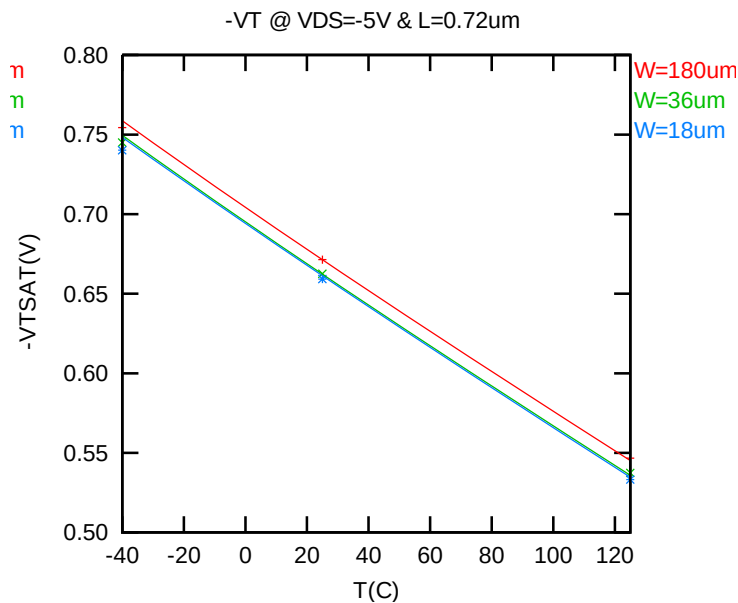


Figure 3-18 V_{TSAT} vs. temperature for PMOS

I_{DLIN} vs. temperature

The following plots show the temperature dependence of I_{DLIN} extracted at $V_{DS} = -0.1$ V, $V_{GS} = -3.3$ V, $V_{BS} = 0$ V at $W_{DES} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

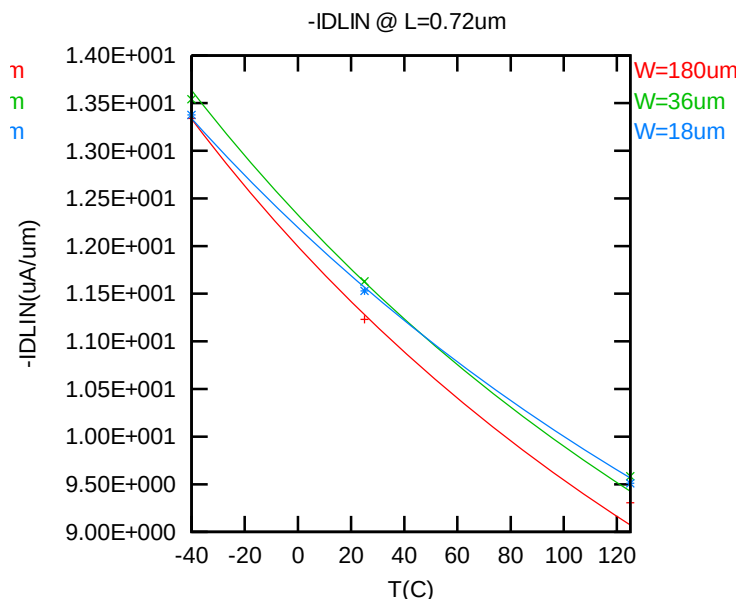


Figure 3-19 Linear current vs. temperature for PMOS

I_{DSAT} vs. temperature

The following plots show the temperature dependence of I_{DSAT} extracted at $V_{DS} = -5.0$ V, $V_{GS} = -3.3$ V, $V_{BS} = 0$ V at $W_{DES} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

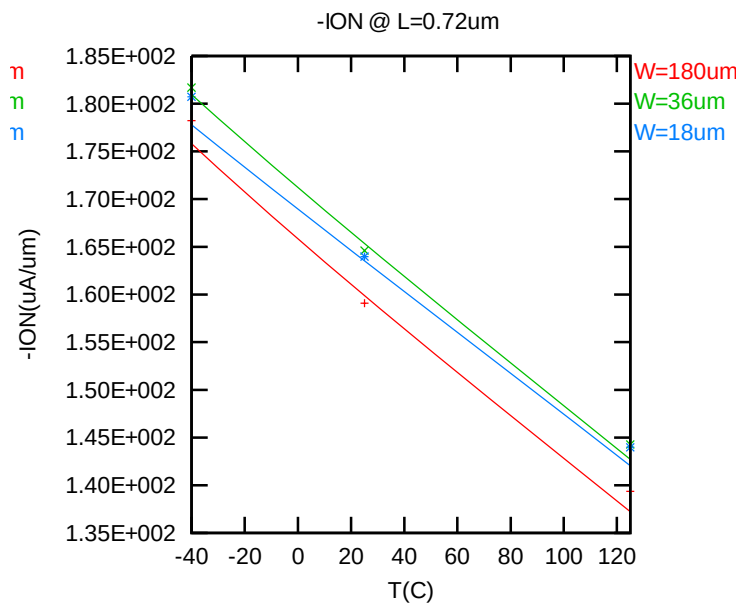


Figure 3-20 Saturation current vs. temperature for PMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{\text{DS}} = -5.0 \text{ V}$, $V_{\text{GS}} = V_{\text{BS}} = 0 \text{ V}$ at $W_{\text{DES}} = 20\mu\text{m}$, $40\mu\text{m}$, $100\mu\text{m}$, $160\mu\text{m}$, and $200\mu\text{m}$.

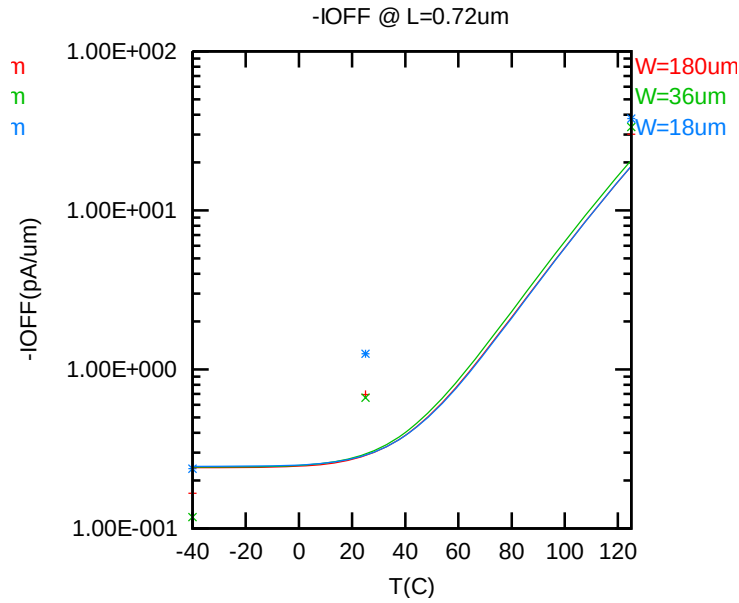


Figure 3-21 I_{OFF} vs. temperature for PMOS

3.3.4 Model fitting accuracy

Model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- g. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
 - h. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
 - i. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$
 - j.
- , where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- k. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- l. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 3-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

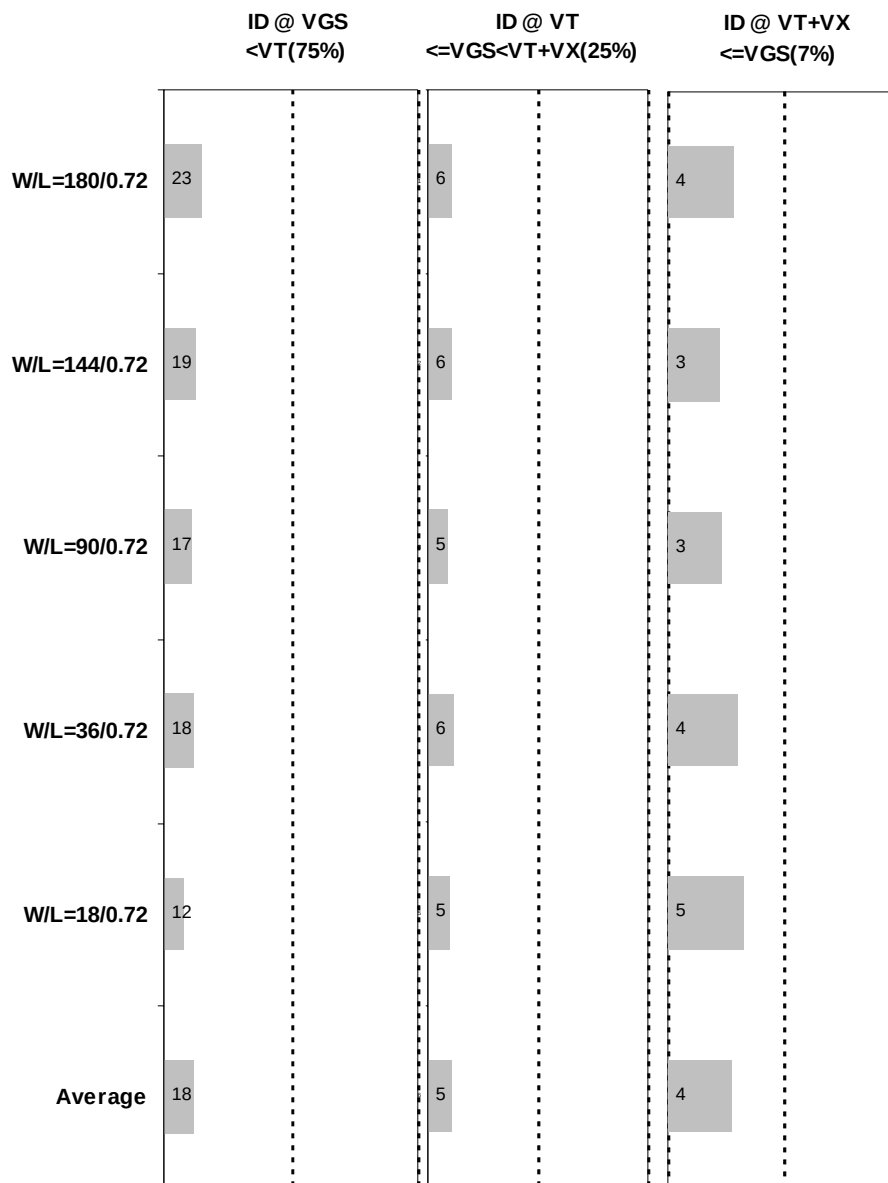


Figure 3-22 I_D accuracy at 25°C for PMOS

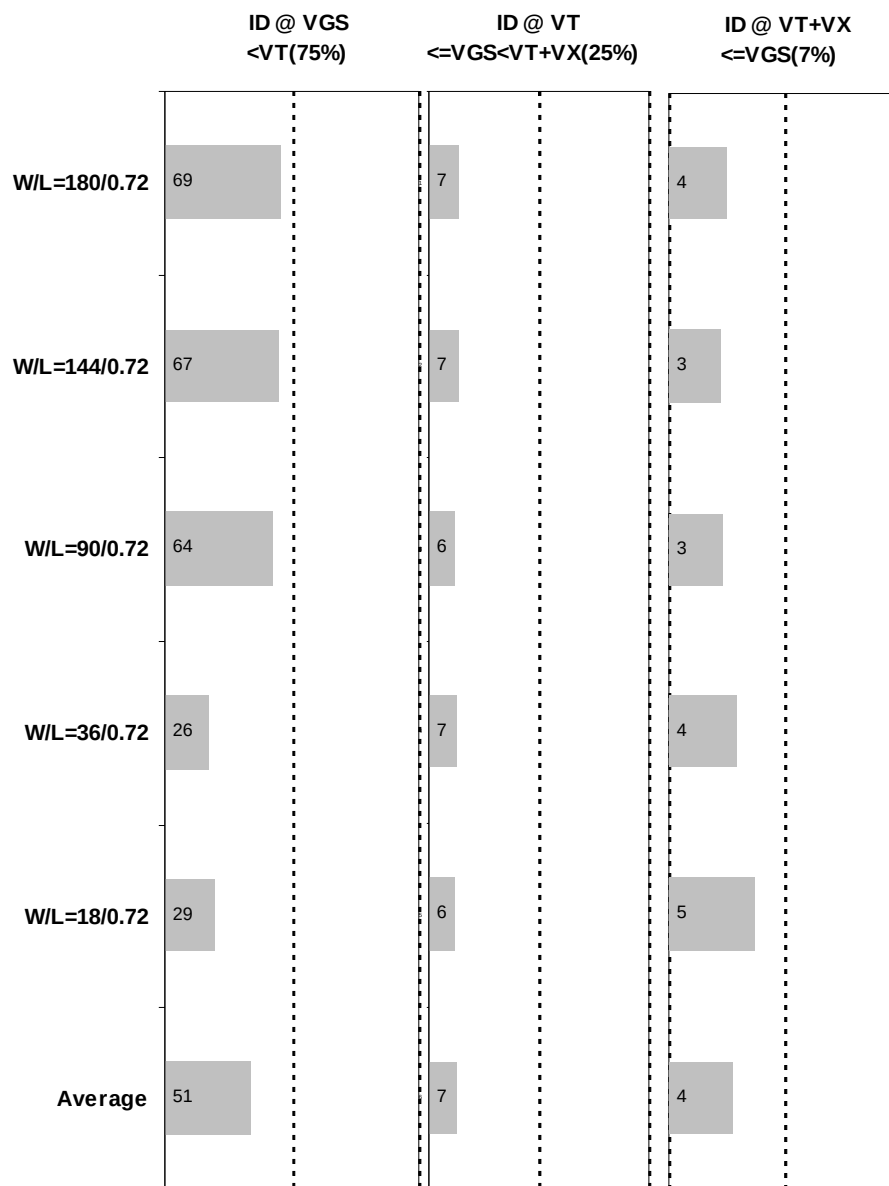


Figure 3-23 I_D accuracy at 125°C for PMOS

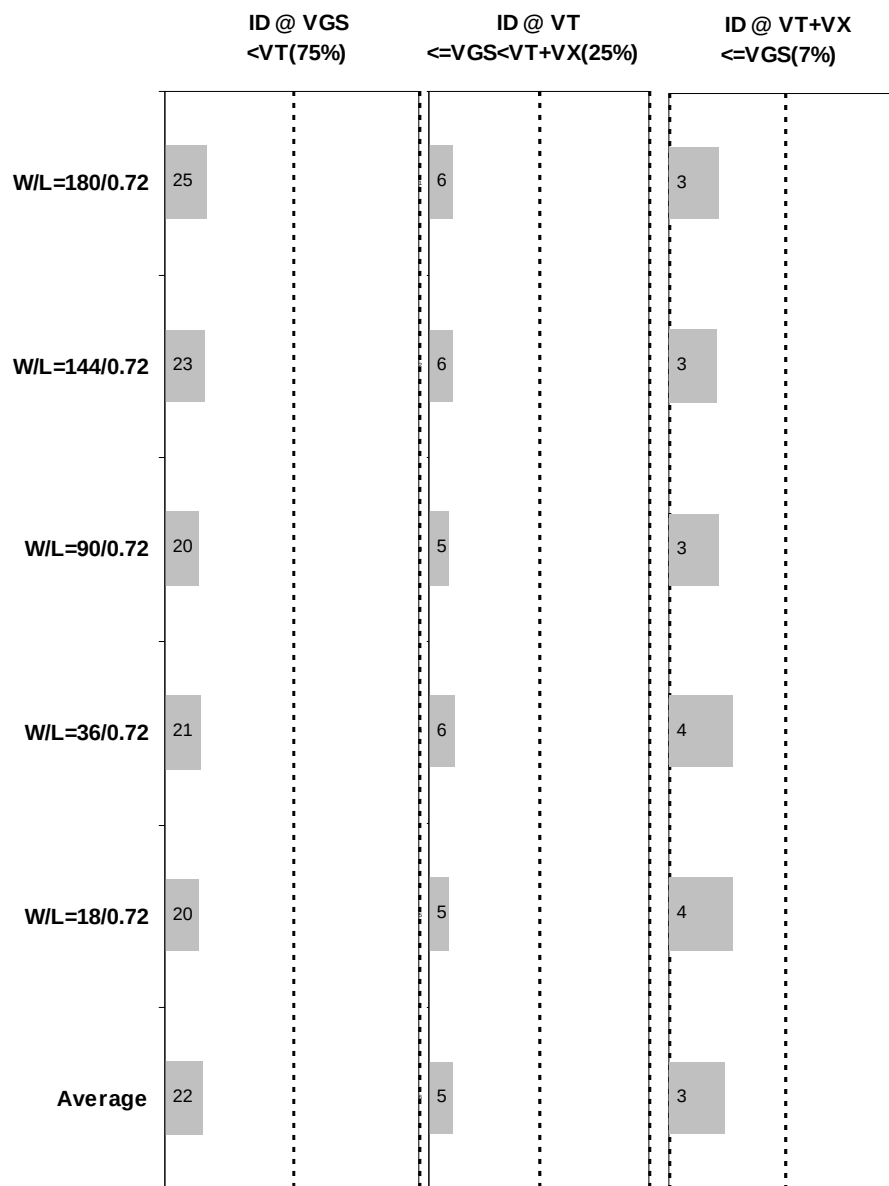


Figure 3-24 I_D accuracy at -40°C for PMOS

G_M and G_{DS} accuracy

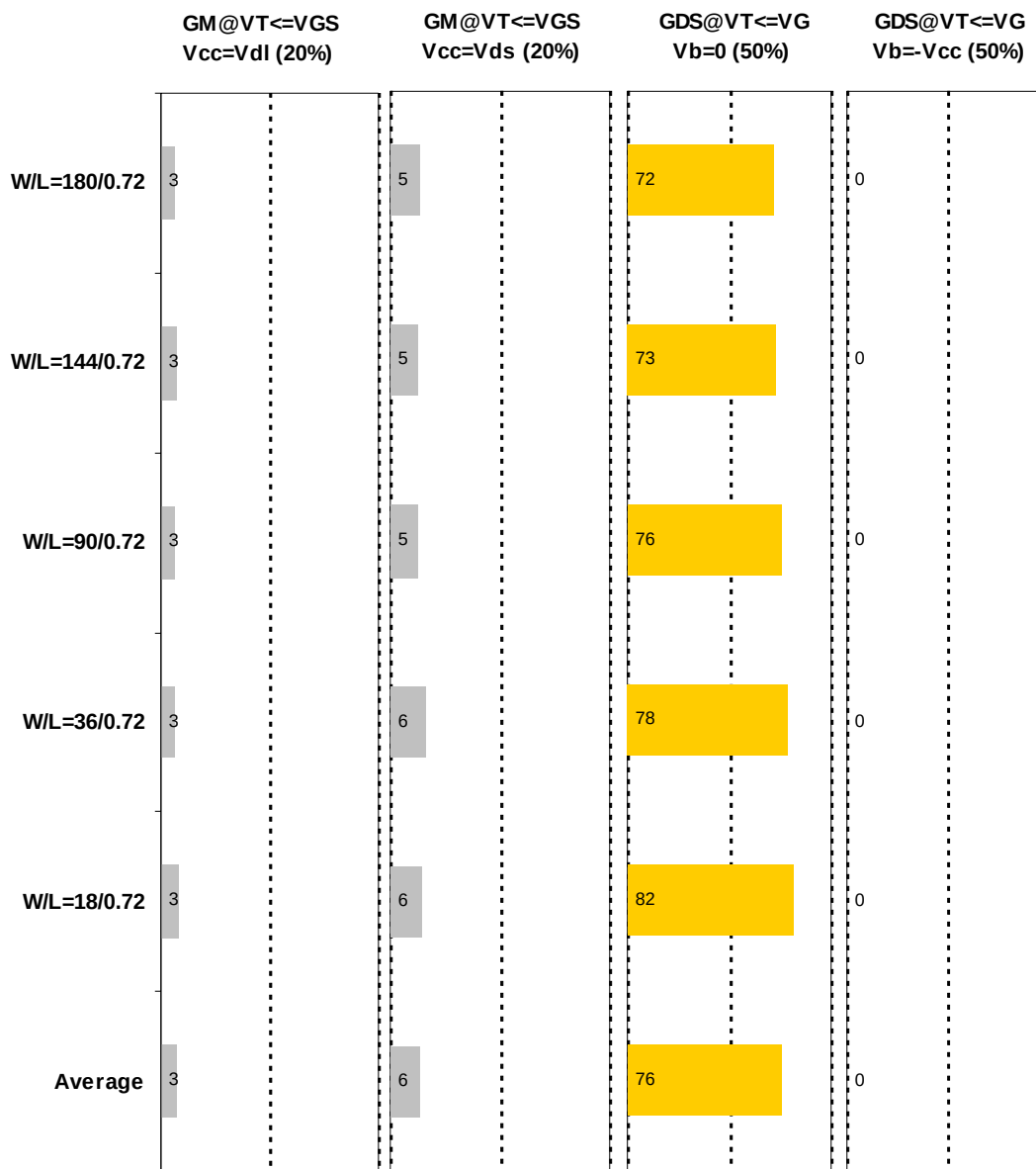


Figure 3-25 G_M and G_{DS} accuracy at 25°C for PMOS

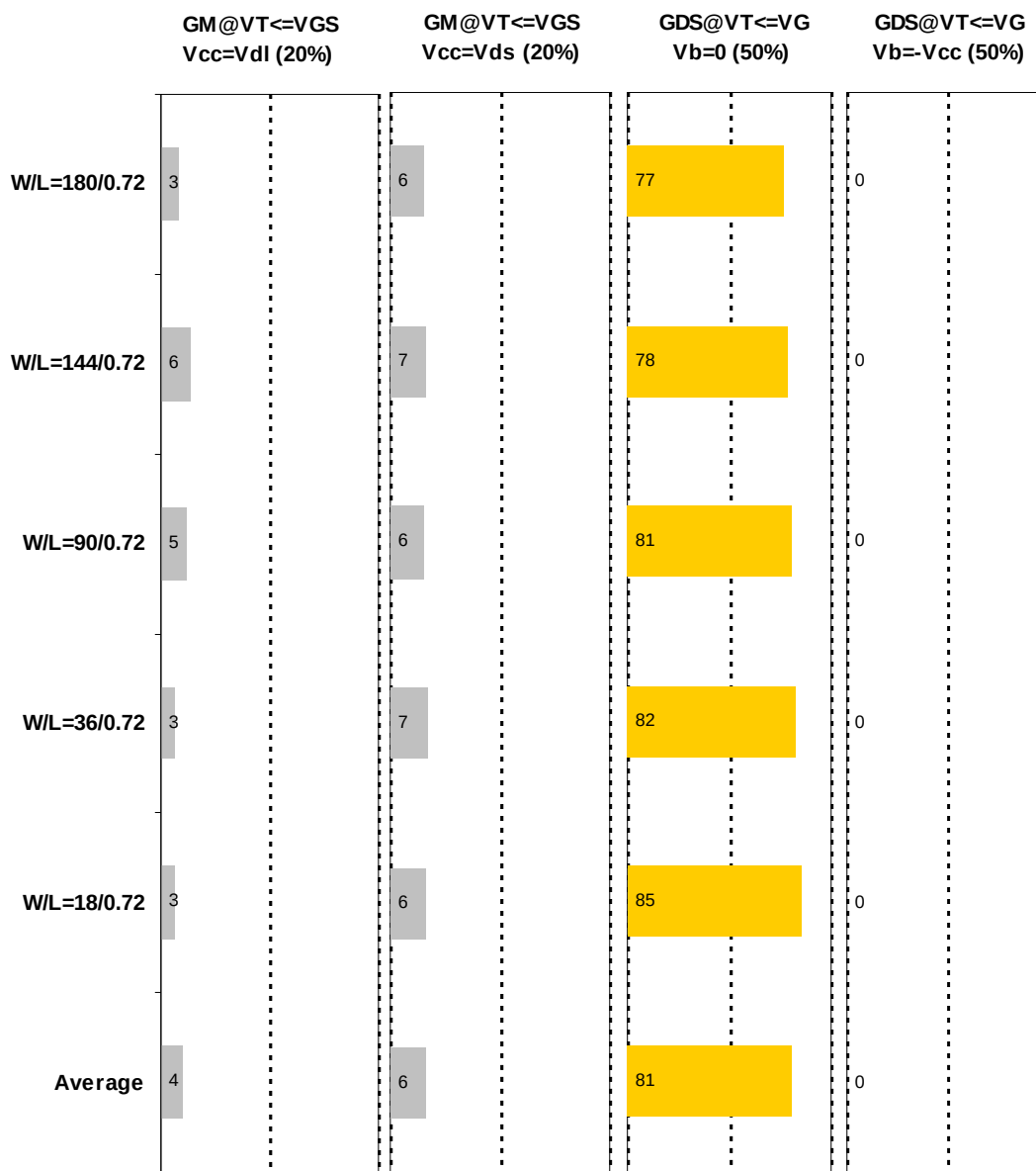


Figure 3-26 G_M and G_{DS} accuracy at 125°C for PMOS

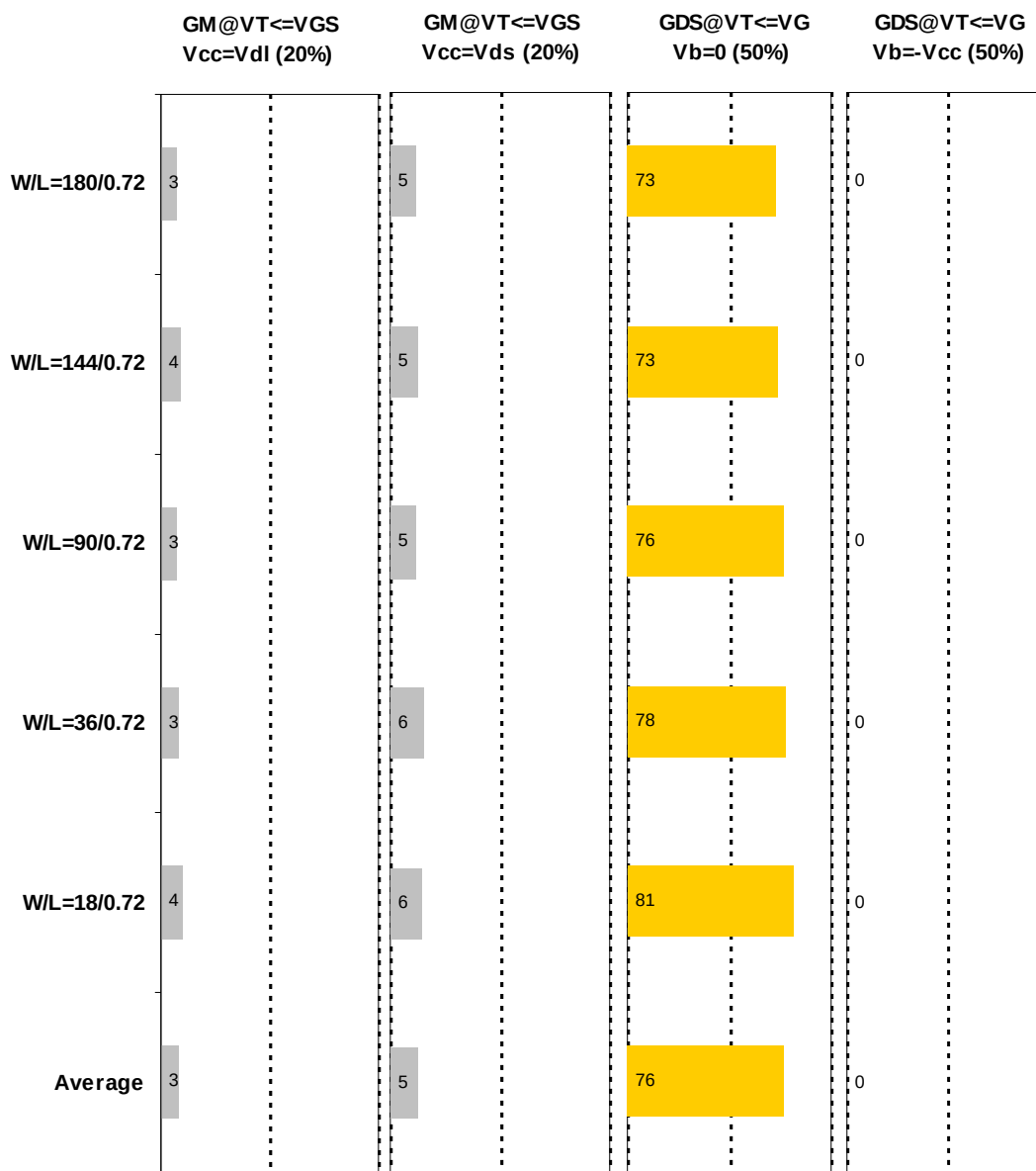


Figure 3-27 G_M and G_{DS} accuracy at -40°C for PMOS

3.4 MOSFET Capacitance Model

3.4.1 Test devices

Following test structures were used for MOSFET capacitance model parameter extraction.

Table 3-6 Large area measurement structure for PMOS Cgc capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	0.8	10
	0.8	50
	0.8	100

3.4.2 Measurement conditions

Measurement conditions for MOSFET capacitance are summarized in Table 3-8 below.

Table 3-7 Measurement conditions for PMOS capacitance model

V _{BIAS} [V]	From -3.6 to 3.6 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	25

3.4.3 Width and length AC model accuracy

Width and length AC model accuracy

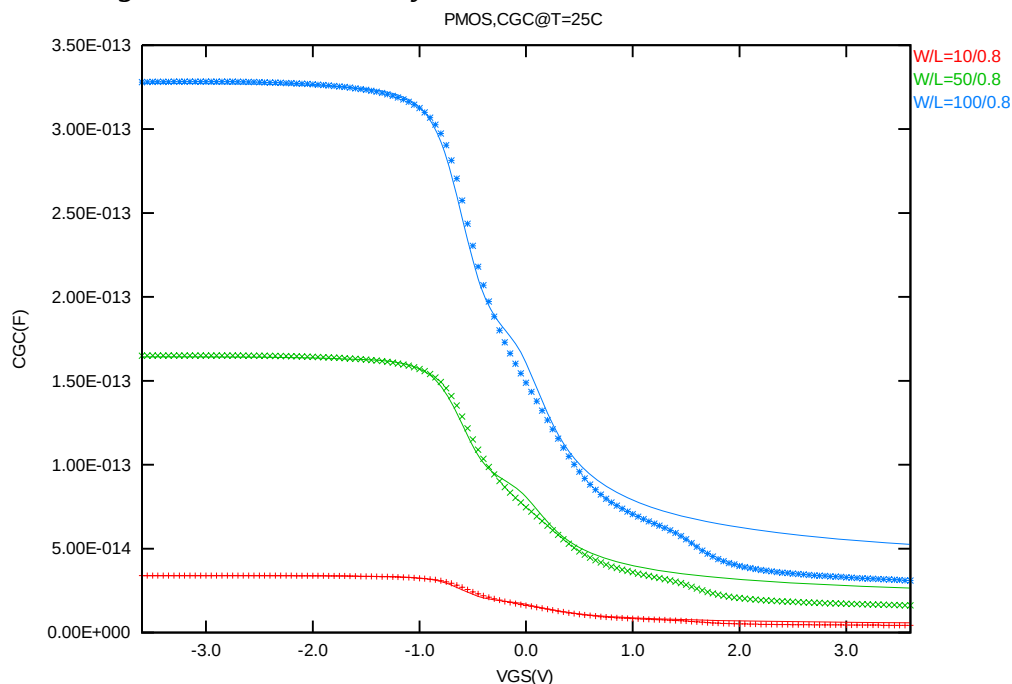


Figure 3-28 Measured and modeled C_{GC} curves for pMOSFET test structures

3.5 Parasitic Source / Drain Junction Capacitance Model

3.5.1 Test devices

The following test structures were used for parameter extraction.

Table 3-8 Junction capacitor structures for PMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	3.11E-08	4.08E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.11E-08	2.84E-02	Finger diode

3.5.2 Measurement conditions

Junction capacitance was measured under the bias conditions summarized in the following table.

Table 3-9 Junction capacitor measurement conditions for PMOS

V _{BIAS} [V]	VH: nwell & VL: p+ From 0 to 8.5 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,150

3.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from silicon data.

Table 3-10 Extracted parameter from junction capacitor measurement for PMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	1.18E-03	CJSW	F/m	3.33E-10	CJSWG	F/m	3.61E-10
PB	V	7.79E-01	PBSW	V	7.39E-01	PBSWG	V	8.45E-01
MJ	--	3.58E-01	MJSW	--	3.73E-01	MJSWG	--	7.70E-01
TCJ	1/K	1.03E-03	TCJSW	1/K	1.19E-03	TCJSWG	1/K	1.42E-03
TPB	V/K	1.72E-03	TPBSW	V/K	1.98E-03	TPBSWG	V/K	2.58E-03

C_J , C_{JSW} vs. V_J plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

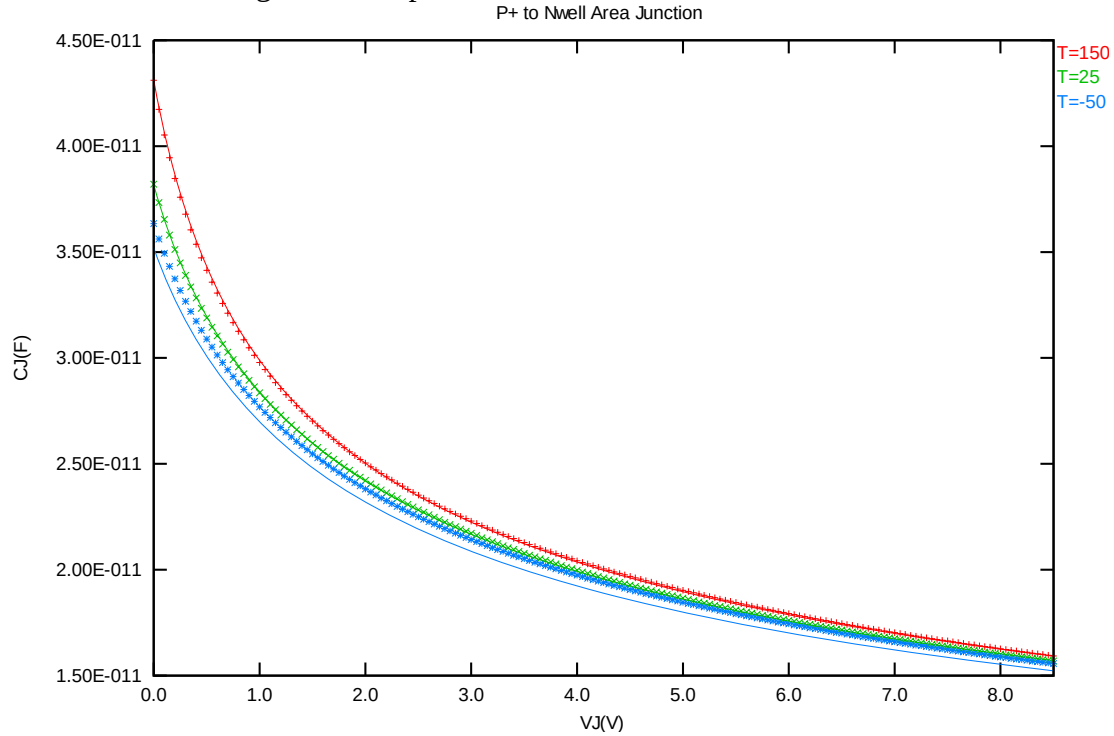


Figure 3-29 Bottom area junction capacitance at -50°C , 25°C and 150°C for PMOS

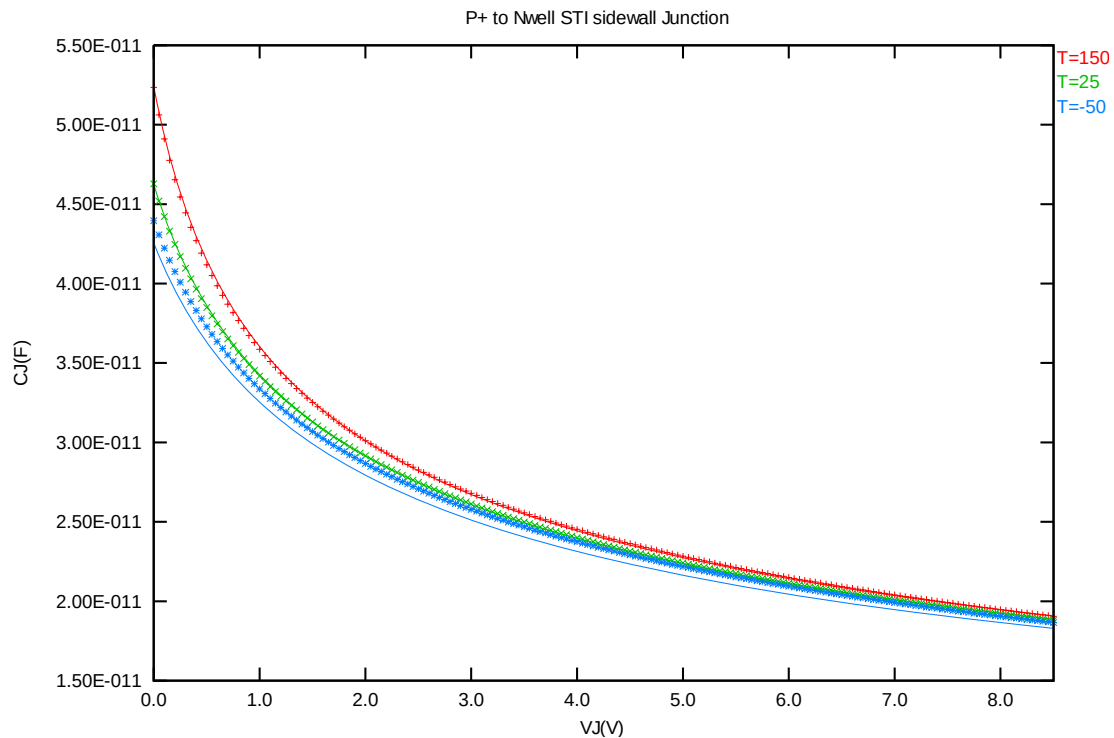


Figure 3-30 STI bounded sidewall junction capacitance at -50°C , 25°C and 150°C for PMOS

4 Worst Case Model

4.1 Parameter Variations

Table 4-1 Corner parameters

	n_ldcd_d50g33_l110ae	TT	SS	FF	SNFP	FNSP
1	tox	7.49E-09	+1.50E-10	-1.50E-10	+0.00E+00	+0.00E+00
2	xl	-5.43E-07	+2.00E-08	-2.00E-08	+1.00E-08	-1.00E-08
3	xw	1.00E-09	-1.00E-06	+1.00E-06	-5.00E-07	+5.00E-07
4	vth0	7.05E-01	+7.05E-02	-7.45E-02	+3.59E-02	-3.67E-02
5	u0	3.18E-02	-7.11E-03	+7.56E-03	-3.67E-03	+3.80E-03
6	vsat	9.72E+04	-9.36E+03	+6.59E+03	-5.05E+03	+4.46E+03
7	rdsw	1.24E+03	+0.00E+00	+0.00E+00	+0.00E+00	+0.00E+00
8	cgso	5.00E-11	-10.0%	+10.0%	+0.0%	+0.0%
9	cgdo	5.00E-11	-10.0%	+10.0%	+0.0%	+0.0%
10	cf	6.83E-11	-6.83E-12	+6.83E-12	+0.00E+00	+0.00E+00
11	cgsi	1.80E-10	-10.0%	+10.0%	+0.0%	+0.0%
12	cgdl	2.48E-09	-10.0%	+10.0%	+0.0%	+0.0%
13	cj	1.10E-04	+10.0%	-10.0%	+10.0%	-10.0%
14	cjsw	9.36E-10	+10.0%	-10.0%	+10.0%	-10.0%
15	cjswg	1.68E-09	+10.0%	-10.0%	+10.0%	-10.0%
16	agidl	1.90E-06	-1.62E-06	+8.73E-06	-1.79E-06	+1.88E-05

	p_ldcd_d50g33_l110ae	TT	SS	FF	SNFP	FNSP
1	tox	7.68E-09	+1.50E-10	-1.50E-10	+0.00E+00	+0.00E+00
2	xl	-3.63E-07	+2.00E-08	-2.00E-08	-1.00E-08	+1.00E-08
3	xw	4.00E-09	-1.00E-06	+1.00E-06	+5.00E-07	-5.00E-07
4	vth0	-6.60E-01	-8.43E-02	+8.74E-02	+4.32E-02	-4.25E-02
5	u0	1.08E-02	-7.68E-04	+4.97E-04	+2.99E-04	-3.63E-04
6	vsat	9.22E+04	-2.52E+03	+1.30E+03	+1.60E+03	-1.86E+03
7	rdsw	1.61E+03	+0.00E+00	+0.00E+00	+0.00E+00	+0.00E+00
8	cgso	9.11E-11	-10.0%	+10.0%	+0.0%	+0.0%
9	cgdo	1.50E-11	-10.0%	+10.0%	+0.0%	+0.0%
10	cf	6.78E-11	-6.78E-12	+6.78E-12	+0.00E+00	+0.00E+00
11	cgs1	2.02E-10	-10.0%	+10.0%	+0.0%	+0.0%
12	cgdl	1.56E-09	-10.0%	+10.0%	+0.0%	+0.0%
13	cj	1.18E-03	+10.0%	-10.0%	-10.0%	+10.0%
14	cjsw	3.33E-10	+10.0%	-10.0%	-10.0%	+10.0%
15	cjswg	3.61E-10	+10.0%	-10.0%	-10.0%	+10.0%
16	agidl	6.00E-06	-5.02E-06	+2.97E-05	+5.92E-05	-5.60E-06

4.2 Wafer Data vs. Corner Plots

The L and W in the corner plots refer to $0.9 \cdot L_{DES}$ and $0.9 \cdot W_{DES}$, respectively.

Threshold voltage in linear region V_{TLIN}

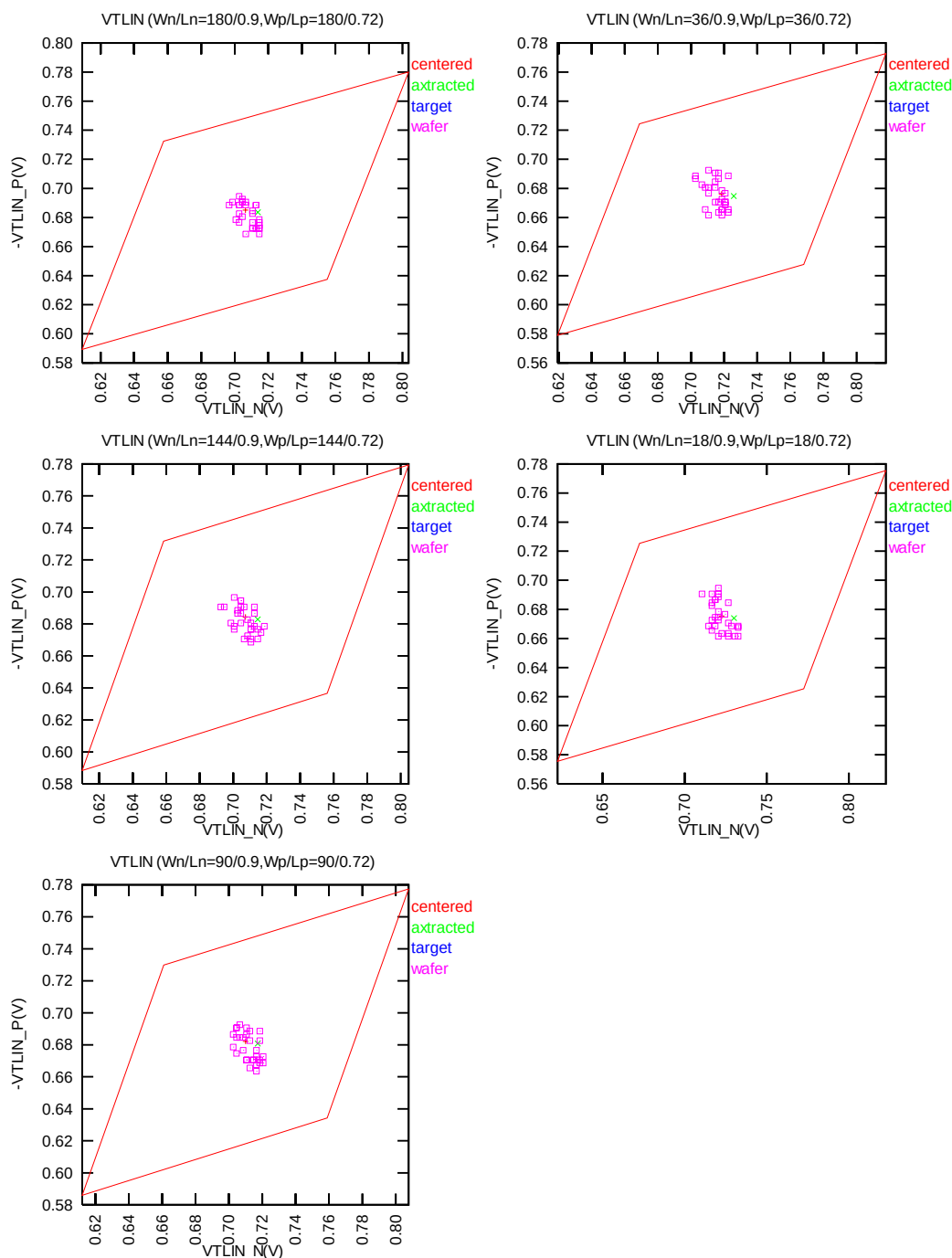


Figure 4-1 Corner plots of threshold voltage at low drain bias

Threshold voltage in linear region V_{TSAT}

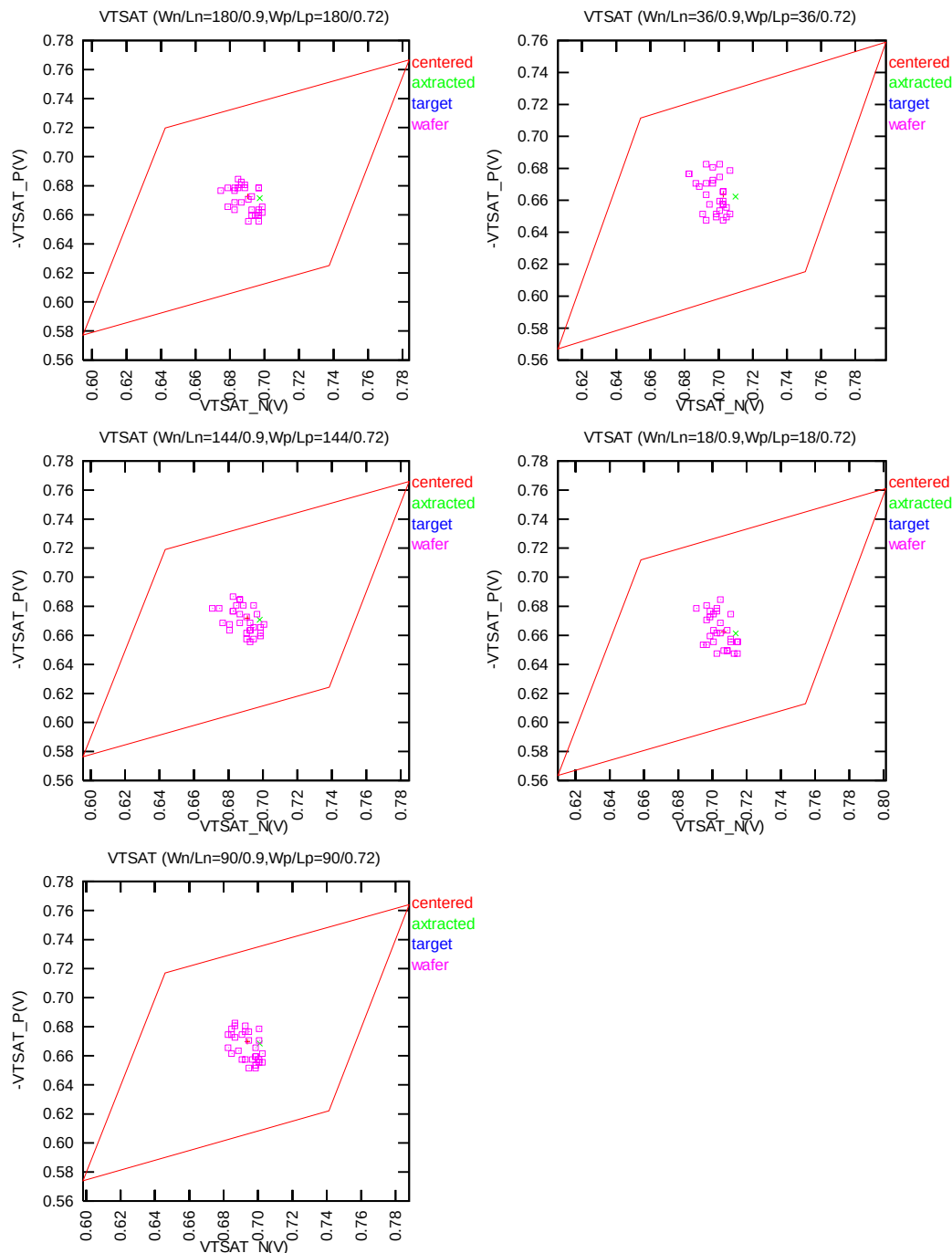


Figure 4-2 Corner plots of threshold voltage at high drain bias

Linear current I_{DLIN}

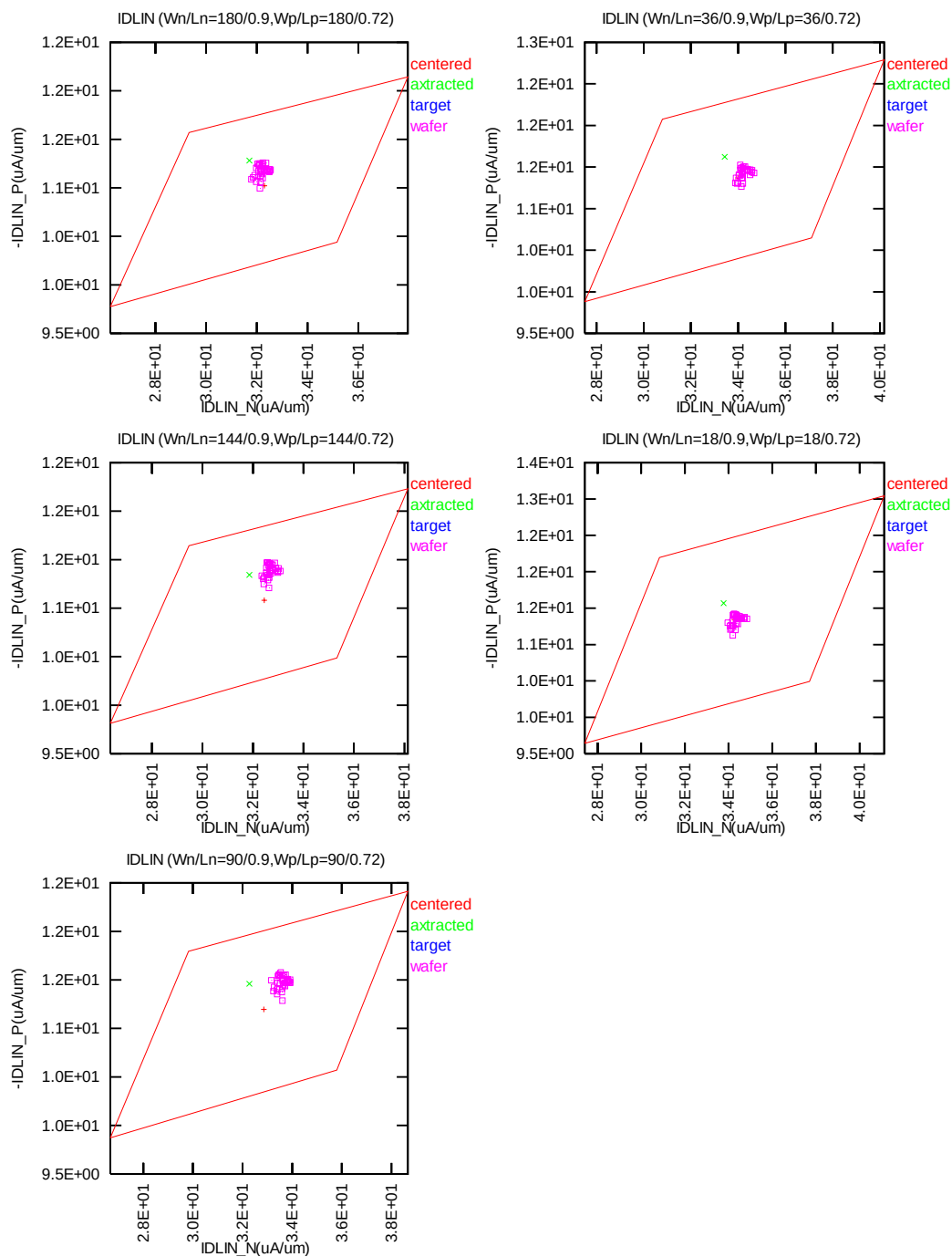


Figure 4-3 Corner plots of linear current

Saturation current I_{ON}

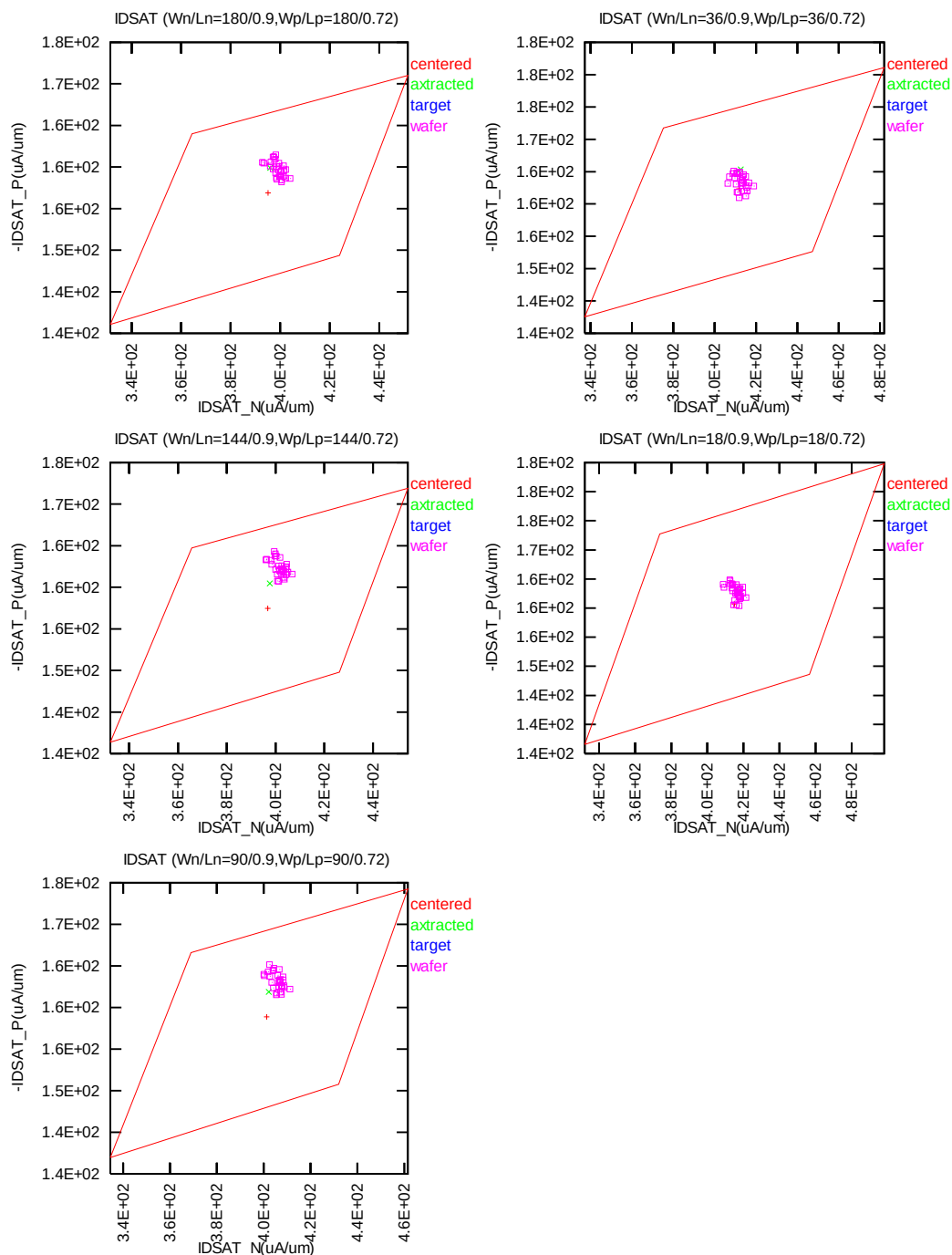


Figure 4-4 Corner plots of on current

Off current I_{OFF}

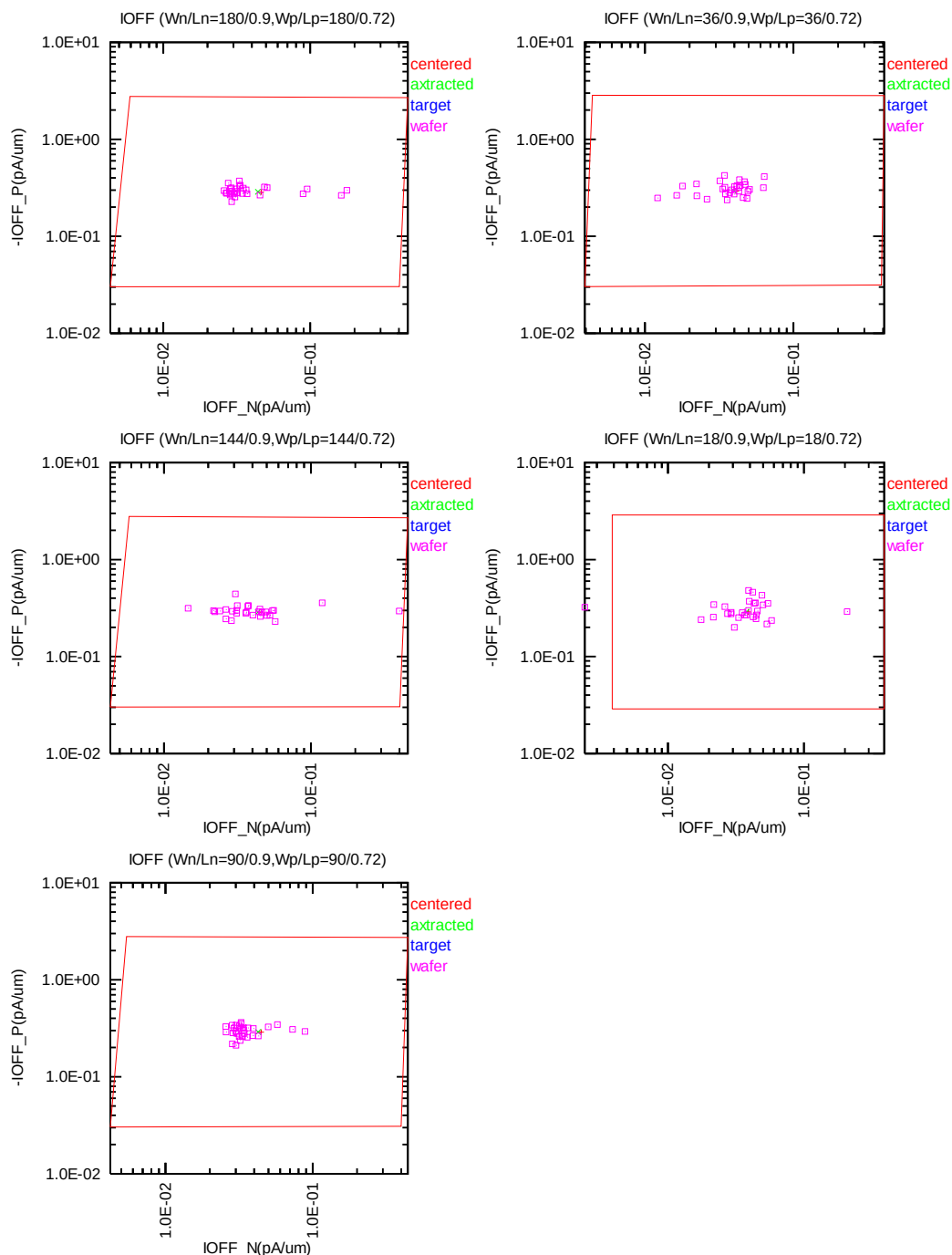


Figure 4-5 Corner plots of off current

4.3 Device Performance Reference Tables

Table 4-2 NMOS I_D current at $V_{GS}=V_{DS}=V_{CC}/2$ and $V_{GS}=V_{DS}=V_{CC}$.

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IDSAT (μA)	VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	18/0.9	8323.40	8415.40	7374.60	7470.90	6008.70	6114.40
SS	18/0.9	6837.00	6909.40	5899.20	5976.70	4633.60	4722.00
FF	18/0.9	9785.40	9897.70	8849.90	8965.00	7425.60	7546.40
SNFP	18/0.9	7592.00	7674.20	6636.90	6723.80	5306.10	5403.20
FNFP	18/0.9	9048.50	9150.50	8112.20	8218.00	6723.00	6836.70

Table 4-3 NMOS I_D current density at $V_{GS}=V_{DS}=V_{CC}/2$ and $V_{GS}=V_{DS}=V_{CC}$.

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IDSAT (μA/um)	VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	18/0.9	462.41	467.52	409.70	415.05	333.82	339.69
SS	18/0.9	379.83	383.86	327.73	332.04	257.42	262.33
FF	18/0.9	543.63	549.87	491.66	498.06	412.53	419.24
SNFP	18/0.9	421.78	426.34	368.72	373.54	294.78	300.18
FNFP	18/0.9	502.69	508.36	450.68	456.56	373.50	379.82

Table 4-4 NMOS V_{TLIN} and V_{TSAT} .

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
VTH (V)	VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	18/0.9	0.786	0.776	0.723	0.707	0.633	0.606
SS	18/0.9	0.880	0.867	0.823	0.801	0.743	0.705
FF	18/0.9	0.691	0.682	0.623	0.610	0.525	0.504
SNFP	18/0.9	0.833	0.822	0.773	0.754	0.687	0.656
FNFP	18/0.9	0.739	0.730	0.673	0.658	0.579	0.555

Table 4-5 NMOS I_D current at $V_{GS}=0$ and $V_{DS}=0.1V$ or $V_{DS}=V_{CC}$ (off current).

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IOFF (A)	VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	18/0.9	2.49E-16	6.02E-13	8.58E-14	7.00E-13	1.39E-10	1.57E-10
SS	18/0.9	2.02E-16	6.16E-14	7.61E-15	7.00E-14	7.24E-11	7.85E-11
FF	18/0.9	1.43E-15	5.68E-12	1.15E-12	7.00E-12	6.34E-10	7.49E-10
SNFP	18/0.9	2.10E-16	4.34E-14	2.34E-14	7.00E-14	9.00E-11	9.91E-11
FNFP	18/0.9	4.58E-16	6.63E-12	3.25E-13	7.00E-12	2.76E-10	3.25E-10

Table 4-6 PMOS I_D current at $V_{GS}=V_{DS}= -V_{CC}/2$ and $V_{GS}=V_{DS}= -V_{CC}$.

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IDSAT (μA)	-VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	18/0.72	3105.20	3151.00	2845.00	2892.30	2457.90	2503.50
SS	18/0.72	2649.20	2688.10	2418.00	2458.40	2080.40	2119.80
FF	18/0.72	3563.60	3616.80	3271.90	3326.10	2830.80	2882.20
SNFP	18/0.72	3332.10	3381.50	3058.40	3109.20	2648.20	2697.00
FNFP	18/0.72	2878.50	2920.90	2631.50	2675.40	2266.90	2309.30

Table 4-7 PMOS I_D current density at $V_{GS}=V_{DS}= -V_{CC}/2$ and $V_{GS}=V_{DS}= -V_{CC}$.

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IDSAT (μA/um)	-VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	18/0.72	172.51	175.06	158.06	160.68	136.55	139.08
SS	18/0.72	147.18	149.34	134.33	136.58	115.58	117.77
FF	18/0.72	197.98	200.93	181.77	184.78	157.27	160.12
SNFP	18/0.72	185.12	187.86	169.91	172.73	147.12	149.83
FNFP	18/0.72	159.92	162.27	146.19	148.63	125.94	128.29

Table 4-8 PMOS V_{TLIN} and V_{TSAT} .

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-VTH (V)	-VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	18/0.72	0.759	0.749	0.675	0.662	0.555	0.536
SS	18/0.72	0.856	0.846	0.775	0.761	0.659	0.638
FF	18/0.72	0.661	0.652	0.575	0.563	0.452	0.435
SNFP	18/0.72	0.710	0.701	0.625	0.613	0.504	0.486
FNFP	18/0.72	0.808	0.798	0.725	0.712	0.607	0.587

Table 4-9 PMOS I_D current at $V_{GS}=0$ and $V_{DS}= -0.1V$ or $V_{DS}= -V_{CC}$ (off current).

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IOFF (A)	-VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	18/0.72	1.07E-15	4.43E-12	6.82E-13	5.20E-12	2.90E-10	3.36E-10
SS	18/0.72	2.48E-16	4.45E-13	6.70E-14	5.20E-13	4.90E-11	5.61E-11
FF	18/0.72	1.73E-14	4.36E-11	7.37E-12	5.20E-11	1.79E-09	2.10E-09
SNFP	18/0.72	4.28E-15	4.93E-11	2.34E-12	5.20E-11	7.39E-10	8.97E-10
FNFP	18/0.72	3.86E-16	2.93E-13	2.01E-13	5.20E-13	1.15E-10	1.31E-10

5 Appendix

5.1 HSPICE Warning Message

Table 5-1 HSPICE warning message

Item	Warning Message	Explanation
1.	None	None

5.2 ELDO Warning Message

Table 5-2 Eldo warning message

Item	Warning Message	Explanation
1.	None	None

5.3 SPECTRE Warning Message

Table 5-3 Spectre warning message

Item	Warning Message	Explanation
1	Parasitic resistors 'rd' and 'rs' have been deleted because its value of 61.7855 mOhm (R/MFactor) was smaller than `minr'	Both rd and rs are defaulted to be zero.
2	Parasitic resistors 'rd' and 'rs' have been deleted because its value of 53.5881 mOhm (R/MFactor) was smaller than `minr'	Both rd and rs are defaulted to be zero.